AvareX User Manual

This manual documents all features in the current AvareX app, including:

- **What features exist**
- **How to reach them in the UI**
- **What each feature does**
- **Platform-specific availability**
- **Annotated screenshots of every major screen**
- **Step-by-step walkthroughs for common workflows**

> Tip: Screenshots in this manual were captured from the macOS desktop build. Mobile builds (iOS/Android) reflow some controls into bottom sheets, but the labels and workflow remain the same.

AvareX Releases

0.0.138

- Added Emergency mode under Map Settings, Profiles, Emergency. It opens a north-up map with the glide ring and four cards for best glide speed, the best runway to land on, a checklist, and the frequencies to declare.
- Pick the Emergency checklist from the Emergency tile on the Profile screen.
- The glide circle, plan cruise, and aircraft icon use the aircraft chosen on Aircraft & Performance.
- Chart downloads work out the chart cycle from the date.

0.0.137

- Built-in aircraft include cruise speed, fuel endurance, best glide, sink rate, and best climb.
- The glide circle and the PLAN altitude icon use those figures when the profile leaves them empty.
- 3D draws each airport's runway diagram at the field.
- 3D leaves out the circle at a surface airport.

- Help copies this version's manual into Documents, named with the version, and drops the copy from an older install.
- The weather altitude slider label is larger.
- Menu `Messages` keeps each Apps4Av message until you swipe it away.
- Select the title, text, or web address in an Apps4Av message to copy it.

0.0.136

- Type an altitude on each PLAN waypoint.
- The altitude icon picks the fastest cruise altitude from winds and the aircraft.
- Add Best Climb Speed and Best Climb Rate. The altitude icon uses them.
- Opening PLAN fills cruise speed and fuel flow from the aircraft.
- A procedure altitude is copied into the plan Alt column.
- 3D draws the plan at each leg's altitude.
- 3D shelf floors sit on the terrain.
- 3D TFR shows AGL or MSL, and the picture includes that ground.
- TFR details show AGL or MSL.
- Onboarding downloads Elevation with the other charts.
- Tap the MT tile to switch North Up and Track Up. Olive means track-up.
- The weather altitude slider sits under the Pro icon.
- Map and Plate stay loaded when you switch between them.
- 3D Wind starts off. Turn it on for colored winds-aloft streaks.
- Wind streaks in 3D stay above the terrain.
- The plan wind chart keeps the last forecast while a new one downloads.

0.0.135

- Share the active plan from Menu `Link`. One device is Master. Slave devices copy that plan and the active waypoint.
- Sync plates to the active airport waypoint.

0.0.134

- Open the PFD from `Menu`.
- Show an Apps4Av message when you open the app. Tap `OK`.

0.0.133

- Open 3D Class B, C, D, and SUA from the destination `3D` tab.
- Shelf labels such as `25–43` (`SF` is surface).
- Airspace above 10,000 ft is cut open.
- Vector Map labels such as `SF–50`. Zoomed out, Class B is drawn first.

- Menu `3D` adds terrain and ADS-B dots.
- Menu `Cloud` is `Internet`.
- 3D draws the plan at its altitude.
- 3D draws the glide ring, and TFR or Game TFR when on.
- Ownship pulses once a second.
- 3D chips hide B, C, D, SUA, terrain, TFR, Game TFR, the plan, or obstacles.

0.0.132

- Draw a holding pattern from the Hold button beside a fix or navaid name on the Main tab. A dialog sets the inbound course, turn direction, and a 1.5, 2, 3, or 4 NM leg.
- A map profile saves whichever items are checked: layers, weather, traffic volume, instruments, plan, plate, chart type, aircraft, and checklist. Applying the profile restores those items.

0.0.131

- Draw Elevation terrain in 3D behind the PFD horizon.
- Draw the lined-up runway in 3D on the PFD within 10 NM.
- Hold the PFD OBS knobs to set a course; the Nav course line and CDI follow, and a double-tap returns to the GPS course.
- Coast ownship after the last GPS fix until the No GPS lock warning, then freeze.

0.0.130

- Fixed ETA and ETE to use live ground speed.

0.0.129

- Update Flight Intelligence to Gemini 3.8 Flash with Google Search.

0.0.128

- Add Van's RV-9A to built-in aircraft performance.
- Translate one NOTAM from the destination NOTAM tab.
- Delete a map profile from its three-dot menu.
- Applying a map profile keeps the current chart, north up, and audible alerts.
- Bug fixes.

0.0.127

- Save and apply named map profiles from Map Settings.
- Show RP under the runway number for right traffic.

- Translate a single NOTAM with Flight Intelligence from the NOTAM tab.
- Bug fixes.

0.0.126

- Fetch airport NOTAMs from the FAA production NOTAM service.
- Bug fixes.

0.0.125

- Update the Terms of Use and require a new signature.
- After signing updated terms, skip setup if onboarding was already finished.
- Bug fixes.

0.0.124

- Bug fixes.

0.0.123

- Bug fixes.

0.0.122

- Bug fixes.

0.0.121

- Bug fixes.

0.0.120

- Enable Cloud features on Windows and macOS.
- Fix Plan waypoint reorder so drops land on the intended index.
- Fix sign-in.
- Rearrange weather, traffic, and nav buttons.
- Bug fixes.

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1) Before You Start

1.1 Safety and intended use

AvareX displays an in-app Terms of Use during onboarding. You must **Sign** before you can continue. The terms state that AvareX is **not an FAA-certified GPS** or approved navigation, weather, traffic, terrain, or flight-planning system; that you must assume the software and its data will fail or be wrong; that **you as pilot in command** are solely responsible for the flight; and that **Apps4Av Inc.** and the people who write, publish, or distribute the software have no liability for its use. The same screen includes the account privacy notice.

1.2 Internet and GPS requirements

- Internet is required for:
  - Chart/database downloads
  - Weather/doc products
  - Flight Intelligence (AI)
  - FAA plan services / cloud backup
- GPS can come from:
  - Internal device GPS
  - External ADS-B / GPS streams over **UDP ports 4000, 43211, 49002**
  - External Bluetooth stream (Android IO screen)

1.3 First-run onboarding checklist

On first run, complete onboarding pages (titles match the screens):

1. **Sign the Terms of Use** (required to continue; swipe is blocked until you sign).
2. **Welcome to AvareX!**:
   - **Display Theme**: `Night` / `Day`
   - **Measurement Units**: `NM / Knots` (Maritime) or `SM / MPH` (Imperial)
3. **Internet and GPS**: confirm GPS permissions (`GPS Permissions`) and note that GPS can come from the device or an external ADS-B/GPS stream.
4. **Databases and Maps**: AvareX **automatically** downloads what is missing for your current GPS region — the **DatabasesX** package plus the region's **Sectional**, **Plates**, **CSUP**, and **Elevation** — in one blocking step. Nothing to tap. Use **Choose Manually** to open the full `Download` screen if you want to pick specific data.
5. **Keep Warnings in Check**: when the red warning icon appears, tap it for troubleshooting.
6. **Optimize Performance**: turn off unused map layers (layers icon on the MAP screen) for best performance.
7. **File Flight Plans** (optional): enter the same email you use at 1800wxbrief.com to register for FAA flight-plan workflows.
8. **Join the Community**: Apps4Av forum at `https://groups.google.com/g/apps4av-forum` .

You can reopen onboarding later from the drawer header icon (`Show intro screens`).

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2) Platform Feature Availability

Feature	Availability
Core app (Map/Plate/Plan/Find, downloads, documents, aircraft & performance, checklist, logbook)	All supported platforms
Track Viewer with 2D/3D views (KML files)	All supported platforms
Vector map layer with airspace	All supported platforms
CAP Grid layer	All supported platforms
Bluetooth IO screen	**Android only**
Device Link (one device shares the plan with the others over Wi-Fi)	Not available on web
Plan Transfer to/from Avidyne IFD (Wi-Fi)	All supported platforms
Avidyne IFD ADS-B traffic & weather (Capstone GDL90 over Wi-Fi)	All supported platforms
Flight Intelligence (Pro)	**iOS and Android only**
Internet drawer section (Account, Backup/Sync, Community, Aircraft Scheduler)	**iOS, Android, Windows, and macOS**
Apps4Av messages (popup, and Menu `Messages` until you delete one)	**iOS, Android, Windows, and macOS**
Airport Businesses & Reviews (free, sign-in required)	**iOS, Android, Windows, and macOS**
PDF viewing in Documents/Help	Not available on Linux
File sharing from Documents/Logbook export	Not available on Linux

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3) Main Navigation

After onboarding, the app opens to `MainScreen` with:

- **Bottom tabs**: `MAP`, `PLATE`, `PLAN`, `FIND`
- **Left drawer menu** (opened by `Menu` button on map)
- **Warnings drawer** (right side, opened by red warning icon when issues exist)

3.1 Bottom tabs

1. **MAP** — moving map, overlays, weather, traffic, route visualization, tools.
2. **PLATE** — approach plates/airport diagrams/CSUP with ownship overlay.
3. **PLAN** — build, edit, brief/file, and manage flight plans.
4. **FIND** — search destinations, recent list, nearest-airport shortcuts.

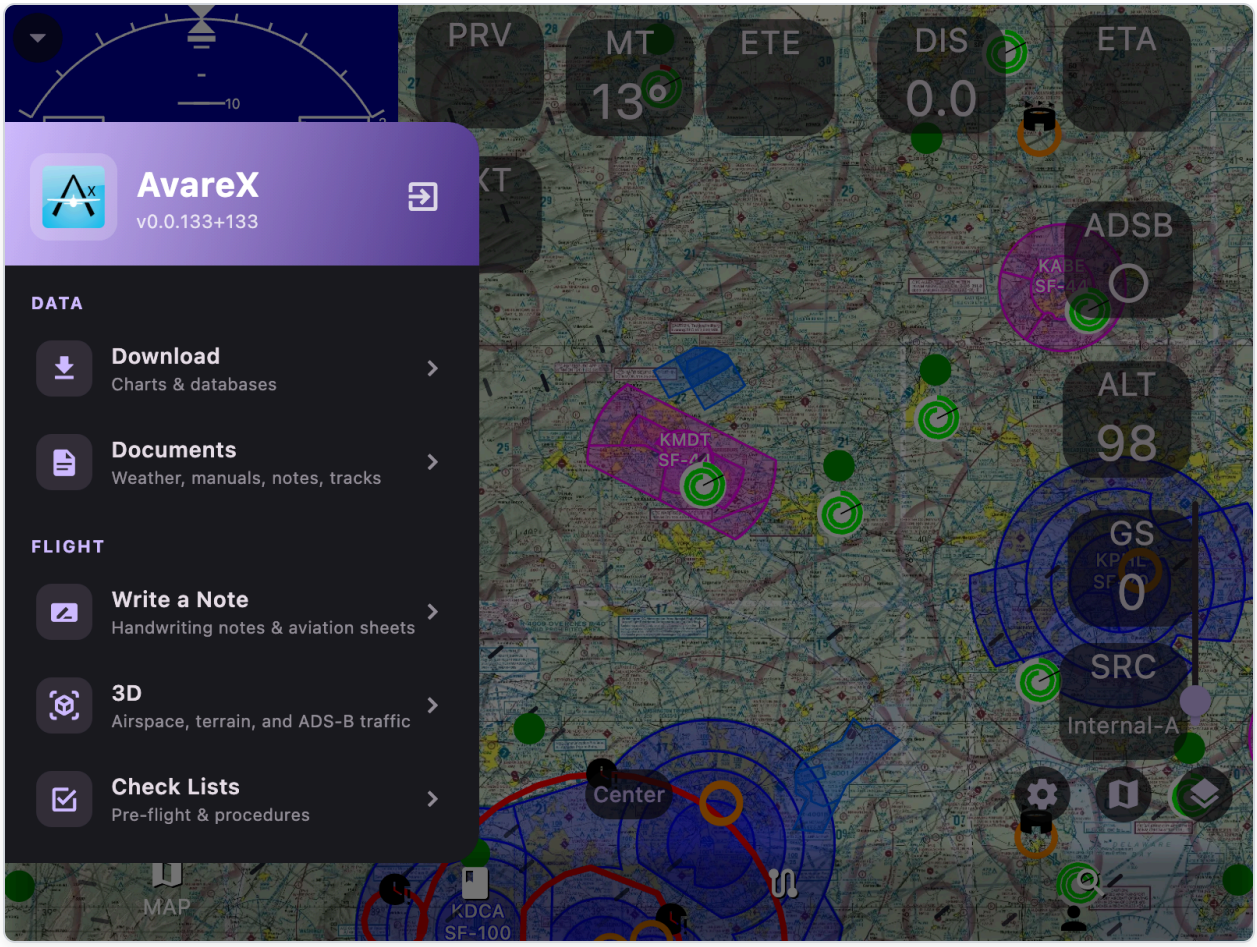
3.2 Drawer menu entries

Open from MAP with **Menu** button (bottom-left). Items are grouped by section headers:

- **DATA**: `Download` (Charts & databases), `Documents` (Weather, manuals, notes, tracks)
- **FLIGHT**: `Write a Note`, `3D`, `3D Weather`, `PFD`, `Check Lists`, `Aircraft & Performance`
- **RECORDS**: `Log Book`
- **INTERNET** (iOS/Android/Windows/macOS): `Account` (Sign in, profile, sign out), `Backup/Sync`, `Community` (unread notifications bell), `Scheduler`, and `Flight Intelligence` (iOS/Android only; Pro)
- **CONNECTIVITY**: `Link` (share the active plan with other AvareX devices on the same Wi-Fi). `IO` (Android only; Bluetooth & connections)
- **SUPPORT**: `Messages` (Notices from Apps4Av; iOS, Android, Windows, and macOS), `Help` (User manual PDF; not available on Linux)

`3D`, under `Write a Note`, opens the same airspace view as the destination popup, centered on the aircraft, without the yellow long-press mark. Downloaded Elevation tiles are drawn as terrain (brown, and red within 500 ft of altitude). Shelf floors sit on that surface instead of extending below it. ADS-B traffic inside 25 NM is drawn as dots. The active plan is drawn at each leg's altitude. The glide ring, and `TFR` and `Game TFR` volumes when those layers are on, are drawn here too. Each airport inside 25 NM is drawn as its runway diagram at the field. See Section 9.8. For winds aloft and 3D radar, use `3D Weather` (Section 9.8.1).

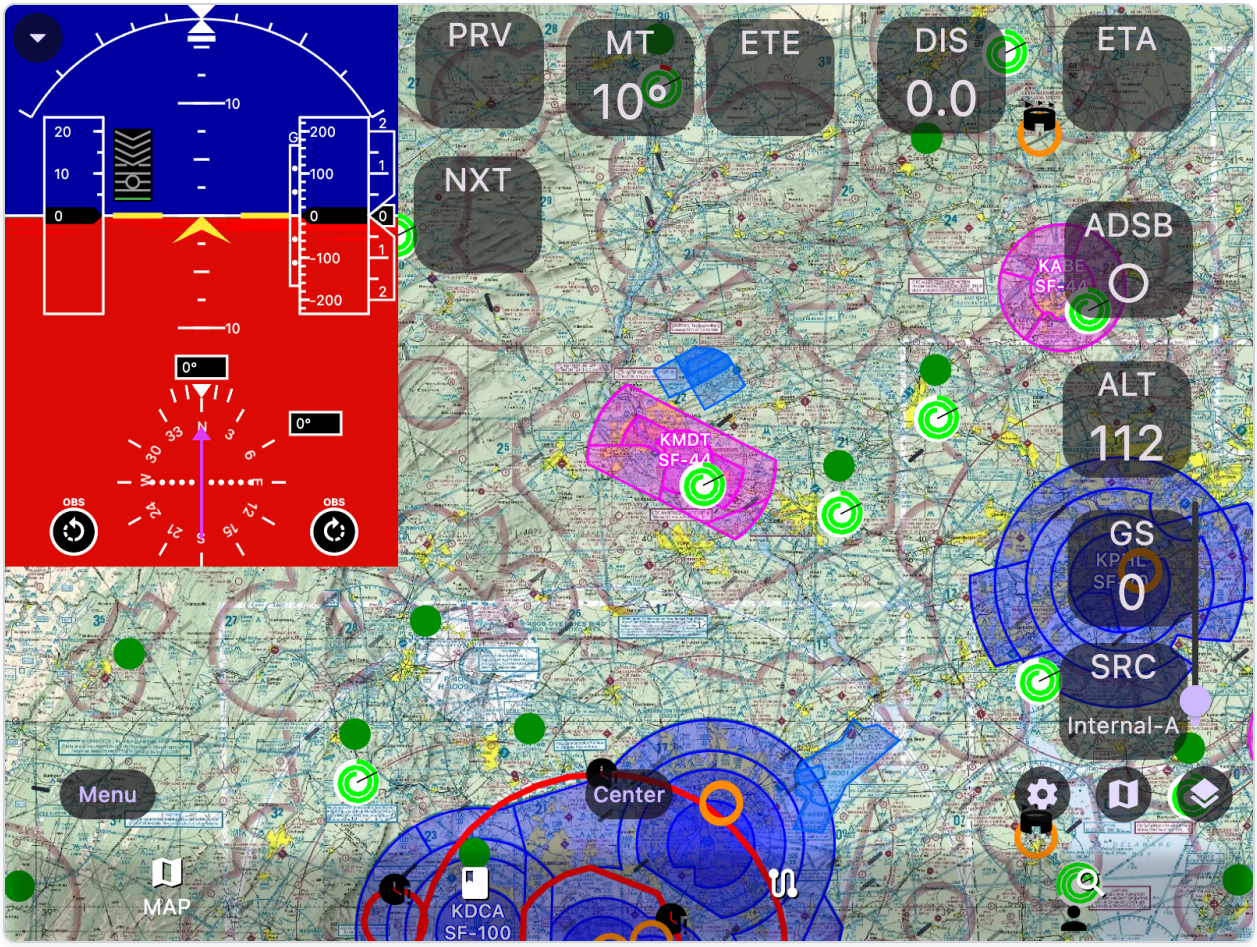
`PFD`, under `3D`, opens the same Primary Flight Display as the map `PFD` layer, on its own screen. The map layer is unchanged. See Section 9.9.



4) MAP Tab (Primary Flight Display + Map Tools)

4.1 How to access

Tap bottom tab **MAP**.



The **instrument tiles** float as a movable overlay over the map (GS, ALT, MT, PRV, NXT, DIS, BRG, GEL, ETA, ETE, VSR, UPT…); each tile can be dragged anywhere and its position is remembered. All map controls sit on a single bottom row: **Menu** (drawer) at the left, **Center** in the middle, and **Map Settings, Chart Type, Layers** at the right, plus the warning icon top-right.

4.2 Core interactions

- Pan/zoom map
- Track-up or North-up orientation toggle (`Map Settings` , or tap the `MT` tile)
- Long-press behavior:
  - If ruler mode OFF: opens destination popup near pressed location
  - If ruler mode ON: drops ruler points for distance/bearing measurement
- Single tap dismisses pop-up toasts

4.3 Map controls (buttons and menus)

Top-right

- **Pro icon** (iOS/Android): opens `Flight Intelligence` when you are signed in, or the `Account` sign-in screen if you are not. Manage the account from `MAP → Menu → Internet →`

`Account`.

- **Altitude slider** (under the Pro icon, when Cloud tops, Icing, Turbulence, Ceiling, or Wind vectors are on): vertical slider, 0 to 30,000 ft.
- **Red warning icon** (if active): opens troubleshooting drawer.

Bottom center

- **Center** button:
  - Tap: recenter on ownship (current zoom)
  - Long press: recenter and zoom to chart max zoom

Bottom-left

- **Menu** button: opens drawer list (Download, Documents, etc.)

Bottom-right control row (scrollable)

- **Map Settings popup** (gear icon): opens the `Map Settings` list described below.
- **Chart type popup**: selects chart source type.
- **Layers popup**: per-layer on/off via opacity slider.

Map Settings popup

Tap the gear icon in the bottom-right control row to open the `Map Settings` list. Each label shows that setting's current state, and active settings are highlighted. Tap the close (X) icon or outside the popup to dismiss it.

- **Measure Distance On / Measure Distance Off**: toggles measure mode; long-press map to add points.
- **Rubber Banding On / Rubber Banding Off**: toggles dragging route waypoints directly on map.
- **Audible Alerts On / Audible Alerts Off**: toggles audible alerts (traffic, GPWS, runway awareness).
- **North Up / Track Up**: shows the current orientation mode; tap to switch. Tapping the **MT** instrument tile does the same thing. The tile is olive in track-up and the normal tile color in north-up.
- **Profiles**: tap the `Profiles` heading to open the Profile screen, where a name can be saved and a profile deleted. Saved profiles are listed under the heading by name. Tap a name to apply the items that were checked for that profile. The red `Emergency` profile is always first. It opens the Emergency map instead of applying settings (see 9.4.2).

Top-left

- **Instrument tiles menu** (arrow dropdown): tile sizing, lock/reset layout, and show/hide individual instrument tiles (including the `ADSB` tile).

Nav sub-layers (in the Layers popup)

- With the **Nav** layer above 0%, **Obstacles**, **Tape**, **Circles**, and **CAP Grid** appear indented under it, each with its own opacity slider.
- Each one's effective opacity is its own slider multiplied by the **Nav** slider, so turning **Nav** off hides the route and all four.

Weather products (in the Layers popup)

- With the **Weather** layer above 0%, the weather products appear indented under it, each with its own opacity slider. Enable any combination of **ADS-B Radar**, **ADS-B Cloud Tops**, **ADS-B Icing**, **ADS-B Turbulence**, **ADS-B Lightning**, **Radar**, **Ceiling**, and **Wind Vectors**. ADS-B items show an antenna icon; Radar, Ceiling, and Wind Vectors show an internet/radar icon. **Radar** is internet Mesonet radar animation.
- Each product's effective opacity is its own slider multiplied by the **Weather** layer slider, so turning **Weather** down dims all products.

Traffic volume (in the Layers popup)

- With the **Traffic** layer above 0%, a **Volume** row appears indented under it with three choices:
  - `S`: 20 aircraft, 3000 ft, 10 NM
  - `M`: 200 aircraft, 6000 ft, 50 NM
  - `L`: 1000 aircraft, 30000 ft, 500 NM

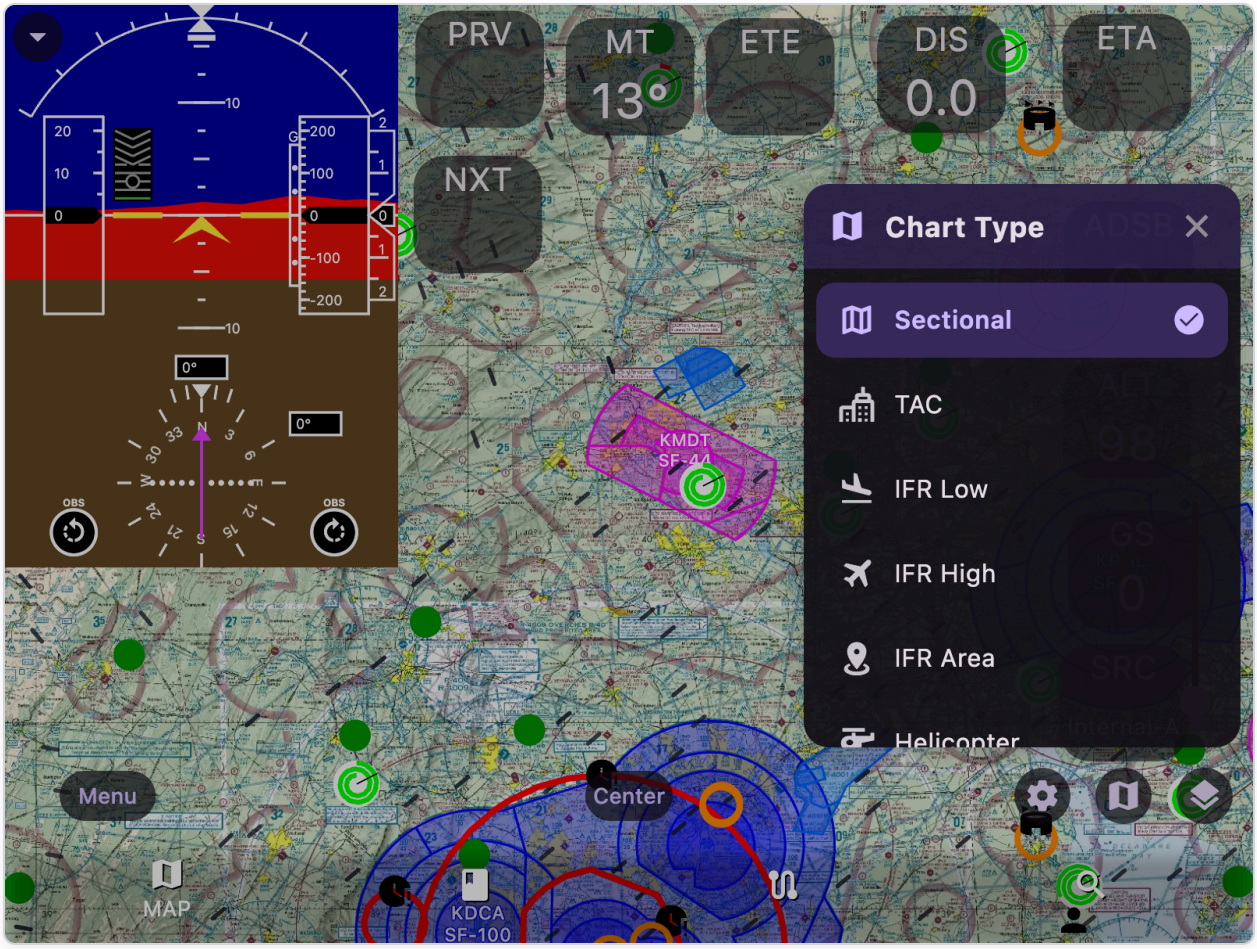
Altitude slider (under the Pro icon)

- Appears on the right, under the Pro icon, with the **Weather** layer when a product needing an altitude is on: Cloud tops, Icing, Turbulence, Ceiling, or Wind vectors.
- Range: 0 to 30,000 ft (1,000 ft increments). The altitude label is larger while you drag.

4.4 Map chart types

Selectable map chart types with max zoom levels:

Chart Type	Max Zoom	Description
Sectional	10	VFR sectional charts
TAC	11	Terminal Area Charts (metro areas)
IFR Low	10	IFR Enroute Low Altitude charts
IFR High	9	IFR Enroute High Altitude charts
IFR Area	11	IFR Area charts
Helicopter	12	Helicopter route charts
Flyway	11	VFR Flyway planning charts

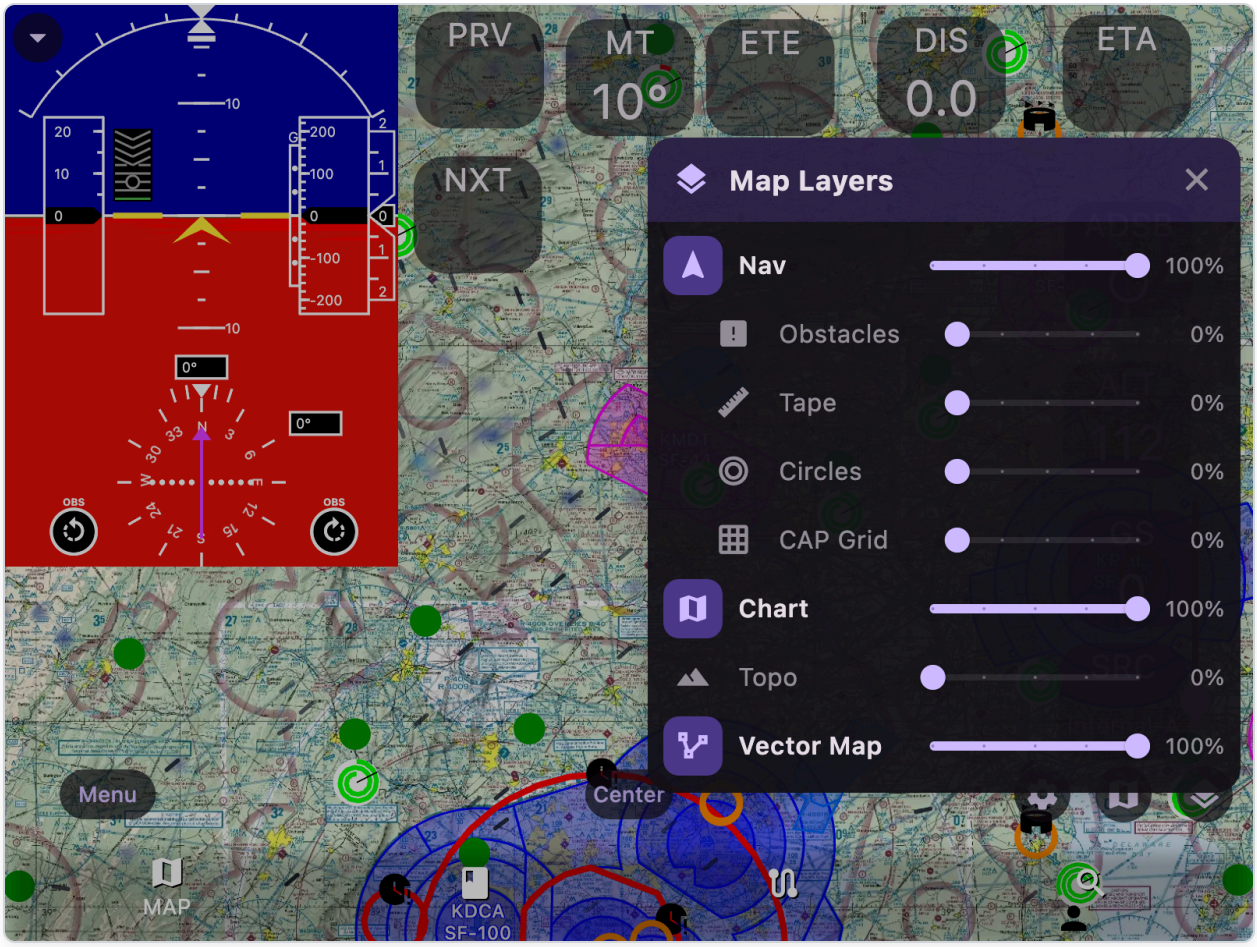


4.5 Map layers and what they do

Layer list from settings (with per-layer opacity 0-100%):

Layer	Description
Nav	Route lines (cyan=passed, purple=current, gray=next), runway depiction, waypoint markers/labels, ownship symbol, wind barb, north indicator. **Obstacles**, **Tape**, **Circles**, and **CAP Grid** are listed under **Nav** in this popup, each with its own opacity slider.
Circles (under Nav)	Range rings (10/5/2 NM black rings), speed ring (blue with a yellow halo, 1-minute travel distance), glide circle (white with an orange halo, power-off glide range at the selected aircraft's best glide speed) + labels. The 10 NM ring carries a magnetic compass rose with ticks and labels at the eight major points: `N`, `NE`, `E`, `SE`, `S`, `SW`, `W`, `NW`.
Chart	Downloaded FAA chart tiles (offline chart base).
Vector Map	NASR vector tiles from MBTiles files, with Class B/C/D airspace and SUA (MOA, Restricted, Warning, Alert, Prohibited, NSA) rendered with standard aviation colors. Zoomed out, only Class B is drawn. MOAs, restricted areas, and warning areas appear next, then Class C. Class D, smaller special-use areas, and each sector altitude (`25–43`, `05` is 500 ft, `SF` is the surface) appear up close. The airport name, such as `KMHT`, is at the center.
CAP Grid (under Nav)	Civil Air Patrol grid overlay with grid identifiers (e.g., BOS42, SEA123). Only visible at zoom level 9+.
Topo	USGS topo online base map (max zoom 16).
Elevation	Downloaded elevation tiles with color-coded terrain.
Weather	METAR/TAF/PIREP/AIRMET/SIGMET symbols plus selectable weather products (multiple at once) listed under **Weather** in this popup: ADS-B Radar, ADS-B Cloud Tops, ADS-B Icing, ADS-B Turbulence, ADS-B Lightning, Radar (internet), Ceiling, and Wind Vectors — each with its own opacity slider like map layers. Use the altitude slider when tops/icing/turbulence/ceiling/winds are on.
TFR	Temporary flight restriction shapes/markers with time validity. Red=active, orange=future.
Game TFR	Stadium / motor speedway TFRs (14 CFR 99.7): orange 3 NM rings and a stadium marker at NFL, MLB, NCAA Division I football, and major speedway

Layer	Description
	venues. Applies when the event is in progress, SFC–3000 ft AGL. Tap a marker for the venue name.
Plate	Georeferenced plate overlay on map (when loaded).
Traffic	Avare-style simple traffic dots with a 1-minute projection line drawn in the direction of travel. Cyan circle (with black outline) for normal/proximate traffic, red circle (with black outline) for threat traffic (advisory or resolution alert). Each dot is annotated with the relative flight-level offset, vertical trend arrow, and callsign. Integrates with audible alerts.
Obstacles (under Nav)	Obstacle markers (red squares) in your vicinity.
Tape (under Nav)	Distance tape labels from ownship upward in NM.
GeoJSON	Imported user GeoJSON polygons/markers.
PFD	Inset Primary Flight Display panel (artificial horizon, speed/altitude tapes, VSI, compass, CDI/VDI, AOA indicator, turn coordinator). Requires AHRS data. Downloaded Elevation tiles are drawn in 3D behind the horizon (brown ground, red within 500 ft of altitude like the map Elevation layer). Within 10 NM of a runway you are lined up with, the runway is drawn in 3D on top of that terrain (pavement, edges, displaced threshold arrows and bar, threshold stripes, 1000-ft aiming point markers, centerline dashes, runway numbers).
Tracks	Ownship breadcrumb/polyline track recording (green line).



Notes:

- Turning **Tracks** layer OFF saves a KML track file into Documents/tracks and clears active track recording.
- **PFD** requires AHRS data from GDL90-compatible devices to show attitude. Two black **OBS** knobs sit on the bottom of the PFD under the speed and altitude tapes (white rings, circular arrows). Hold the right knob to turn the selected course clockwise, or the left knob counterclockwise (the course box and CDI needle follow; the knobs turn white). The brown plane-to-destination line on the **Nav** map layer rotates with that course through a full 360° and runs from the waypoint toward an aircraft flying inbound (it does not extend past the waypoint). Double-tap either knob to return to the GPS/route course. OBS clears when the active waypoint changes.
- The **PFD** synthetic runway is picked automatically from the closest airport (or the plan's active airport waypoint): the runway end whose heading is within 45° of your track, with you inside 35° of its extended centerline, up to 10 NM out. It is positioned from GPS position and altitude and the runway's database coordinates, elevation, and displaced threshold. With live AHRS it moves with pitch, roll, and heading exactly like the horizon line; without AHRS it is drawn level along the GPS track. It is an aid, not a substitute for an approved approach.

- **PFD** terrain uses the same Elevation tiles as the map. Ground is the usual brown; terrain is red when it is within 500 ft of your altitude or higher — the same red threshold as the map Elevation layer. Sky stays blue. Download Elevation charts or the mesh is omitted and the brown ground remains. It is an aid, not a TAWS.

4.6 Instrument overlay (movable tiles)

Instrument tiles float as a full-screen overlay on top of the map. Each tile can be **dragged independently** to any position on the screen, and its position is remembered across app restarts (separate layouts are kept for portrait and landscape orientation). By default only the first few tiles are shown; every tile has a row in the menu with a **+/–** toggle to show or hide it. Available tiles include:

Tile	Description	Tap Action
GS	Ground speed (knots/mph)	—
ALT	GPS altitude (feet)	—
MT	Magnetic track (degrees). Olive while the map is track-up; normal tile color while north-up.	Toggle North Up / Track Up
PRV	Previous waypoint identifier	Jump to previous waypoint
NXT	Next waypoint identifier	Jump to next waypoint
DIS	Distance to next waypoint (NM/mi)	—
BRG	Bearing to next waypoint (magnetic)	—
GEL	Ground elevation (requires Elevation charts)	—
ETA	Estimated time of arrival (HH:MM)	—
ETE	Estimated time en-route (HH:MM)	—
VSR	VSI required to arrive 1000ft above destination	—
UPT	Up timer (count up, green when running)	Start/stop timer
DNT	Down timer (count down from 30min, red when expired)	Start/stop timer

Tile	Description	Tap Action
UTC	Current UTC time (HH:MM)	—
SRC	GPS source mode. Shows `Internal`, `External`, `Internal-A`, or `External-A` (A=Auto mode)	Tap to cycle modes: Auto, Internal, External. Green=Internal, Blue=External
FLT	Total flight time in hours	Reset timer
ADSB	ADS-B receiver status (from GDL90 heartbeat + FIS-B uplinks). Shows your own tail number when the receiver reports it; otherwise a status circle — green filled circle ● = connected with GPS, yellow filled circle ● = connected without GPS (partial), empty circle ○ = disconnected. Only the text/circle is colored; the tile background is unchanged. Enable from the tiles menu.	Open the **ADS-B Status** screen

Interactions:

- **Drag** any tile to move it anywhere on the screen; its position is saved automatically.
- **Dropdown arrow** (top-left menu): **Expand/Contract** tile sizes, **Lock Tiles/Unlock Tiles** to prevent or allow moving tiles (unlocked by default), a row for every tile with a **+** (show) / **–** (hide) toggle, **Reset Layout** to restore the default tiles and positions, and a **Help** entry with tile descriptions.
- **Tap the `ADSB` tile** to open the **ADS-B Status** screen (back button in the app bar). It shows live receiver status (connected, GPS position valid, UTC timing OK, UAT initialized, ground stations received) and a scrolling log of the **last 50 received messages**, newest first. Use the **filter** button (funnel icon) to choose which message types appear in the log — toggle Heartbeat, Uplink, Ownship, Traffic, Basic/Long report, AHRS, Device, and more. **AHRS is off by default** because it arrives many times per second and would crowd out other messages. For FIS-B uplinks, the collapsed row also shows a short summary of the products carried (e.g. `uplink (0x07) – METAR, NEXRAD, TFR`). **Tap any message** to expand its decoded fields along with the **raw bytes** (selectable hex). Decoded detail per message type includes:
  - **Heartbeat**: GPS position valid, maintenance required, GPS battery low, UTC timing OK, UAT initialized, receiver UTC time, and uplink/traffic message counts.

    - **Traffic**: alert status, address type (ADS-B/TIS-B/surface/beacon), NIC/NACp integrity and accuracy, emergency/priority code, callsign, ICAO, position, altitude, ground speed, track/heading, vertical speed, emitter category, and airborne state.
    - **Basic/Long report**: target type, NIC, address qualifier, plus the common traffic fields above.
    - **Ownship**: ICAO, position, altitude, ground speed, track/heading with track type, vertical speed, NIC/NACp, report (updated/extrapolated), and airborne state.
    - **FIS-B uplink**: ground station position, position-valid flag, slot ID, TIS-B site ID, and each weather product carried in the frame, decoded individually:
        - **METAR/SPECI/TAF/PIREP/Winds Aloft**: the full textual report.
        - **NOTAM / NOTAM-TFR**: NOTAM text, or for TFRs the report number, polygon vertex count, altitude band (`AGL` or `MSL`), and effective/expiry times.
        - **D-ATIS / TWIP**: airport digital ATIS and terminal weather text (also stored for the airport **NOTAM** tab).
        - **NEXRAD radar** (regional or CONUS): the bin block number and whether the block carries precipitation or is empty.
        - **Icing / Turbulence / Cloud tops / Lightning**: block number, scale, altitude band (icing and turbulence), and whether the block carries data or is empty.
        - **SIGMET / AIRMET / SUA**: location, polygon vertex count, effective times, and any embedded text.
    - **Device report**: battery charge percentage and charging state.

  Each message type has its own **color**, so you can tell types apart at a glance. The same colors are used in the **filter** list (funnel icon) next to each message type, so the log and the filter selector stay in sync. When a traffic target is suppressed, the row also carries a text tag — `filtered: ownship` (its ID matched yours, so it was treated as "us") or `filtered: out of range` (outside the current S/M/L traffic distance/altitude window). This makes it easy to confirm whether a nearby aircraft (for example one received via **TIS-B**) is being received but suppressed rather than never received at all.

  Use **Pause**/**Resume** to freeze or continue the live log. Leaving the screen automatically pauses the log (it resumes when you reopen the screen).

- **Ground stations**: under the **Ground stations received** count, each currently-heard FIS-B ground station (tower) is listed, nearest first, showing its identity (**TIS-B site** number when it provides TIS-B, plus its UAT **slot**), its **distance and bearing** from your position, its latitude/longitude, and how many seconds ago it was last heard. Distance/bearing are shown only when your position is known.

- **Stratus Open ADS-B Mode**: a one-shot command via the antenna icon (top-right of the ADS-B Status screen) that broadcasts the command to put an Appareo **Stratus 3 / 3i** into

Open ADS-B (standard GDL90) mode so AvareX can receive its traffic and weather. Join the Stratus Wi-Fi network first (and grant **Local Network** on iOS). Tap the icon once; a toast confirms the command was sent. Official configuration is also available in Appareo's **Stratus Horizon Pro** app.

- **Diagnostics** (expandable card under the receiver status): reception-quality counters to help debug ADS-B problems. It shows the total decoded messages and average message rate, per-type counts (Heartbeat, Ownship, Traffic, Uplink), how long ago the last heartbeat/ownship/traffic message was seen, the uplink/traffic counts reported by the latest heartbeat, how many aircraft are currently tracked, how many traffic targets were filtered (ownship-match vs out-of-range), and error counts (**CRC**, **frame**, and **parse** errors — highlighted red when non-zero). Tap **Reset** to zero the counters. Use these to tell, for example, whether traffic is simply not being received versus being received but filtered.

4.7 GPWS (Ground Proximity Warning System)

- Monitors terrain ahead in direction of flight
- Predicts collisions within 3-minute lookahead
- Audible "PULL UP" alerts
- Requires: 30+ knot ground speed, Elevation data downloaded

4.7.1 Runway awareness (crossing alert)

Announces "approaching runway" when your ground track is about to meet a runway centerline, so you get a callout before taxiing onto an active runway.

Runways come from the **closest airport** to ownship (updated with the normal area refresh).

The callout fires when all of these hold:

- Ownship flight phase is **Taxi** (not Airborne) and moving at least 3 knots.
- Within about 250 ft of the runway centerline, or already on the pavement.
- Tracking across the runway rather than along it (heading more than 25° off the runway heading).
- The projected path meets the centerline within about 15 seconds, between the two runway ends.

A takeoff or landing roll — aligned on a runway above 15 knots — suppresses the alert, so rolling through an intersecting runway stays quiet. Once the flight phase is Airborne there is no crossing callout. Each runway announces once per approach and re-arms after you are clear of it.

Alerts share the global **Audible Alerts** control (`MAP → Map Settings`) with traffic and GPWS. This is an advisory aid only, not a substitute for ATC clearances or looking outside.

4.8 Weather marker interactions

- **METAR tap**: Shows full METAR text with flight category icon
- **TAF tap**: Shows TAF forecast
- **PIREP tap**: Shows pilot report
- **AIRMET/SIGMET tap**: Shows advisory text. **Long-press**: toggles shape visibility on map
- **TFR tap**: Shows TFR details, including whether each altitude is `AGL` or `MSL`
- **Game TFR tap**: Shows venue name and `Game TFR (3 NM, SFC–3000' AGL when active)`

4.9 Ruler/measurement tool

- Activate via `Measure Distance Off` in the `Map Settings` popup (the row reads `Measure Distance On` and is highlighted when active)
- Long-press map to add measurement points
- Shows distance (NM) and bearing (degrees) between points
- Multiple segments supported

4.10 Rubber banding

- Activate via `Rubber Banding Off` in the `Map Settings` popup (the row reads `Rubber Banding On` and is highlighted when active)
- Long-press and drag waypoint icons to reposition
- On release, snaps to nearest navaid/fix from database
- Must be explicitly enabled to prevent accidental edits

---

5) Destination Popup (Map Long-Press or Find Tap)

5.1 How to access

- Long press on map (when ruler mode is off), or
- Tap an item in FIND list

5.2 Top action buttons

- **→D**: set Direct-To destination, center map there, return to map

- **+Plan**: insert destination into current plan at the current waypoint position
- **↓Plan**: append destination to the end of the plan
- **Plates** (airport only): jump to the `PLATE` tab for that airport, even when it is not the active waypoint

5.3 Information pages (shown when data exists)

Possible tabs:

- **Main**: destination summary (airport info, runway diagram, or VOR/navaid info with nearby VORs). Each runway end is a circle (green is best for the current wind). Right traffic shows `RP` under the runway number; left traffic has no pattern mark. For a fix or navaid, the name and type (such as `YREPPOINT, GPS`) has **Hold** beside it. That opens a dialog for a holding pattern at this fix: inbound course, **Right** (standard) or **Left** turns, and a straight-leg length of **1.5**, **2**, **3**, or **4 NM** (default 4). **Draw on Map** draws a standard-rate racetrack; the turns use your ground speed (90 kt if you are slower than 60). Arrows on the two turns show the direction of flight. An orange dot on the fix is the entry point. Parallel and teardrop draw that entry from the fix, with arrows along it for the direction of travel. **Clear hold** removes it. One hold at a time.
- **AD**: airport diagram/runway depiction (interactive viewer with zoom)
- **METAR**: METAR + TAF text with flight category color indicator
- **NOTAM**: fetched NOTAM list from the FAA production NOTAM service (downloaded async). On iOS/Android, each NOTAM has a **Translate** icon that opens `Flight Intelligence` with only that NOTAM, asking for a plain-English translation (Pro).
- **SUA**: special-use airspace data for the area
- **Wind**: nearest winds-aloft station data at multiple altitudes
- **3D**: Class B, C, and D shelves, plus restricted, warning, prohibited, and MOA areas, within 25 NM of this place, from the Vector Map database. Drag to rotate, pinch to zoom, and drag with two fingers to pan. Class colors match the Vector Map (B blue, C magenta, D light blue). Restricted areas are red, warning areas are brown, prohibited areas are purple, and MOAs are olive. Each shelf is labeled with its altitude range, for example `25–43`. An airport with surface airspace is named on the ground, such as `KMHT`. Each airport inside 25 NM is also drawn as its runway diagram on the ground at the field. The pavement follows the runway ends. A field with one runway and no ends uses that runway's length and heading. Pavement matches the AD tab, with the runway numbers at the ends. Areas without an airport, such as `R–4101A SF–25`, keep the name on the shelf. Altitudes are hundreds of feet MSL (`25–43` is 2500–4300; `05` is 500; `SF` is the surface; `UNL` is unlimited). Anything above 10,000 ft is cut there so a tall area does not cover the airport. The cut is an open dashed top with a fading outline and upward marks, and the label still shows the real ceiling (`R–5203 SF–500`). Obstacles from the main database are red lines

from the ground to the obstacle top. A yellow line marks the long-press point. Downloaded Elevation tiles are drawn as terrain, brown and red within 500 ft of altitude. A shelf floor that is below that terrain, including surface Class D, is drawn from the ground up so the volume does not continue under the surface. The ownship is a larger white dot with a black ring that pulses once a second, so it is not a traffic dot, and it moves with the aircraft while it is inside this 25 NM area. The active plan is drawn at each leg's altitude: cyan for the part already flown, purple for the current leg, and grey ahead, matching the map. The line leaves a departure airport at field elevation and arrives at a destination airport at field elevation. The glide ring is a white circle with an orange halo on the surface, and an orange cone from the aircraft down to that circle. When the `TFR` layer is on, areas in effect are red volumes and later ones are brown. A TFR whose floor is above the picture is a dashed outline at the top of the scale with arrows pointing up. When `Game TFR` is on, each venue is an orange cylinder 3 NM across, from the surface to 3000 ft above the ground. Chips along the bottom show or hide Class `B`, `C`, and `D`, `SUA` (restricted, warning, prohibited, and MOA), terrain, `TFR`, `Game TFR`, the plan, and obstacles. **Reset** restores the view. If Vector Map is not downloaded, the page says so.

- **3DW**: the same 3D box as 3D Weather, for weather instead of airspace, centered on this place, with the yellow long-press mark. It shows the terrain, the runway diagrams, the aircraft, and the winds aloft. Use `Build NEXRAD` for 3D radar and `dBZ` for the weakest echo shown. See Section 9.8.1.
- **ST**: sounding chart/image for the area (Skew-T diagram)
- **Business**: for airports, lists the crowd-sourced businesses/FBOs for that airport inline (free; sign-in required) with details and reviews shown in place — see Section 11.7

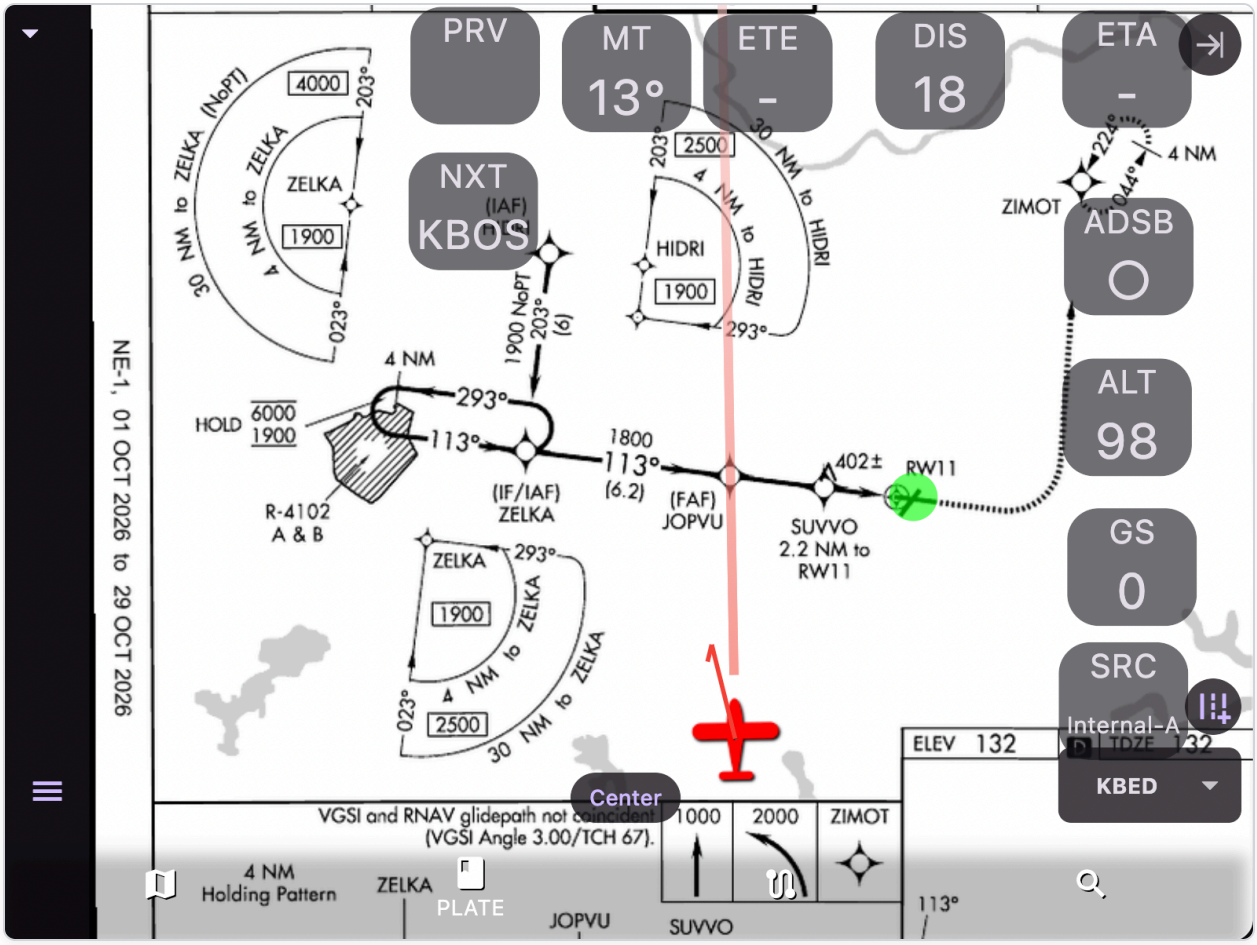
If nearby alternatives exist, a horizontal "Nearby" selector appears.

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6) PLATE Tab

6.1 How to access

- Bottom tab **PLATE**, or
- From destination popup using **Plates**



6.2 What it does

- Displays downloaded plates/diagrams/CSUP
- Draws ownship and heading on georeferenced plates
- Can overlay a selected business marker on airport diagrams (from the crowd-sourced Airport Businesses directory; iOS/Android/Windows/macOS, sign-in required)
- Supports zoom/pan via InteractiveViewer (up to 8x)

6.3 Controls

- **Airport selector** (bottom-right): choose airport from recent airports list
- **Plate selector** (bottom-left, always shown): choose plate within airport (color-coded by type). Always contains an `APS-AERIAL VIEW` entry — once downloaded it loads the georeferenced Google Maps satellite picture like any other plate; until then it appears as an `APS-AERIAL VIEW (Get)` placeholder that downloads the image when selected. When downloaded, a trash icon on that entry **deletes** the stored picture immediately; you can download it again later. The picture is stored in the airport's plates folder as `APS-AERIAL VIEW.png` with ownship drawn on it
- **Procedure menu** (plus icon near bottom-right):

    - Tap a procedure to show its profile, draw the CIFP route on georeferenced plates (progress-colored), and add its waypoint sequence to the plan
- **Instrument overlay show/hide** toggle (top-right arrow icon)
- **Airport diagram on landing** (top-right, airport diagram only): landing icon under the instrument toggle. On shows the nearest airport diagram after landing. Off leaves the current plate in place.
- **Sync plates** (directly under the landing icon, airport diagram only): on follows the active waypoint when that waypoint changes to a different airport, including while you are on another tab. The plate becomes that airport's diagram when one is downloaded, otherwise its first plate. A plate already open for that same airport stays. Off leaves whatever plate is showing. Opening `Plates` for a long-pressed airport still shows that airport.
- **Profile close (X)** closes procedure profile card and clears the route overlay
- **IAF chips** (black with white text, above profile when multiple transitions exist): switch IAF without re-adding to the plan

6.3a Procedure route on plate

When you select a procedure from the plate procedure menu (`+` / road icon):

- The vertical profile card appears
- On **georeferenced** plates, the CIFP route is drawn with colors counted back from the final fix (same on the vertical profile):
    - **Green** — final leg
    - **Yellow** — 1 before final
    - **Brown** — 2 before final
    - **Blue** — 3 before final
    - **Red** — 4 before final
    - **Orange** — 5+ before final
- Fix names are not drawn on the plate (keeps chart symbology clear)
- When the chart has multiple IAFs, **black chips** (white text) above the profile switch transitions (without re-adding to the plan)
- Closing the profile (X), or changing airport, removes the overlay
- Non-georeferenced plates show the profile only (no route overlay)

Route geometry follows the CIFP fix sequence in the databases (great-circle chords between fixes). Path-terminator arcs/holds are not drawn until those ARINC fields ship in a future DatabasesX cycle.

6.4 Plate type color coding

Color	Plate Type
Green	Airport Diagram (APD)
Blue	CSUP
Pink	Departure Procedure (DP)
Purple	Instrument Approach (IAP)
Cyan	STAR
Brown	Minimums (MIN)
Red	Hot Spots (HOT), Land and Hold Short (LAH)
Teal	Satellite View (`APS-AERIAL VIEW`)

6.5 Terrain caution overlay on plates

When map **Elevation** layer has opacity > 0, plate rendering can include colored terrain risk cells:

- **Yellow**: terrain within 500-1000 ft below aircraft
- **Red**: terrain within 500 ft or above aircraft

6.6 Business selector

On airport diagrams, when businesses with a known location exist for the airport, a right-side selector (business icon) lets you:

- Select a business from the crowd-sourced Airport Businesses directory
- See its location marked on the airport diagram, with its name labeled next to the marker

The businesses come from the shared Firestore directory (see Section 11.7), so this selector appears only on iOS/Android/Windows/macOS and when you are signed in.

6.7 Automatic plate changes

Two controls on an airport diagram, stacked under the instrument toggle, can change which airport is shown. Anything else leaves the current plate alone.

- **Landing** (landing icon): after landing, switch to the nearest airport diagram when that airport has one.

- **Sync** (sync icon): when the active waypoint changes to a different airport, switch to that airport. This runs on the `PLATE` tab and while another tab is showing. A fix, navaid, airway, or a plate already open for that same airport does not change. `Plates` on a long-pressed airport still opens that airport.

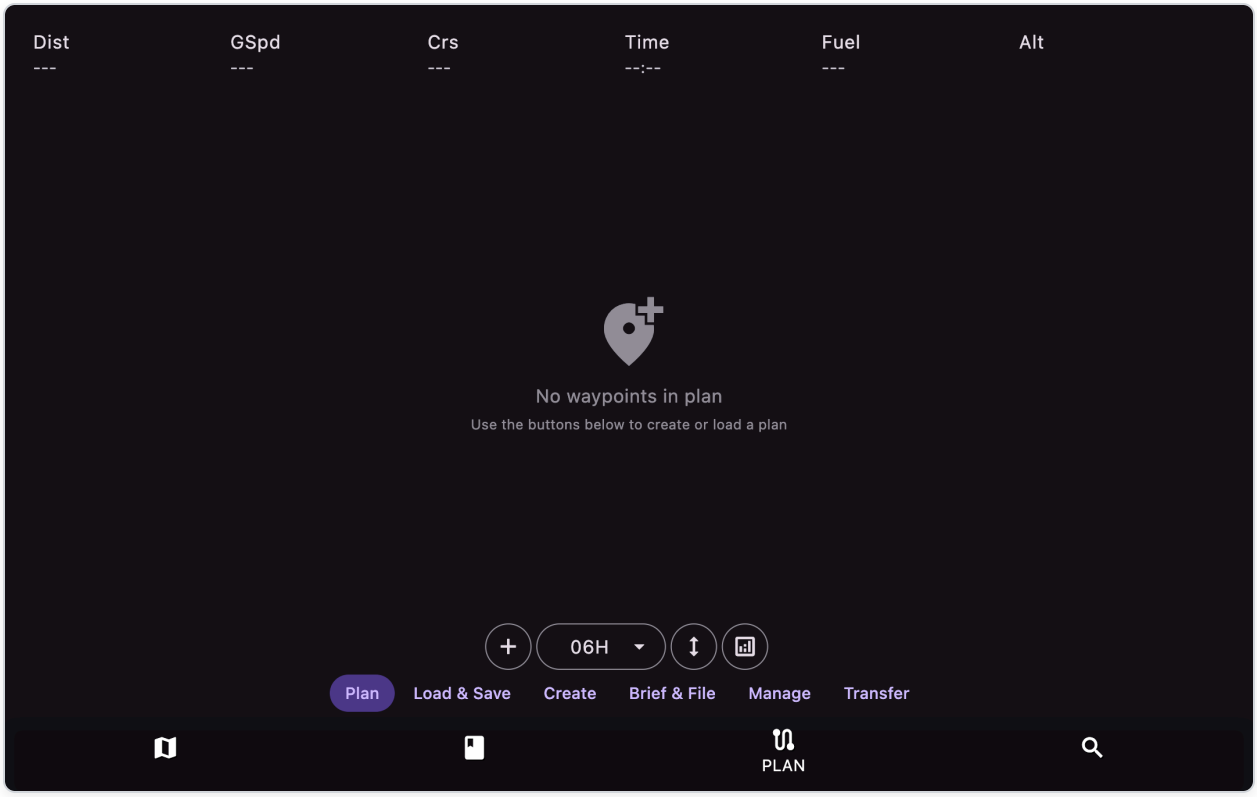
Turning a control off stops only that automatic change. The airport selector, plate selector, and long-press `Plates` button still work.

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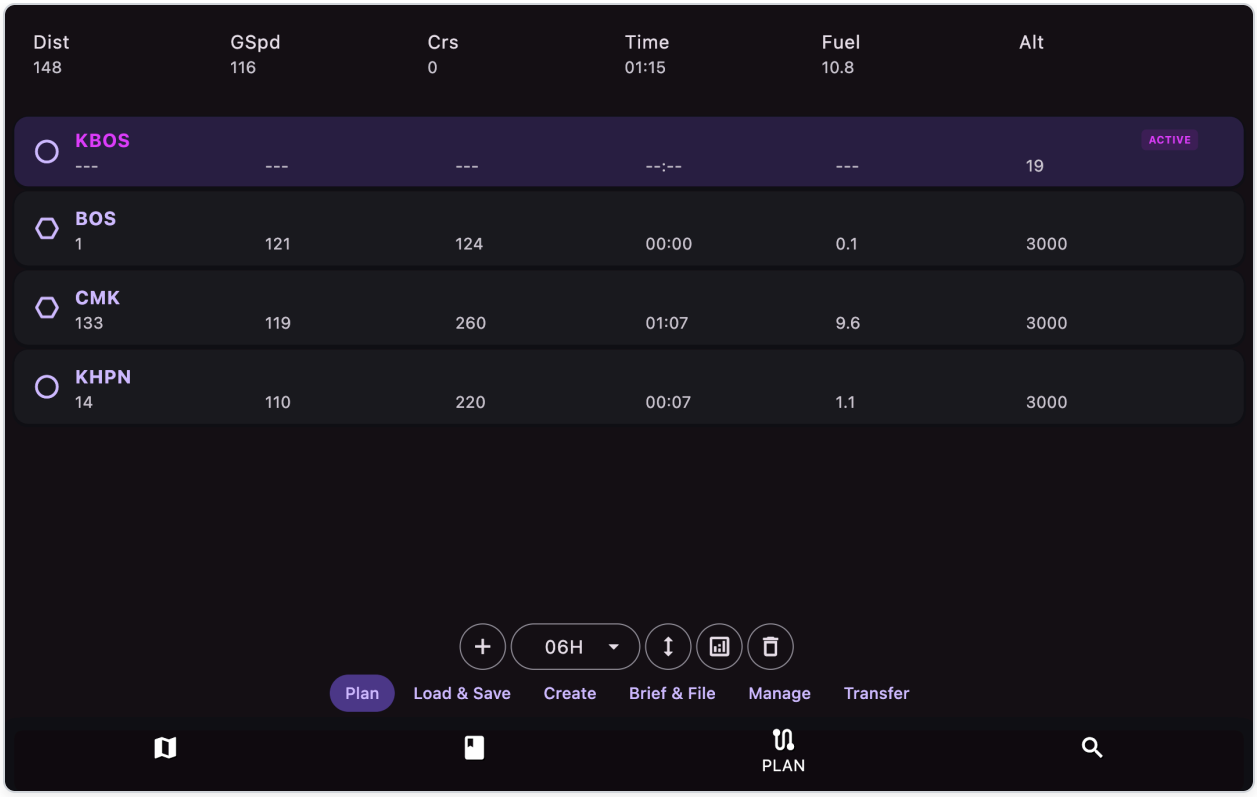
7) PLAN Tab

7.1 How to access

Tap bottom tab **PLAN**.



After waypoints are added (e.g. `KBOS BOS CMK KHPN`), each row shows distance/ground speed/course/time/fuel calculations:



7.2 Main plan editor

- Reorderable waypoint list (long-press and drag to reorder)
- Swipe row right-to-left to delete waypoint
- Tap row to set current waypoint (highlighted in purple with "ACTIVE" badge)
- When this device is a **Slave** in `Link`, the plan and the active waypoint follow the master. Edits made on the slave are replaced by the master's next update.
- **Insert waypoint**: tap a waypoint to select it, then press **+** (left of the wind forecast) to insert a new waypoint after that position (opens FIND tab; next selection inserts after the selected waypoint)
- Long press (grab) a waypoint to move it up and down
- Header row shows plan-wide calculated totals (Distance, Ground Speed, Course, Time, Fuel). Column labels are **Dist**, **GSpd**, **Crs**, **Time**, **Fuel**, **Alt**
- Each waypoint row has an **Alt** field (feet): the altitude at that waypoint. It starts at the plan default (3000 ft, or the map altitude slider). The first waypoint, when it is an airport, uses that airport's field elevation. Moving an airport into the first row sets Alt back to field elevation. After the altitude icon runs, Alt is the altitude the airplane has reached there while it is still climbing. Once it levels at a cruise altitude, Alt stays on that altitude, and the arrival descent does not replace it with an in-between number. Climb and descent follow best climb speed and rate, so a one-mile leg stays near the altitude it started at and the climb continues on the next leg. The last waypoint stays on the cruise altitude that leg leaves; the path still descends to the runway. Type a new altitude and leave the field to aim that leg at it (ground speed, time, and fuel update). The field shows the altitude reached

there. Legs after that one keep the altitudes already chosen. Clear the field to go back to the plan default. Fixes inside an airway or procedure each have their own **Alt**.
- **Delete entire plan** (trash icon, right of the navigation-log button): confirms, then removes all waypoints from the active plan (hidden when the plan is empty)
- Airways and procedures show as expandable items with nested waypoints

Bottom controls are arranged in two rows. The **settings row** (on top) is only shown on the **Plan** tab and holds **+**, the forecast, the altitude icon, the navigation log, and delete:

- Opening PLAN reads cruise speed and fuel flow once from the selected aircraft. That aircraft is the **Performance** screen app-bar **dropdown** (latest selection). Cruise is interpolated at the plan default altitude using the **saved cruise power %** from the Performance **Cruise** tab (ISA deviation 0; custom *My Aircraft* cruise table when that tail is selected, otherwise built-in POH when a template is selected). If no power has been saved yet, **65%** is used. If you have never opened Performance, aircraft selection falls back to the map aircraft setting when set.
- **Altitude icon** (next to the forecast dropdown, tooltip *"Calculate altitudes"*): chooses cruise altitudes for the shortest time from winds aloft and the selected aircraft, at the saved cruise power % and standard-day (ISA deviation 0). The route starts at the departure field elevation and ends with a descent to the arrival field elevation. A climb is the fastest altitude for the rest of the route, at least 1,000 ft above the highest terrain in the next 50 miles, and then it waits 50 miles. A descent can begin 20 miles after the airplane is at that altitude, when the lower altitude is faster over those miles. Climb and descent use **Best Climb Speed** and **Best Climb Rate** from the aircraft. When those fields are empty, the selected built-in aircraft supplies its handbook climb speed and rate. If they are still empty, climb uses three quarters of cruise speed and a rate that slows from about 700 ft/min near the surface to about 100 ft/min at the top of the cruise table. Time and fuel on each leg use climb speed while the altitude is changing and cruise speed while level. Alt at each waypoint is the altitude reached there while the airplane is still climbing, so a short leg stays low while the climb continues. After the airplane levels off, Alt stays on that cruise altitude through the arrival descent. That altitude is a multiple of 500 ft: an odd thousand when the course from the first waypoint to the last is east, and an even thousand when that course is west. Terrain is read every mile, the same spacing as the plan chart. The cruise altitude and the wind used for ground speed stay at least 1,000 ft above the terrain. The altitude line climbs to that clearance at best climb rate, so the first miles of a departure are only as high as that rate allows. The path still meets the runway at the departure and arrival airports. A waypoint **Alt** you type yourself changes that leg to that altitude, using the cruise speed and fuel flow read when PLAN opened. Legs after it keep the altitudes the icon chose. A toast says: Calculated altitudes for shortest flight time

based on winds aloft and performance of the aircraft name. If winds aloft are not available, it still picks the fastest altitudes.

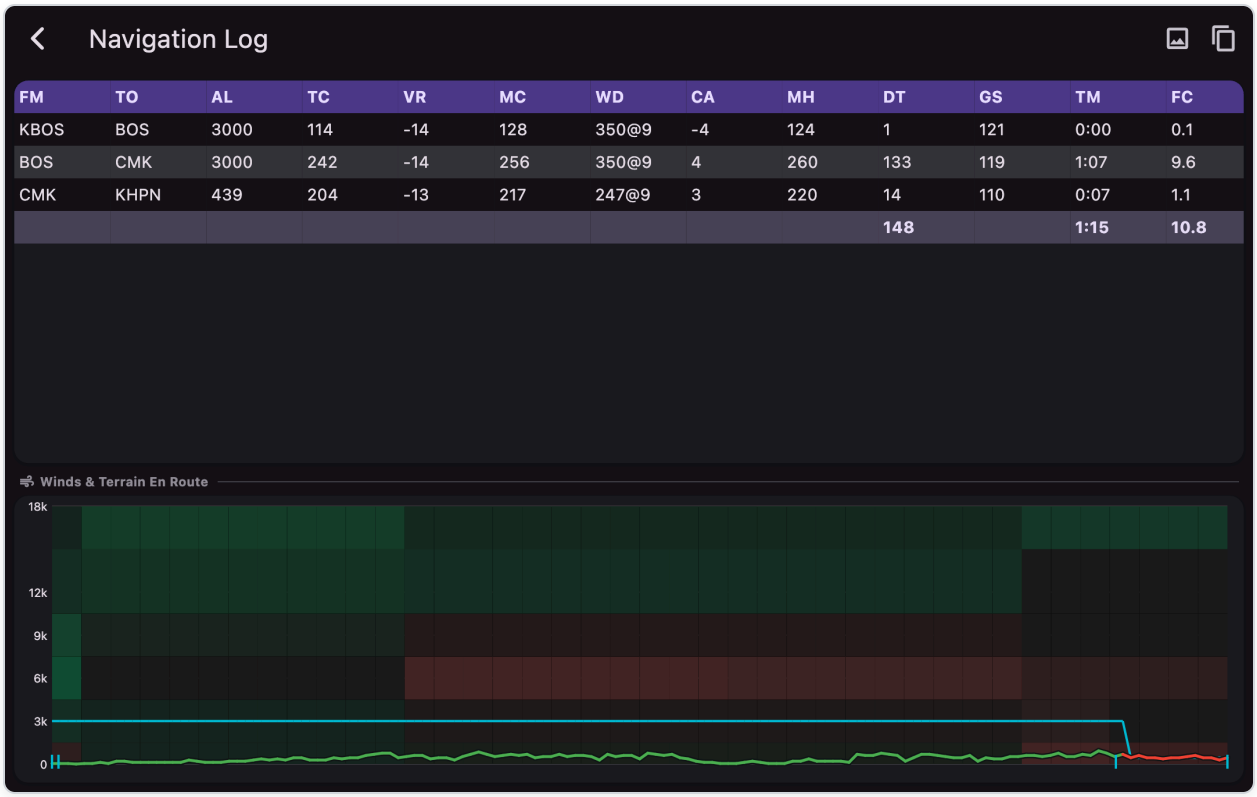
- **Forecast horizon** selector: 06H / 12H / 24H (winds aloft forecast period)
- **Analytics icon** (chart): opens full-screen Navigation Log

The **tab row** (below the settings row) switches what the PLAN tab shows; the active tab is highlighted:

- **Plan** (first tab): the waypoint list / plan editor (default)
- **Load & Save**, **Create**, **Brief & File**, **Manage**, **Transfer**: each shows the matching action sub-page inline; tap **Plan** to return to the waypoint list

7.3 Navigation Log dialog

Open from the PLAN tab by tapping the **analytics** icon (graph, tooltip *"Navigation log and terrain"*) on the settings row, to the right of the altitude icon.



FM	TO	AL	TC	VR	MC	WD	CA	MH	DT	GS	TM	FC
KBOS	BOS	3000	114	-14	128	350@9	-4	124	1	121	0:00	0.1
BOS	CMK	3000	242	-14	256	350@9	4	260	133	119	1:07	9.6
CMK	KHPN	439	204	-13	217	247@9	3	220	14	110	0:07	1.1
									148		**1:15**	**10.8**

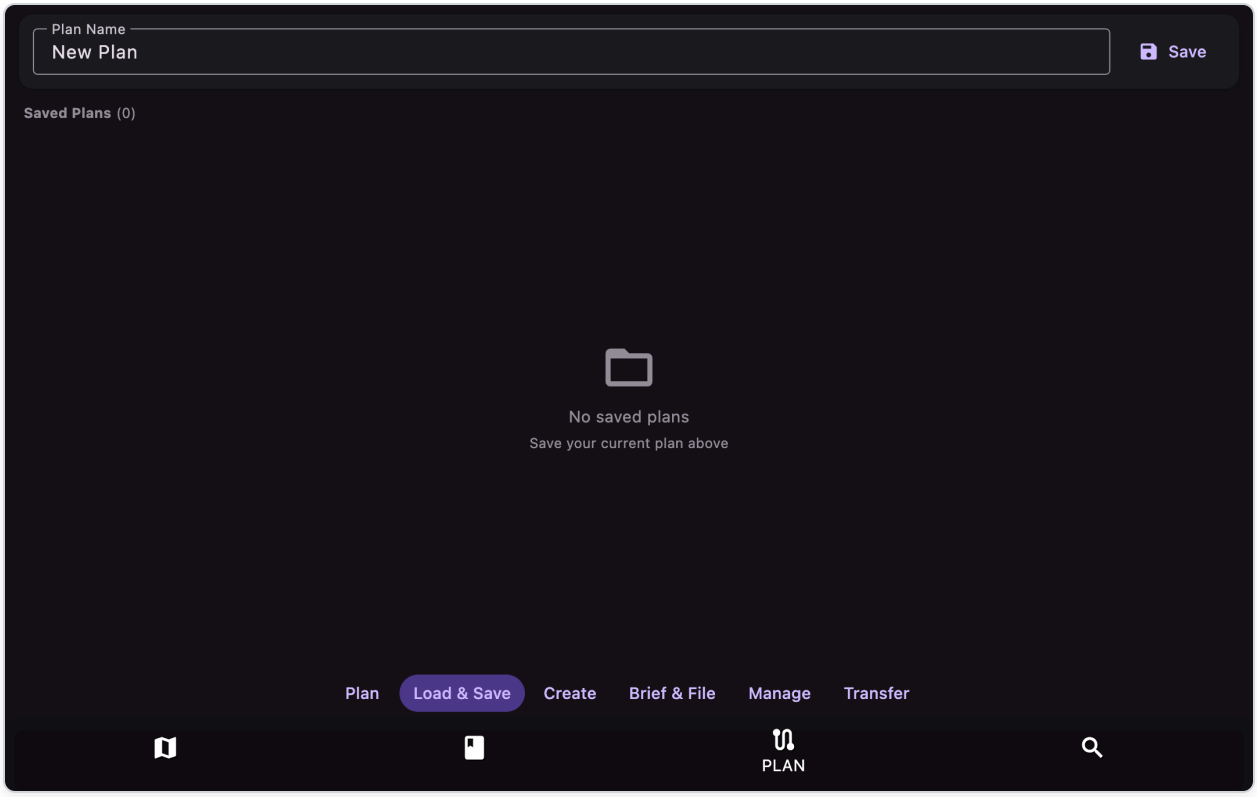
Includes:

- **Per-leg log grid** with columns: FM (from), TO (to), AL (altitude), TC (true course), VR (variation), MC (magnetic course), WD (wind direction@speed), CA (wind correction angle), MH (magnetic heading), DT (distance), GS (ground speed), TM (time), FC (fuel consumption). AL is the altitude at the waypoint you are flying to. On a short leg that is only as high as the climb has reached, and the cyan line keeps climbing after that

waypoint. After the airplane levels off, AL stays on that cruise altitude, and the line still descends to the runway. The last row stays on the cruise altitude its leg leaves. The bottom row of the grid totals DT/TM/FC for the whole plan.
    - Grid is zoomable/pannable; **double-tap** to reset zoom
- **Winds & Terrain En Route diagram**: combined visualization showing wind components and terrain profile:
    - **Wind heat map**: color-coded cells at altitudes 0-18000 ft along route (green = tailwind, red = headwind, darker = lighter winds, brighter = stronger winds up to 50 kt scale). The last forecast stays on the chart while a newer one downloads
    - **Plan altitude line**: cyan line of the altitude flown. After the altitude icon runs, it climbs and descends at best climb speed and rate, levels off at the altitude that is fastest for the rest of the route, and can step down after 20 miles. A short leg rises only as far as that rate allows, and the climb continues after that waypoint. It ends at the arrival field elevation in one steady descent when the wind changes along the way. Typing an altitude moves that leg to the altitude you typed. The line after that leg stays on the altitudes already chosen and still ends at the arrival field elevation
    - **Terrain profile overlay**: terrain elevation curve. Green where terrain is below that leg's altitude, red where it is at or above it
    - **Waypoint markers**: cyan tick marks at the bottom showing waypoint positions along the route
    - **Interactive tap**: tap anywhere on the diagram to display a tooltip with:
        - Nearest waypoint (if within 5% of route position)
        - Altitude at tap position (snaps to altitude bands: 0, 3000, 6000, 9000, 12000, 18000 ft)
        - Terrain elevation at that point
        - Course direction
        - Wind direction and speed
        - Tailwind/headwind component in knots
- **Copy plan to clipboard** button (app bar)

7.4 Plan actions (inline sub-pages)

The action sub-pages are shown inline in the PLAN tab by tapping one of the tab buttons at the bottom. The first tab, **Plan**, is the waypoint list / editor; the remaining tabs are **Load & Save**, **Create**, **Brief & File**, **Manage**, and **Transfer**. The tapped button is highlighted while its sub-page is displayed; tap **Plan** (or complete an action such as loading/creating a plan) to return to the waypoint list.

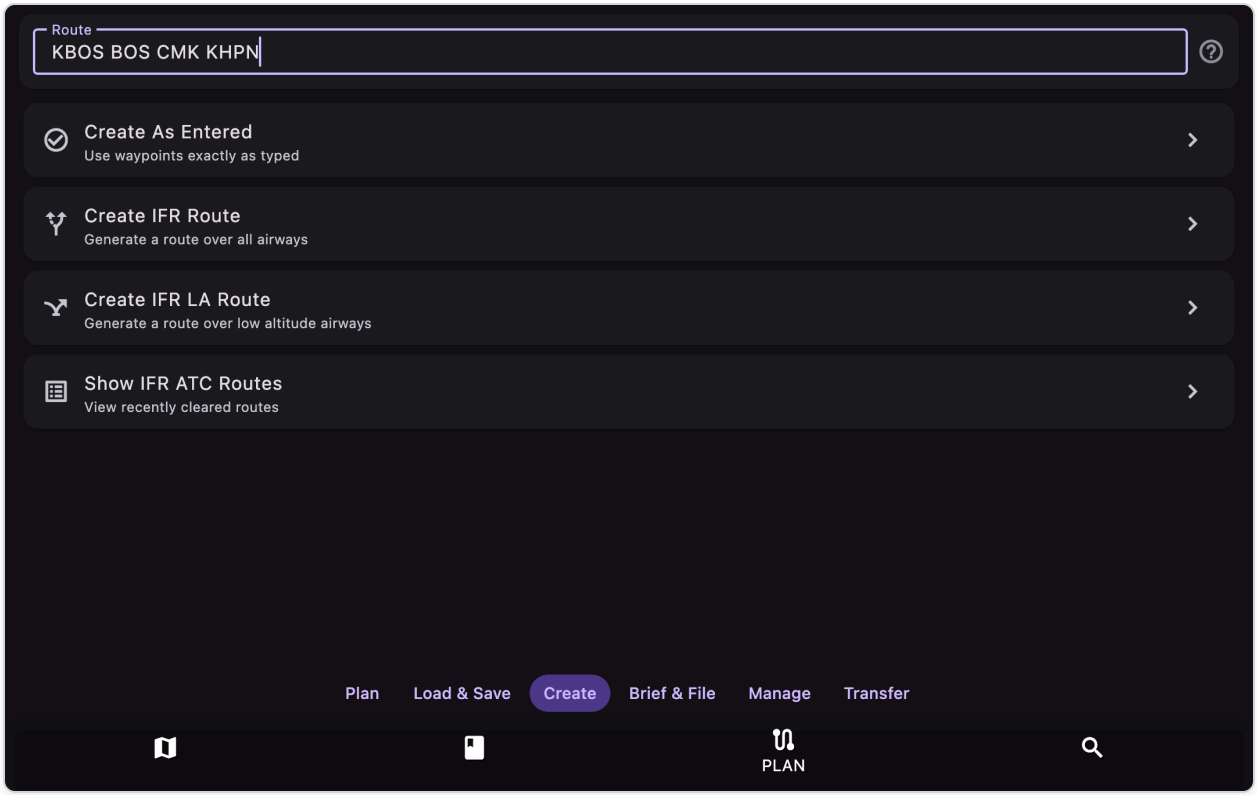


A) Load & Save

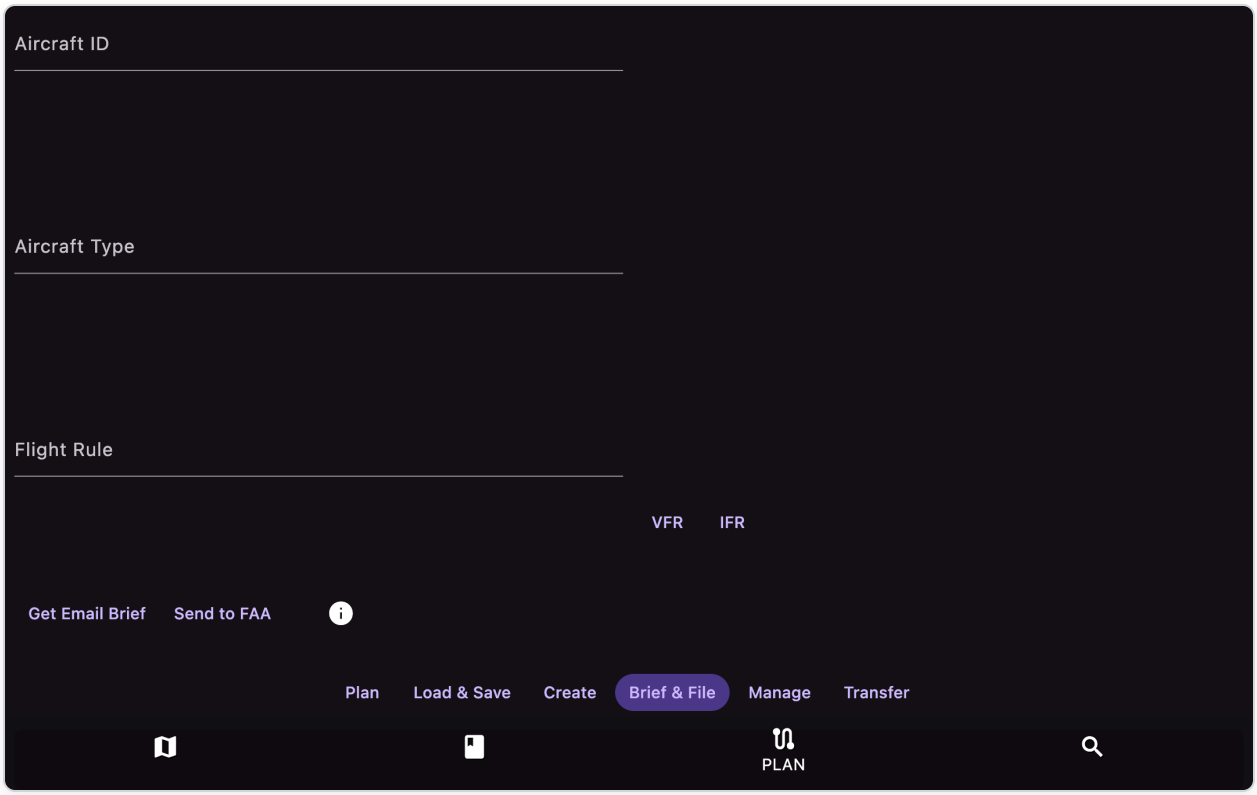
- Save current plan by name
- Load saved plan
- Load reversed (loads plan in reverse order)
- Delete saved plan
- Tap any plan to load immediately

B) Create

- **Create As Entered**: parse typed route string (space-separated waypoints/airways, e.g., `KBOS V123 KHPN`)
- **Create IFR Route**: build a plan between two airports (enter `DEPART DEST`) using the built-in offline route generator, which produces a waypoint route over all airways — joining near the departure, following the airway network, and leaving near the destination
- **Create IFR LA Route**: same as above but restricted to low altitude airways
- **Show IFR ATC Routes**: show recent ATC routes between departure/destination from 1800wxbrief with last departure times



C) Brief & File (FAA)



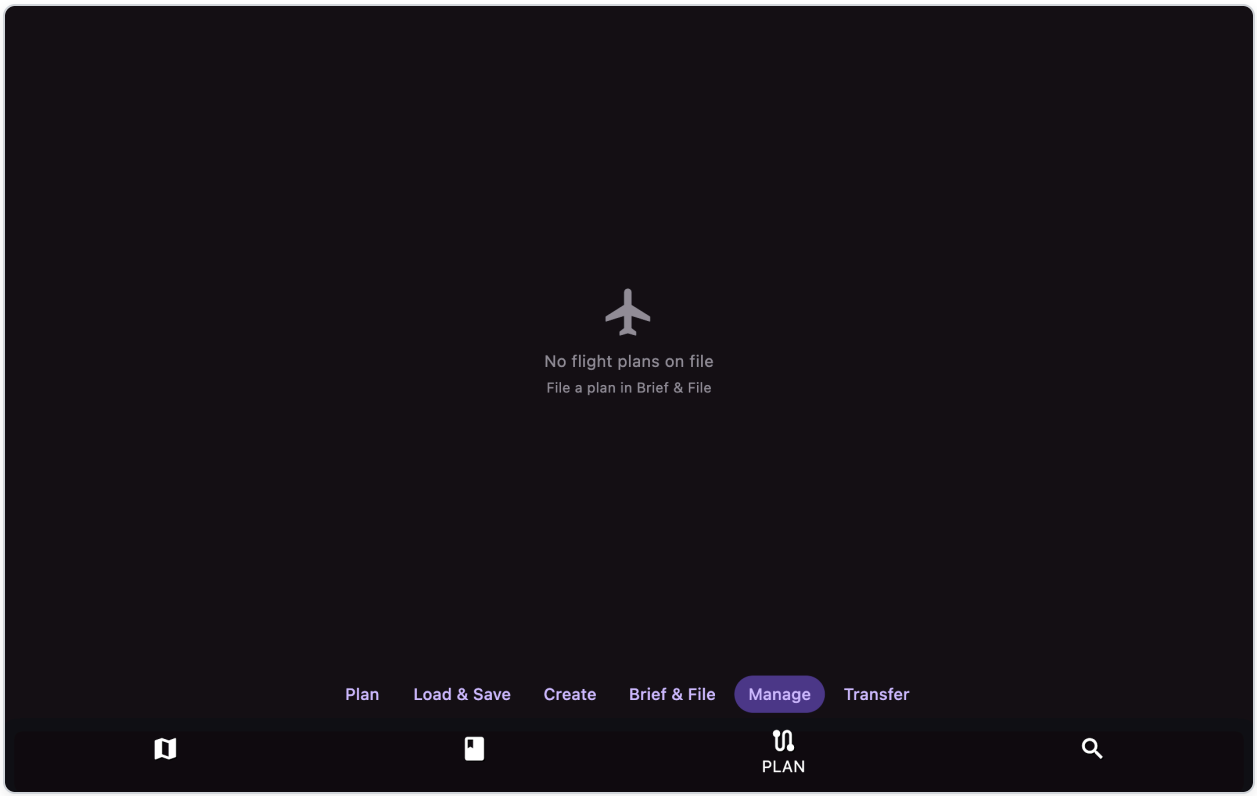
- Detailed ICAO form fields:
  - Aircraft ID, Type, Flight Rule (VFR/IFR), Flight Type, Number of Aircraft
  - Wake Turbulence (LIGHT/MEDIUM/HEAVY)
  - Aircraft Equipment, Surveillance Equipment

    - Departure, Departure Date/Time Zulu
    - Cruising Speed, Altitude, Route
    - Destination, Total Elapsed Time
    - Alternate Airports (1 & 2)
    - Fuel Endurance, People On Board
    - Aircraft Color, Remarks
    - Pilot in Command, Pilot Information
- Quick-fill buttons from current plan and stored aircraft
- Buttons:
    - **Get Email Brief**: sends briefing request to 1800wxbrief (50nm corridor)
    - **Send to FAA**: files flight plan via LMFS API
- Status indicator (check mark=success, info icon with tooltip=error)

Requires configured 1800wxbrief-compatible account email.

D) Manage (FAA)

- Retrieve filed plans
- State-aware actions with color coding:
    - **PROPOSED** (blue): **Cancel** or **Depart** (activate with chosen time - opens time picker with sunrise/sunset)
    - **ACTIVE** (green): **Close**
    - **CLOSED** (grey): view only



E) Transfer

The `Transfer` sub-page exchanges flight plans with a panel-mounted **Avidyne IFD440/540/550** over Wi-Fi, in both directions.

- Connect your device to the **same Wi-Fi network** as the IFD (use the IFD's Maintenance-mode network page to configure it).
- AvareX automatically searches for IFDs; tap the **search icon** (top-right of the Avidyne section) to re-broadcast discovery. Discovered units appear in a list showing their IP address and software version. Each has two buttons:
  - **Send** — uploads the active AvareX plan to the IFD as a stored route. Requires a route with at least 2 waypoints, and the IFD must have flight-plan input enabled (the list notes when it is not). The uploaded route reproduces the AvareX waypoints on the IFD ("stick route"); review and activate it on the IFD.
  - **Get** — downloads the IFD's flight plan and loads it into the active AvareX plan (replacing the current one). Each downloaded waypoint is matched to the AvareX database by identifier (closest to the IFD's coordinate); waypoints that cannot be matched are added as GPS coordinate points. Airway/procedure legs are reproduced as their coordinate waypoints.
- Below the device list, a scrolling **Messages** log shows the last 50 WiFiCrChannel packets exchanged with the IFD during Send/Get — **TX** (AvareX → IFD) and **RX** (IFD → AvareX), newest first. Tap a row to expand the **raw bytes** as selectable hex. Use **Pause** / **Resume** to freeze the log, and **Clear** to empty it.
- **iOS only**: the **Local Network** permission must be granted for discovery and transfer to work.

Avidyne ADS-B (Capstone): While AvareX is running on the same Wi-Fi network as an IFD equipped with an ADS-B receiver, it also receives the IFD's **Capstone** ADS-B stream automatically. Capstone is GDL90-format data, so traffic, ADS-B weather (NEXRAD/METAR/TAF/D-ATIS/TWIP/icing/turbulence/cloud tops/lightning/AIRMET/SIGMET via FIS-B), and receiver status appear the same way as with any other GDL90 device — no separate setup is needed beyond being on the IFD's network (and, on iOS, the **Local Network** permission).

7.5 Route building details

Waypoint types supported:

Type	Description
GPS	User-entered lat/lon coordinates
Airport	Airport from database
Navaid	VOR/NDB
Fix	Named fixes/intersections
Airway	Victor (V) and Jet (J) routes
Procedure	SID/STAR/Approach (format: `AIRPORT.PROCEDURE.TRANSITION`)

Airway processing:

- Automatically finds entry and exit points when airway is between two waypoints
- Max segment length: 500nm (separates AK/HI from lower 48)
- Populates actual fix names asynchronously

7.6 Automatic waypoint passage

When within 2nm of a waypoint and moving away from it, the plan automatically advances to the next waypoint.

7.7 Automatic plan resume after restart or crash

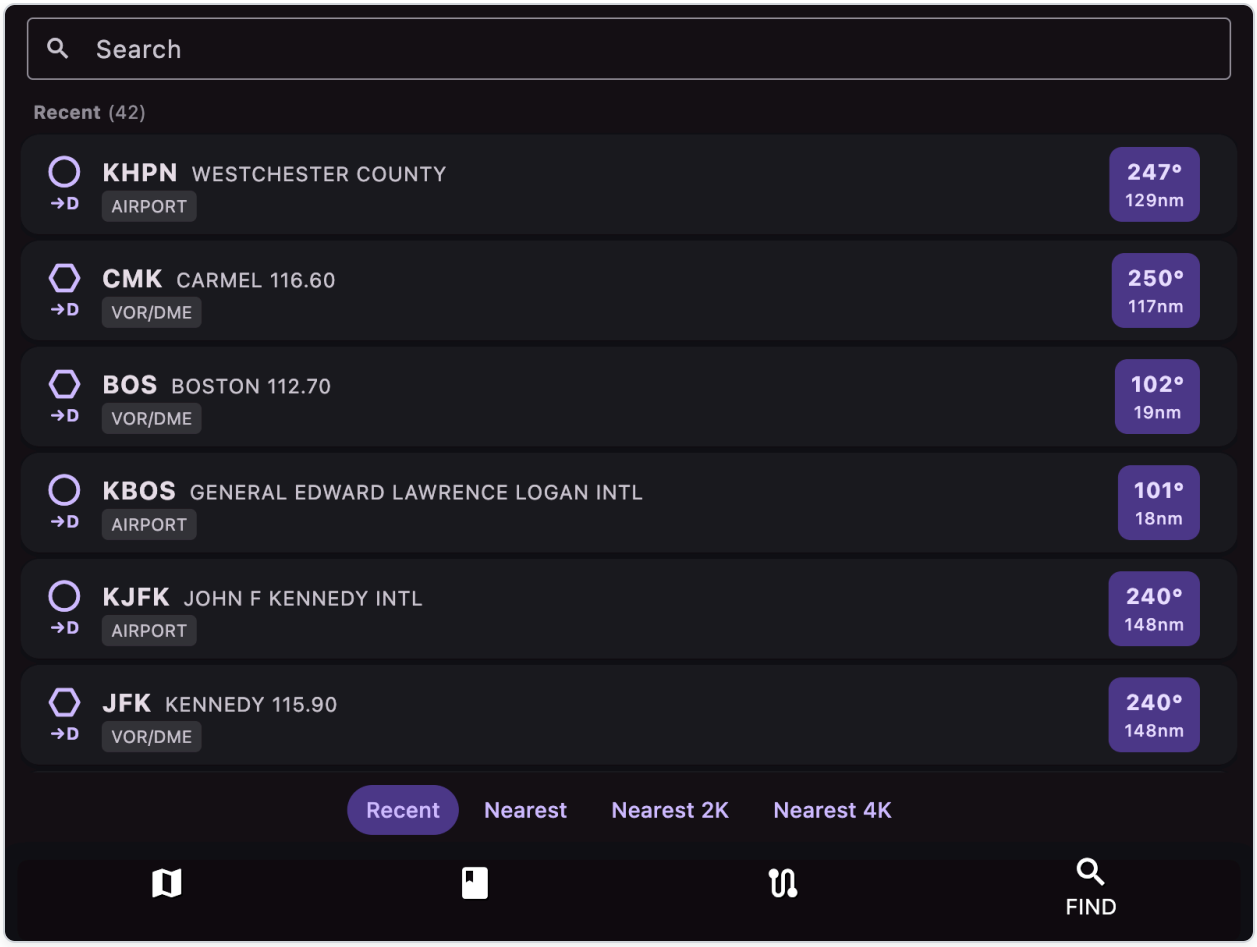
The active plan and the waypoint you are currently flying to are saved automatically as they change. If the app is closed or crashes, the next launch **reloads the same plan and resumes on the waypoint** (and airway/procedure leg) you were on. No action is required; there is no button to press. Clearing the plan clears the saved copy as well.

---

8) FIND Tab

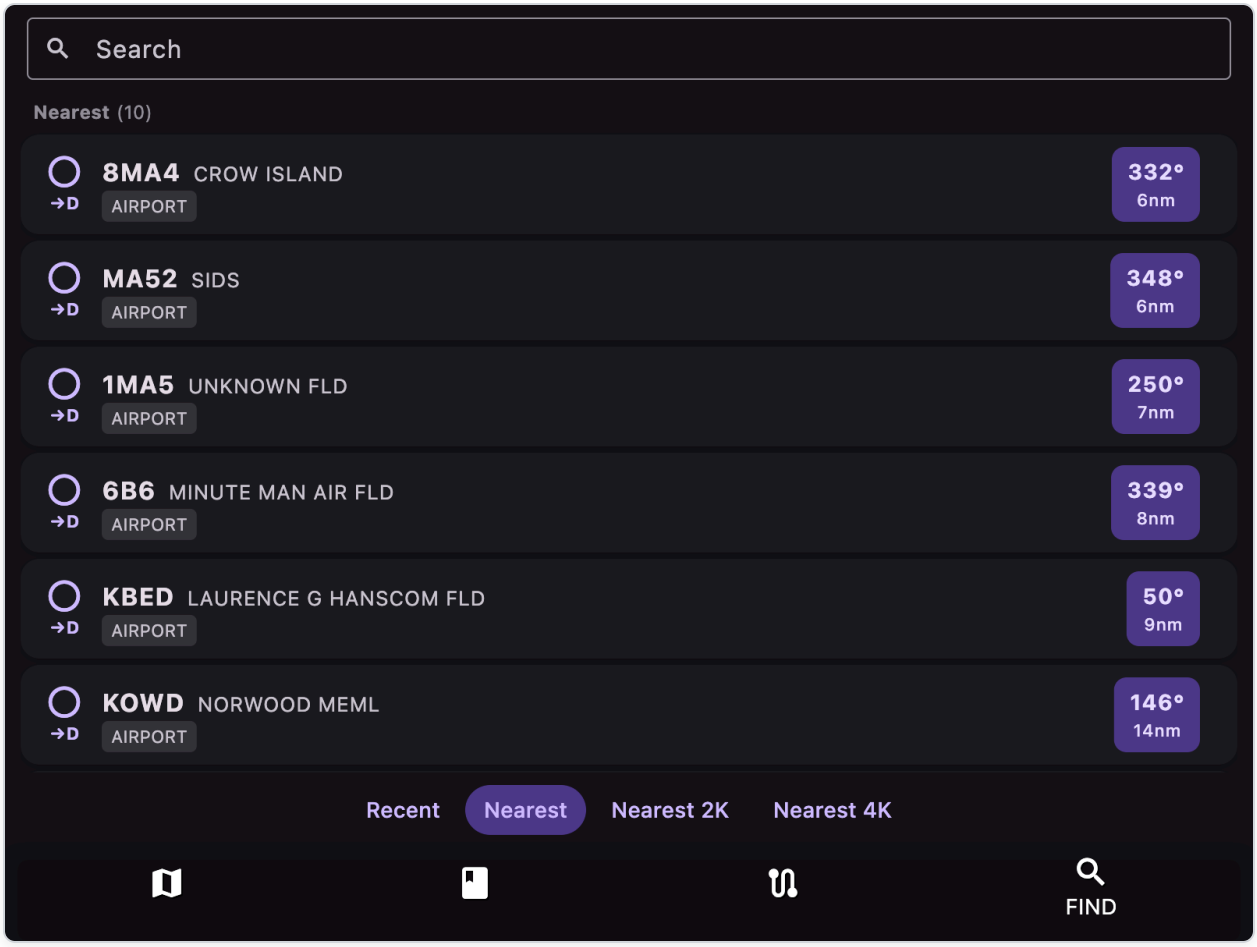
8.1 How to access

Tap bottom tab **FIND**.

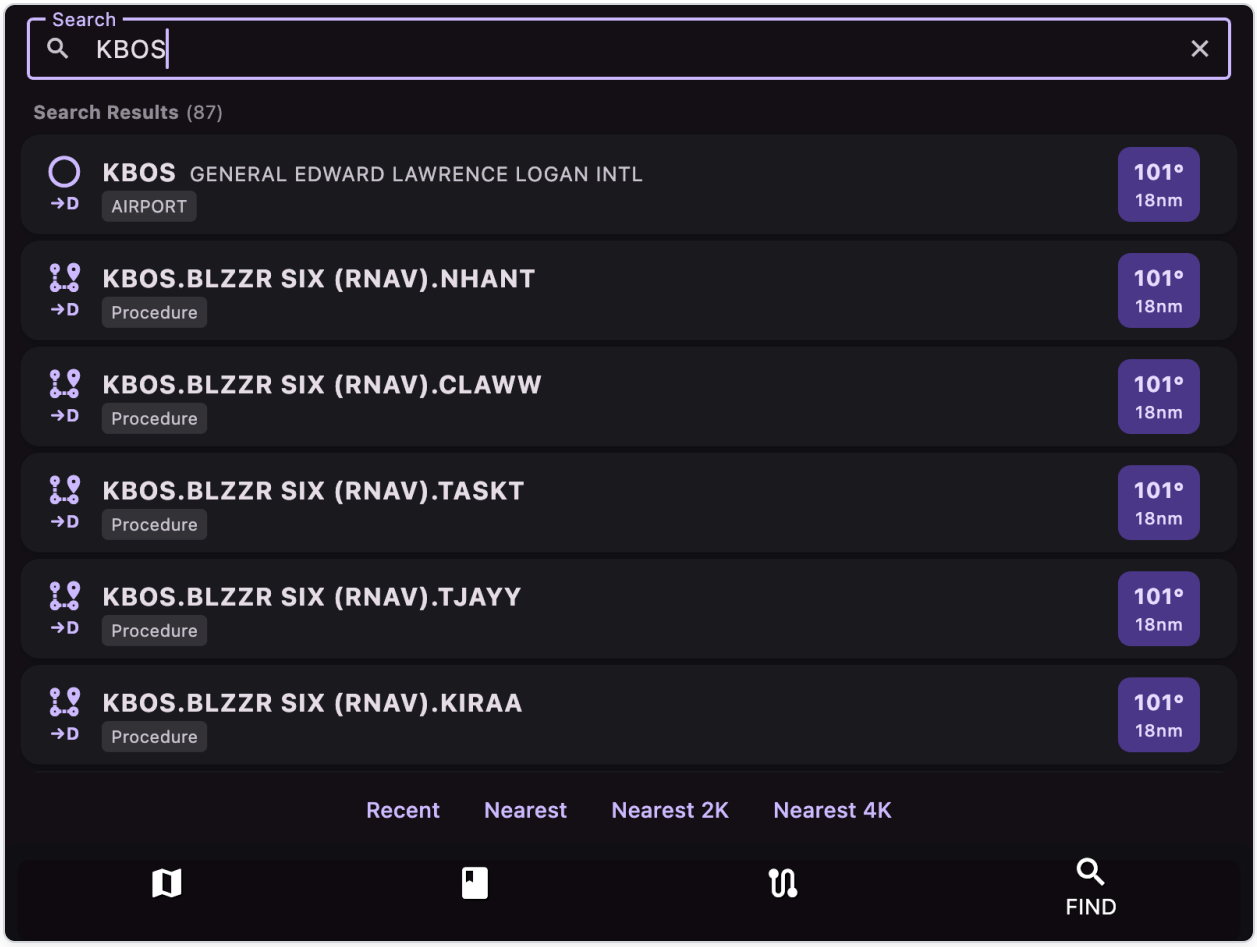


8.2 What it does

- Search destinations by text (airports, navaids, fixes, GPS coordinates)
- Browse:
  - Recent (previously viewed destinations)
  - Nearest (nearest airports)
  - Nearest 2K (runway >= 2000 ft)
  - Nearest 4K (runway >= 4000 ft)



Type into the **Search** box to find airports, navaids, fixes, or procedures by identifier:



8.3 Row actions

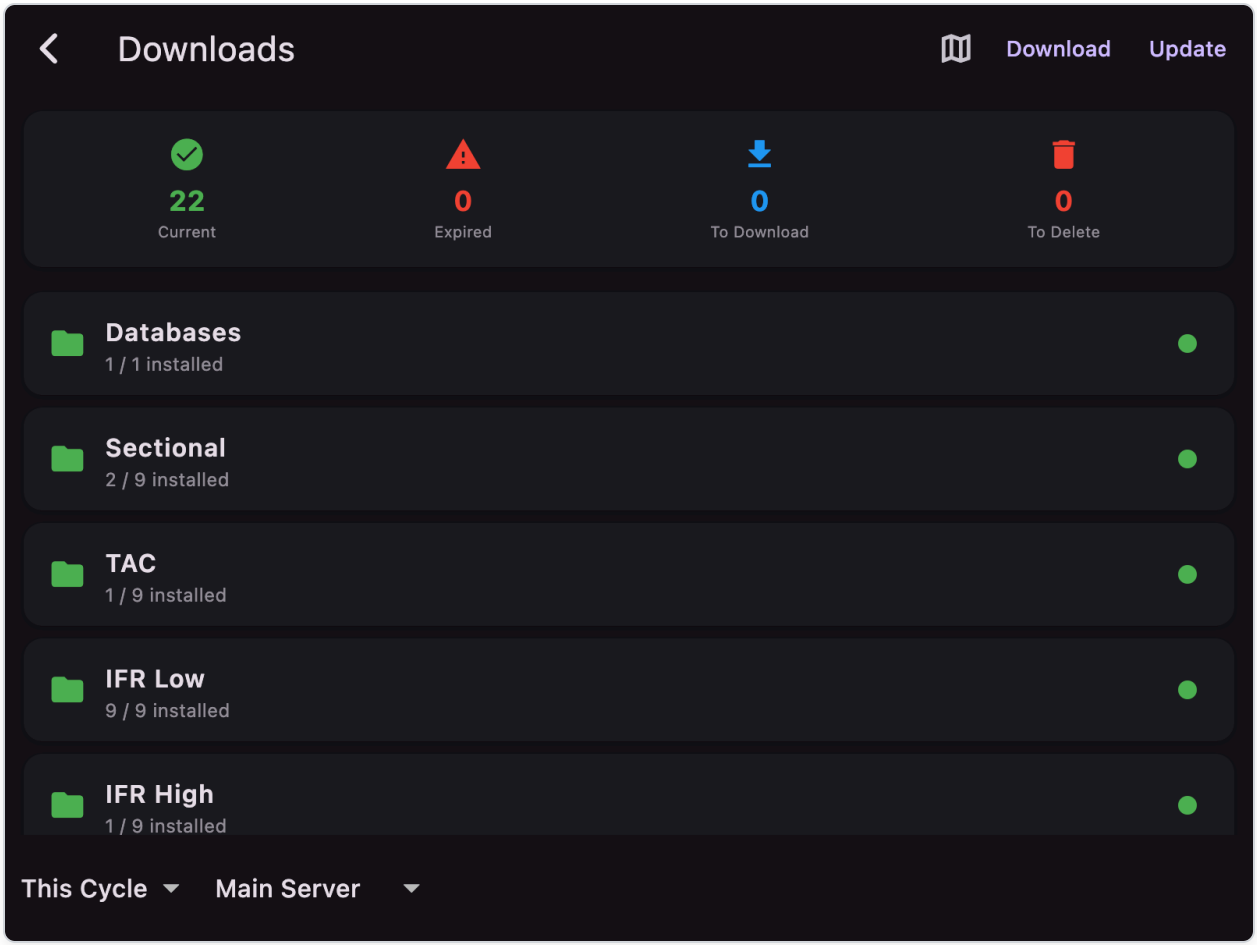
- Tap row: opens destination popup/details
- Tap left icon (marked `→D`): **Direct-To** the destination (same as the `→D` button on the destination popup) — sets it as your active target, shows it on the map, and switches to MAP
- Right-side bearing@distance button: centers map on item and switches to MAP
- Swipe right-to-left to delete recent entry
- For GPS-type recents, tap edit icon to modify custom facility name

---

9) Drawer Features

9.1 Download

Open: **Menu → Download**



Purpose:

- Install/update/delete downloadable data sets by chart category and region
- Monitor progress with per-item progress ring and stop button

Status Summary Card:

- Shows counts of Current, Expired, To Download, and To Delete items

State Indicators:

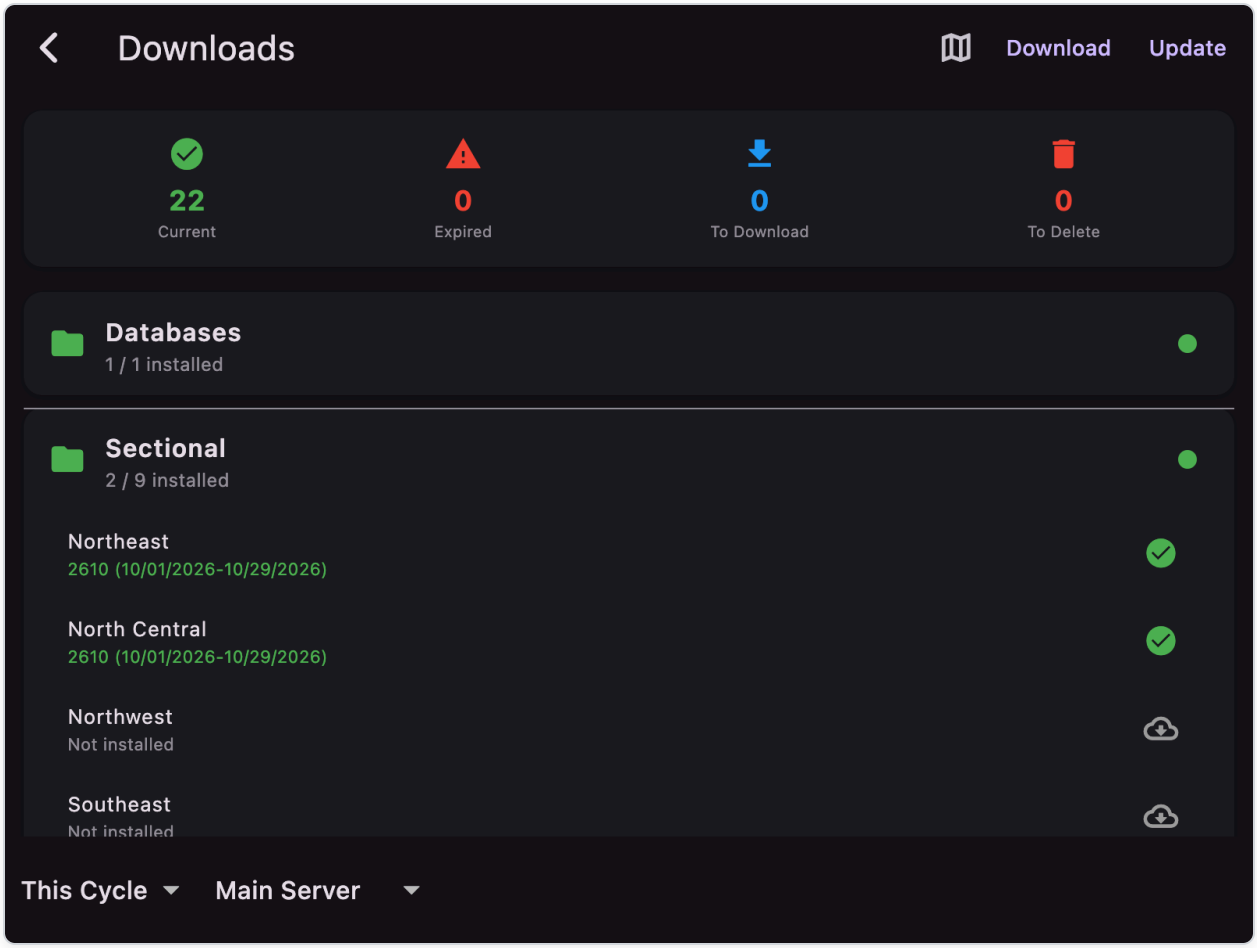
- Green = Current (up-to-date)
- Orange = Expired (needs update)
- Gray = Absent (not downloaded)

Controls:

- Tap chart item to queue for download/update/delete
- **Download** button processes all queued items
- **Update** button automatically updates all expired charts
- Toggle **This Cycle / Next Cycle**
- Toggle **Main Server / Backup Server**

- Info map icon shows regional coverage image

Tap a category (e.g. **Sectional**) to expand and see regional charts that can be selected individually:



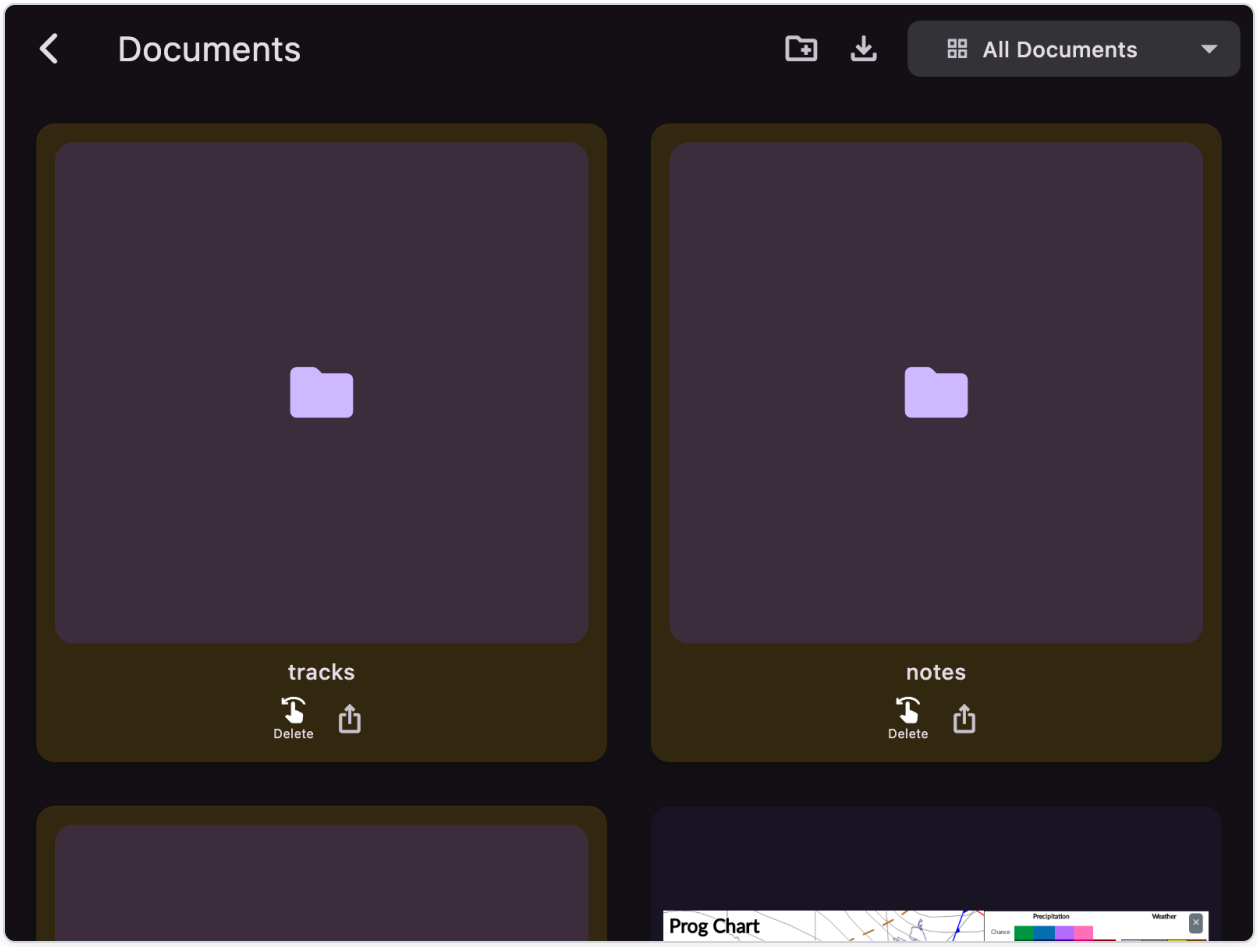
Major download categories include:

- **Databases**: DatabasesX (required for core nav data; fetched automatically during onboarding for your GPS region)
- **VFR Charts**: Sectional, TAC, Flyway, Helicopter
- **IFR Charts**: IFR Low, IFR High, IFR Area
- **Procedures**: Plates, CSUP
- **Terrain**: Elevation (fetched automatically during onboarding for your GPS region)

Regional coverage (9 regions): Northeast, North Central, Northwest, Southeast, South Central, Southwest, East Central, Alaska, Pacific

9.2 Documents

Open: **Menu → Documents**



Includes:

- Built-in weather/prog chart products (fetched from aviationweather.gov):
  - WPC Surface Analysis & Prognostic charts (6/12/18/24/30/36/48/60/72HR)
  - SigWx Low Level & Mid Level charts (00/06/12/18HR)
  - Radar & SIGMETs
  - AIRMET Tango (turbulence), Sierra (mountains/IFR), Zulu (icing) - 00/03/06/09/12HR
  - Surface Forecast charts (03-18HR)
  - Clouds Forecast charts (03-18HR)
  - Winds/Temperature at altitudes 5000-30000ft (06/12HR)
- User documents (imported or generated)
- This version's user manual, after `Help` has been opened (`User Manual` , named with the version). A copy from an older install is removed then
- User database file (`user.db`) entry
- Subfolders for organization:
  - `tracks/` — saved flight tracks (KML files)
  - `notes/` — saved notes screenshots

Folder management:

- **Create folder**: tap folder icon in app bar to create a new subfolder

- **Delete folder**: use delete icon below folder to remove folder and contents
- **Move files**: use move icon on user files to relocate to a folder or back to root
- **Export folder**: use share icon below folder to export all folder contents (not Linux)
- **Navigate**: tap folder to enter, use back arrow to go up

Filter dropdown: Filter by document category (All Documents, User Docs, WPC, SigWx, etc.)

Import button supports:

- `.txt` — text files
- `.adsb` — ADS-B capture logs (daily files written by external ADS-B/GDL90 receivers)
- `.geojson` — GeoJSON geographic data
- `.kml` — KML track files
- `.pdf` (except Linux)
- `user.db` — user database backup

Document behavior:

- Text: in-app text reader
- PDF: in-app PDF viewer (if supported)
- GeoJSON: parsed into map shapes (visible when GeoJSON layer is on)
- KML: opens Track Viewer with 2D map, altitude profile, and 3D terrain view
- Images: zoomable preview
- Share button available where supported (not Linux)
- Swipe delete for user files (preserves User Data entry)

Track Viewer (KML files)

When you tap a KML file in Documents, the Track Viewer opens with three view modes (toggle via app bar icons):

- **2D Map** (map icon): displays the track on a USGS topo background with:
    - Flight track polyline (blue)
    - Start marker (green takeoff icon)
    - End marker (red landing icon)
    - **Log Flight** FAB button to create logbook entry
- **Altitude Profile** (chart icon): shows altitude vs. distance graph:
    - Line chart of altitude (ft) over distance (NM)
    - Min/max altitude display
    - Total distance in nautical miles

    - Shaded area under curve

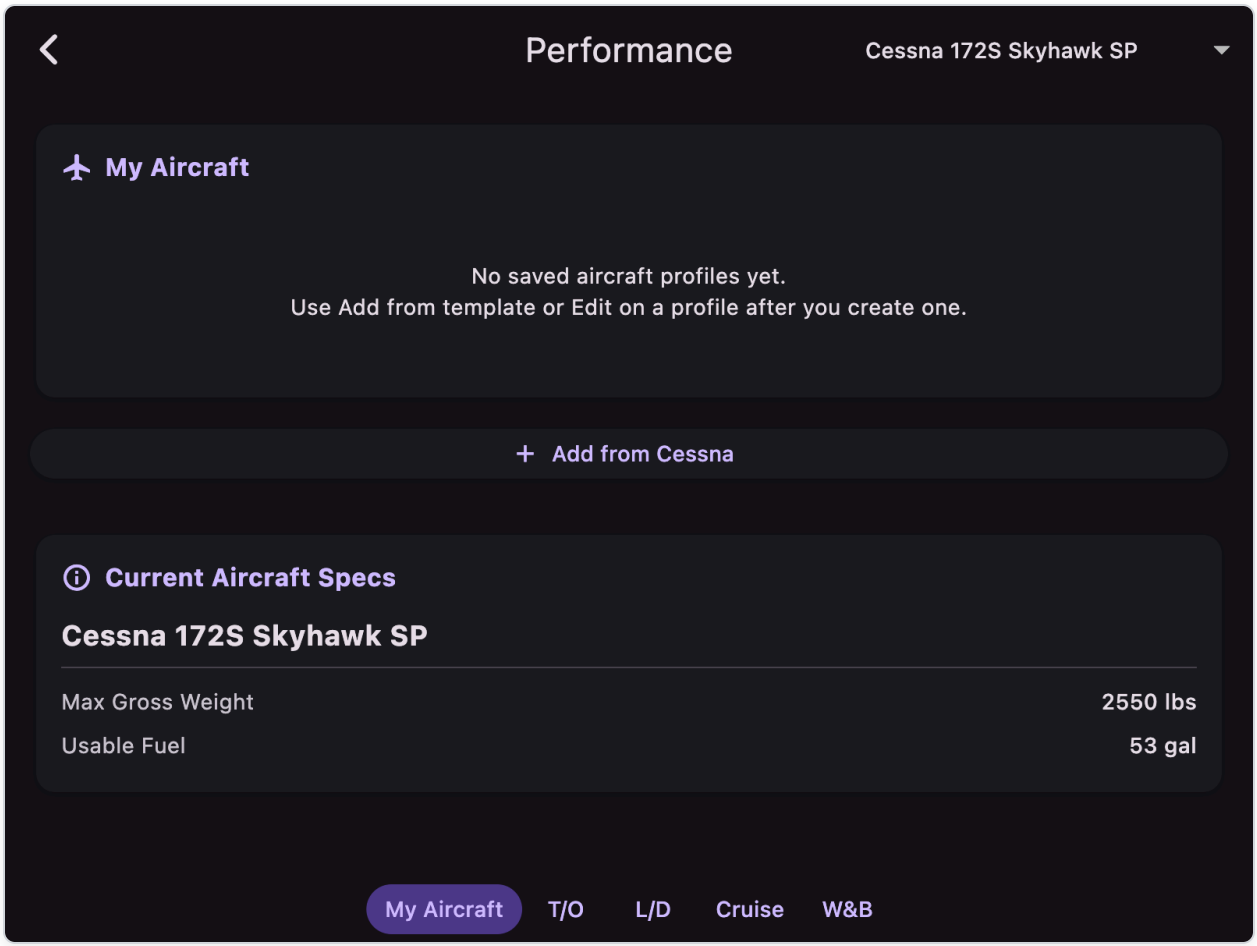
- **3D View** (AR icon): interactive 3D visualization:
    - Terrain grid with elevation data (fetched for 20nm buffer)
    - Topo map texture projected onto terrain mesh
    - Flight path with color gradient (green→yellow→orange→red)
    - Drop lines connecting track to terrain surface
    - Vertical scale with altitude tick marks (1000ft increments)
    - Drag to pan, two-finger rotate, pinch to zoom
    - Reset View button

Auto Logbook Entry Creation (Log Flight button):

- Uses nearest airports for route (takeoff and landing)
- Calculates flight time from track timestamps or from distance/cruise TAS
- Creates entry with aircraft make/model and tail from selected aircraft profile

9.3 Aircraft & Performance

Open: **Menu → Aircraft & Performance**



A unified screen with tabs for managing aircraft profiles, performance calculations, and weight & balance:

Tabs Overview

Tab	Purpose
My Aircraft	Create, edit, and save your own aircraft profiles (POH-style tables and flight-plan fields)
T/O	Calculate takeoff ground roll and 50ft obstacle clearance distances
L/D	Calculate landing ground roll and 50ft obstacle clearance distances
Cruise	Calculate cruise performance (TAS, fuel flow) at altitude/power settings
W&B	Weight and balance calculations with CG envelope

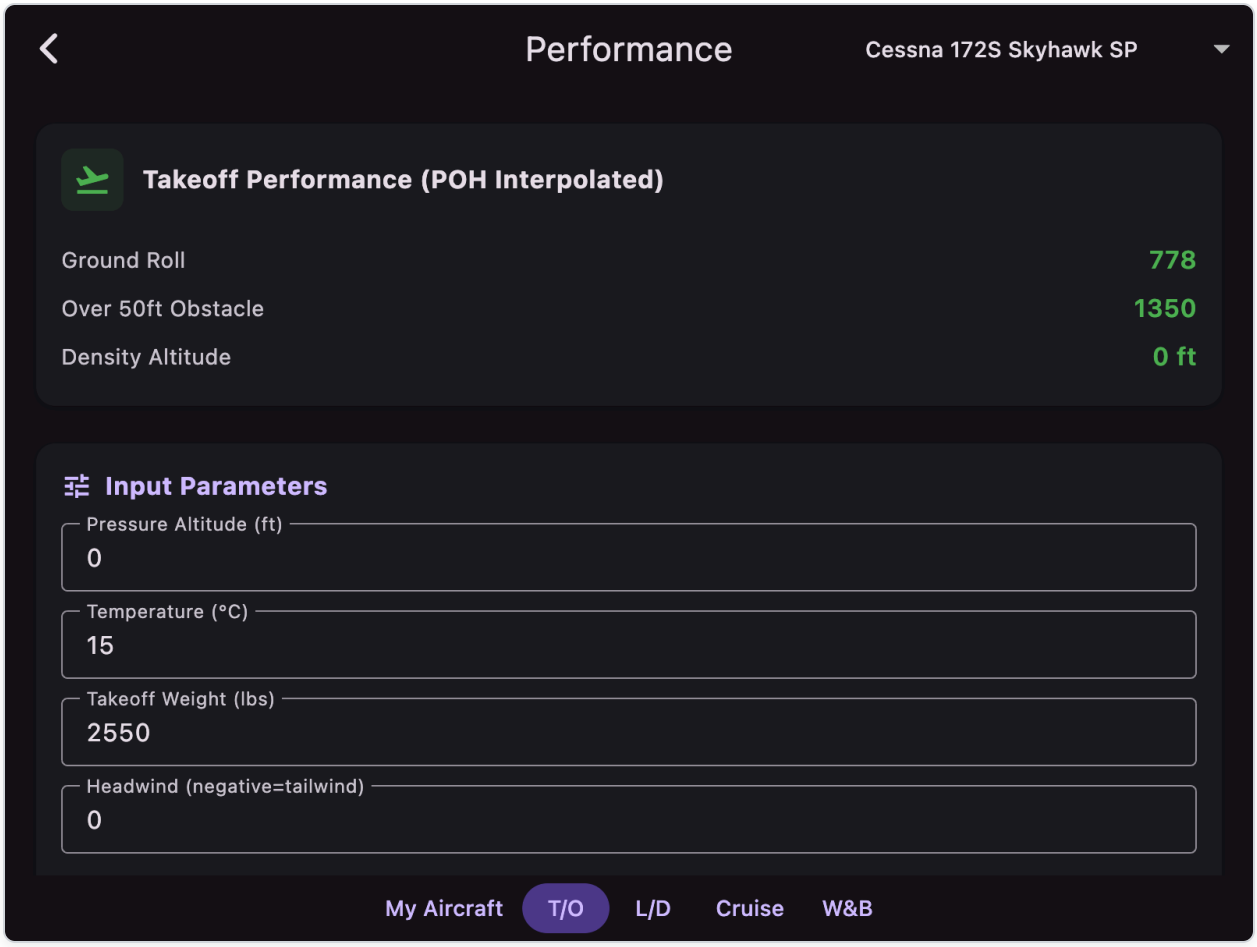
Helicopter (map icon): When the **selected** aircraft uses the helicopter icon (per-aircraft in **My Aircraft**, or the app-wide default in settings if that aircraft has no saved profile row), the **T/O**, **L/D**, and **Cruise** tabs are hidden—only **My Aircraft** and **W&B** remain. The same applies to takeoff, landing, and cruise **POH-style performance tables** when **adding or editing** a profile whose **Aircraft Icon** is **Helicopter**.

Aircraft Selection

Use the dropdown at the top to select from:

- **Built-in aircraft**: C152, C172S, C182T, PA-28-181, PA-28-161, PA-44, Beech A36, Cirrus SR22, Diamond DA40, Van's RV-9A
- **My Aircraft profiles**: User-created profiles saved in your database (same list as in **My Aircraft** tab)

Takeoff Tab



Enter conditions to calculate takeoff performance:

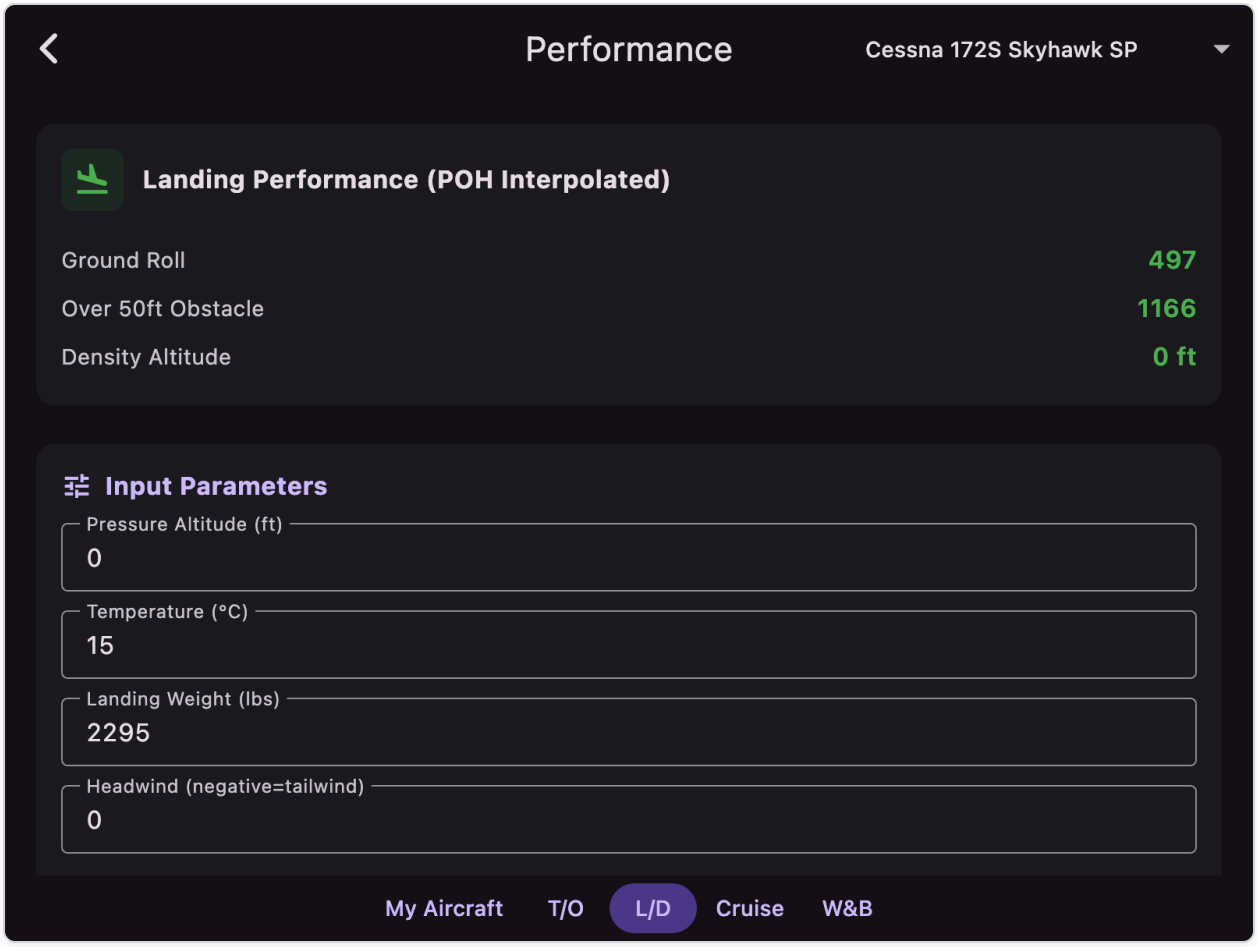
- **Pressure Altitude** (ft)
- **Temperature** (°C)
- **Weight** (lbs)
- **Headwind** (negative for tailwind)

Results show:

- Ground Roll
- Distance to clear 50ft obstacle
- Density altitude (ft) — computed as pressure altitude plus 120 × (OAT – ISA temperature), with ISA 15°C at sea level and –2°C per 1000 ft pressure altitude
- Soft field adjustments (if configured)

Speed and distance values follow the app unit setting (onboarding) and are shown without unit suffixes.

Landing Tab



Enter conditions to calculate landing performance:

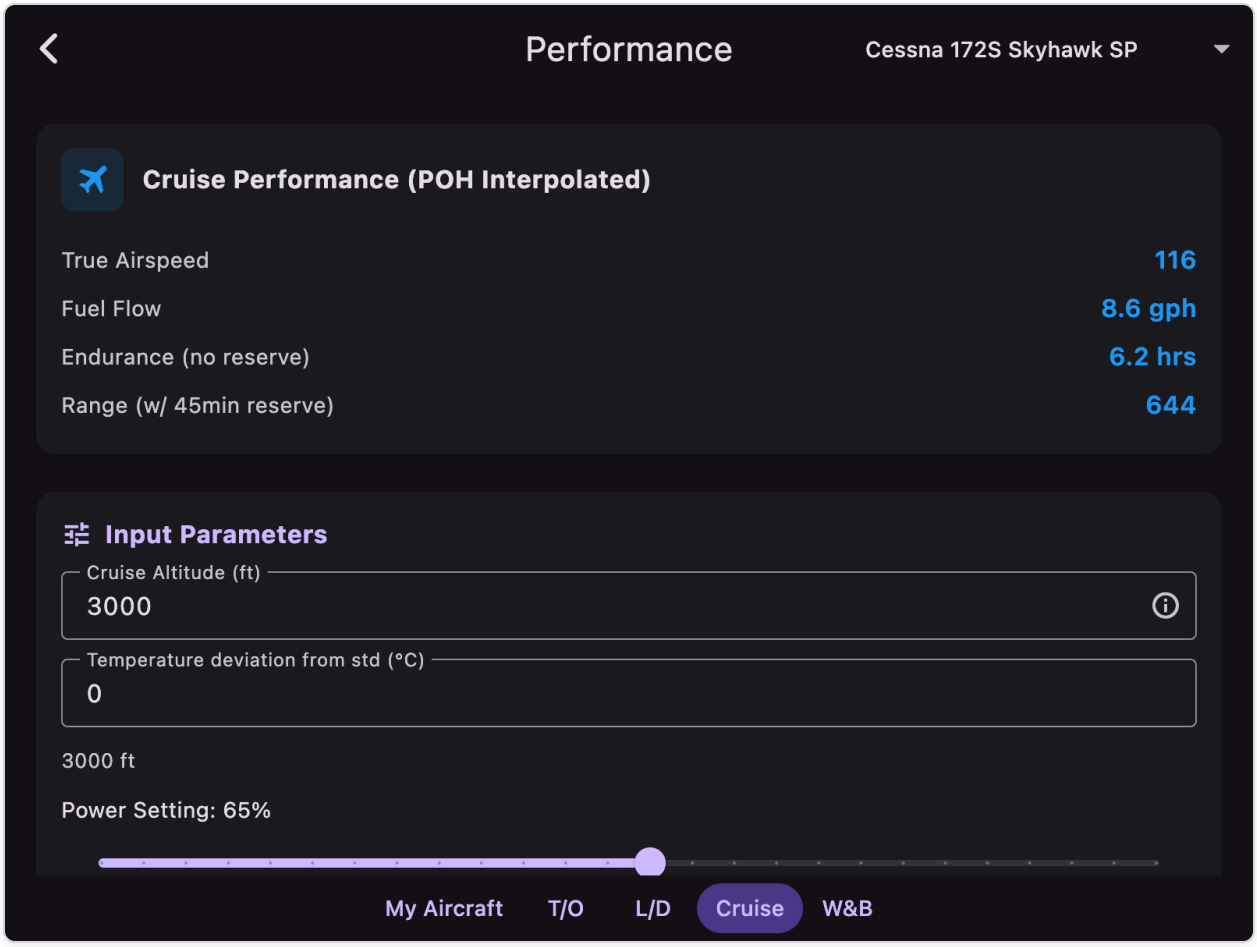
- **Pressure Altitude** (ft)
- **Temperature** (°C)
- **Weight** (lbs)
- **Headwind** (negative for tailwind)

Results show:

- Ground Roll
- Distance to clear 50ft obstacle
- Density altitude (ft) — same formula as takeoff (OAT vs ISA at that pressure altitude)
- Soft field adjustments (if configured)

Speed and distance values follow the app unit setting and are shown without unit suffixes.

Cruise Tab



Enter cruise conditions:

- **Altitude** (ft) — prefilled from the plan default altitude when you open the Cruise tab (or open Aircraft & Performance); you can still edit it for what-if
- **Temperature deviation** (°C from standard)
- **Power setting** (%) — remembered between app sessions in app settings, not the aircraft database table

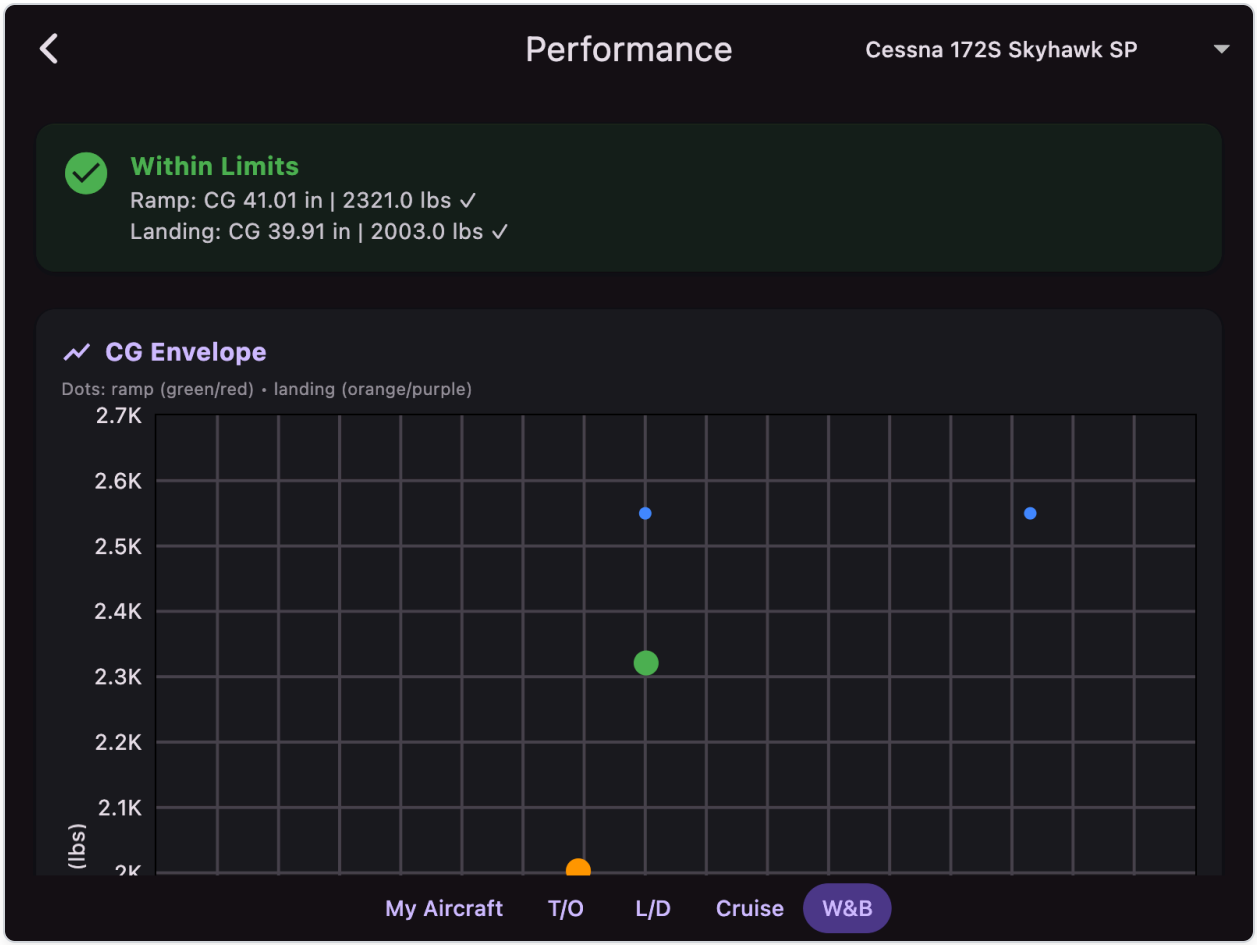
Below altitude and temperature deviation, the screen shows **density altitude** (ft) for those inputs (read-only).

Results show:

- True Airspeed
- Fuel Flow (GPH)
- Range (with 45 min reserve)

Speed and range follow the app unit setting and are shown without unit suffixes.

W&B Tab



Status Card: Green "Within Limits" only when **both** ramp and landing CG are inside the envelope; otherwise red "Outside Limits". Shows ramp and landing lines (CG in inches, weight in lbs, each with ✓ or ✕).

Helicopter map icon: If the aircraft's **map icon** is set to helicopter (per-aircraft in **My Aircraft**, or the app default in settings when no saved profile applies), the W&B tab adds **lateral** CG alongside **longitudinal**: a second status line for ramp and landing, a **Lateral CG** scatter chart (weight vs inches from centerline), and extra table columns (**Lateral** arm, **Longitudinal moment**, **Lateral moment**). Longitudinal behavior is unchanged. "Within limits" requires **all four** checks (ramp longitudinal, ramp lateral, landing longitudinal, landing lateral). Re-open the **W&B** tab after changing the icon so the layout refreshes.

Interactive CG Envelope Chart:

- Scatter plot showing weight (Y-axis) vs arm (X-axis); helicopter mode labels this chart **Longitudinal CG**
- Blue dots: envelope boundary points
- Large **green/red** dot: ramp / takeoff CG (uses all stations except **Landing fuel**)
- Large **orange/purple** dot: landing CG (uses **Landing fuel** instead of **Fuel (lbs)**)
- For custom aircraft: tap chart in edit mode to add/remove envelope points (CG dots are not removable)

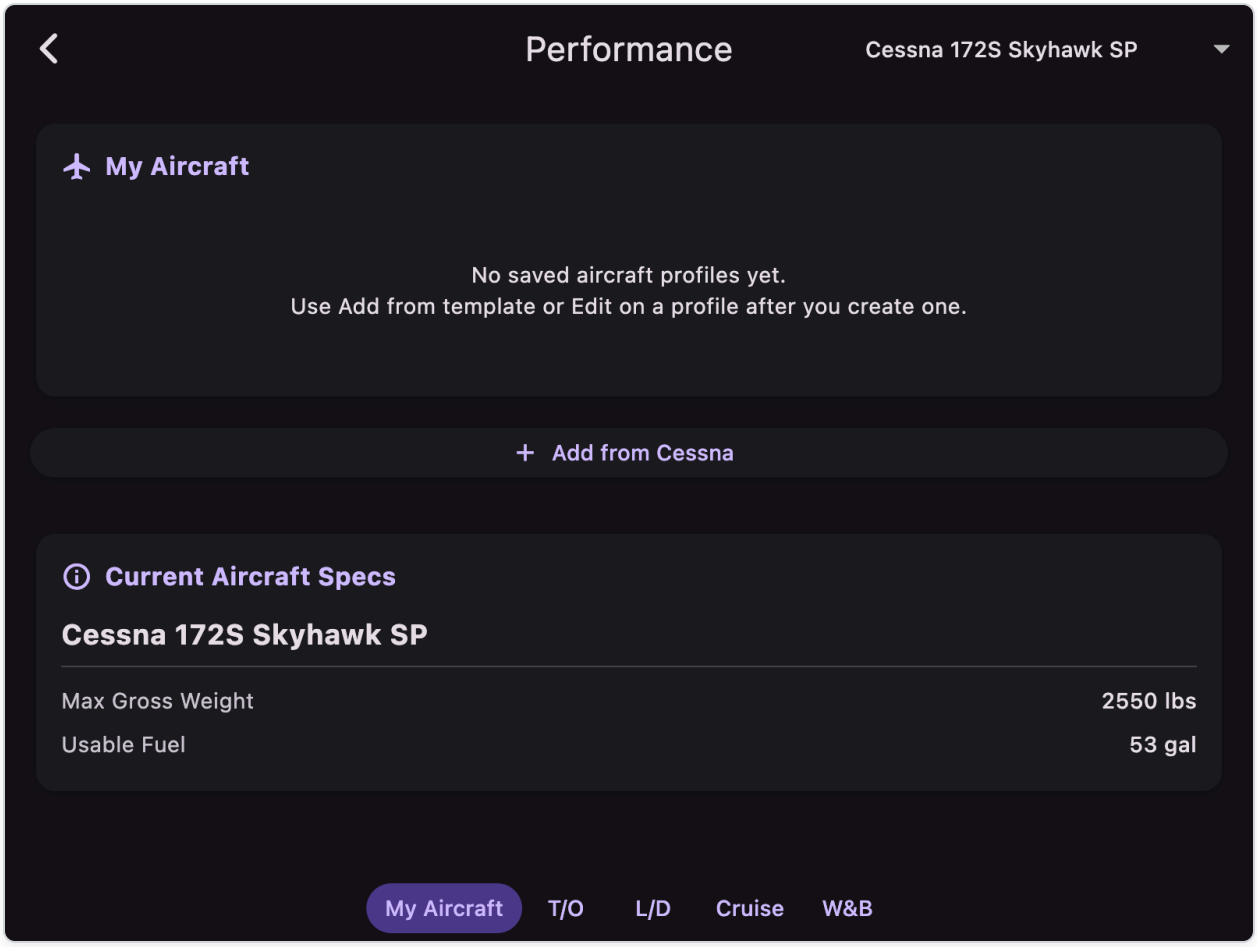
Weight Stations Table:

- Editable weight entries for each station
- Name, weight (lbs), arm (inches); helicopter mode also **Lateral** (lateral arm, inches from centerline) and a **Longitudinal** column for longitudinal arm
- **Landing fuel (lbs)**: expected fuel remaining at landing (same fuel arms as **Fuel (lbs)** by default); excluded from ramp totals
- Auto-calculated **longitudinal** moment (weight × longitudinal arm); helicopter mode also **lateral** moment (weight × lateral arm)
- **TOTAL (ramp)** and **TOTAL (landing)** rows with weight, CG arm(s), and summed moment(s)

For **custom aircraft**: Use **Edit/Save** button to modify envelope points and station definitions.

For **built-in aircraft**: W&B data is from POH (read-only envelope, editable weights).

My Aircraft tab



Create and manage your saved aircraft profiles:

Aircraft list: Tap a row to select that profile for the other tabs. Use the **edit** (pencil) icon to change an existing profile and **Save changes**; use **delete** to remove it from the database.

Add from template: Starts a new profile using the currently selected built-in or saved aircraft as a starting point, then **Save aircraft** to store it. A built-in template fills Cruise Speed, Fuel Endurance, Best Glide Speed, Sink Rate, Best Climb Speed, and Best Climb Rate from that airplane's handbook figures. Speeds use the app speed units from onboarding.

Aircraft Identification:

- Tail Number, Type, Wake Turbulence
- Color & Markings, Mode S Code
- Map icon selection (plane, helicopter, canard)

Pilot Information:

- PIC name and contact info
- Home Base

Flight Plan Fields:

- Equipment codes, Surveillance codes
- Cruise Speed, Fuel Endurance
- Other Information

Glide circle (MAP `Circles` layer):

- **Best Glide Speed** — POH best-glide speed of the **selected** aircraft (Performance dropdown). Used instead of current GPS ground speed. Entered in the same speed units as the rest of the app (onboarding unit setting).
- **Sink Rate** (fpm) — power-off sink rate at best glide
- If Best Glide Speed and Sink Rate are empty, the selected built-in aircraft supplies its handbook best glide and sink rate. If Best Glide Speed is still empty, Cruise Speed is used; if that is also empty, 65 is assumed.
- **Best Climb Speed** — best-rate-of-climb speed, in the same speed units as the rest of the app. The plan altitude icon uses it to climb and descend
- **Best Climb Rate** (fpm) — best rate of climb and descent. The plan altitude icon uses it, and time and fuel use the climb speed while the altitude is changing. If either climb field is empty, the selected built-in aircraft supplies its handbook climb speed and rate. If they are still empty, climb uses three quarters of cruise speed and a rate that slows from about 700 ft/min to about 100 ft/min

Current Aircraft Specs (under the template button) shows max gross weight, usable fuel, and, for a built-in aircraft, cruise speed, fuel endurance, best glide, sink rate, and best climb.

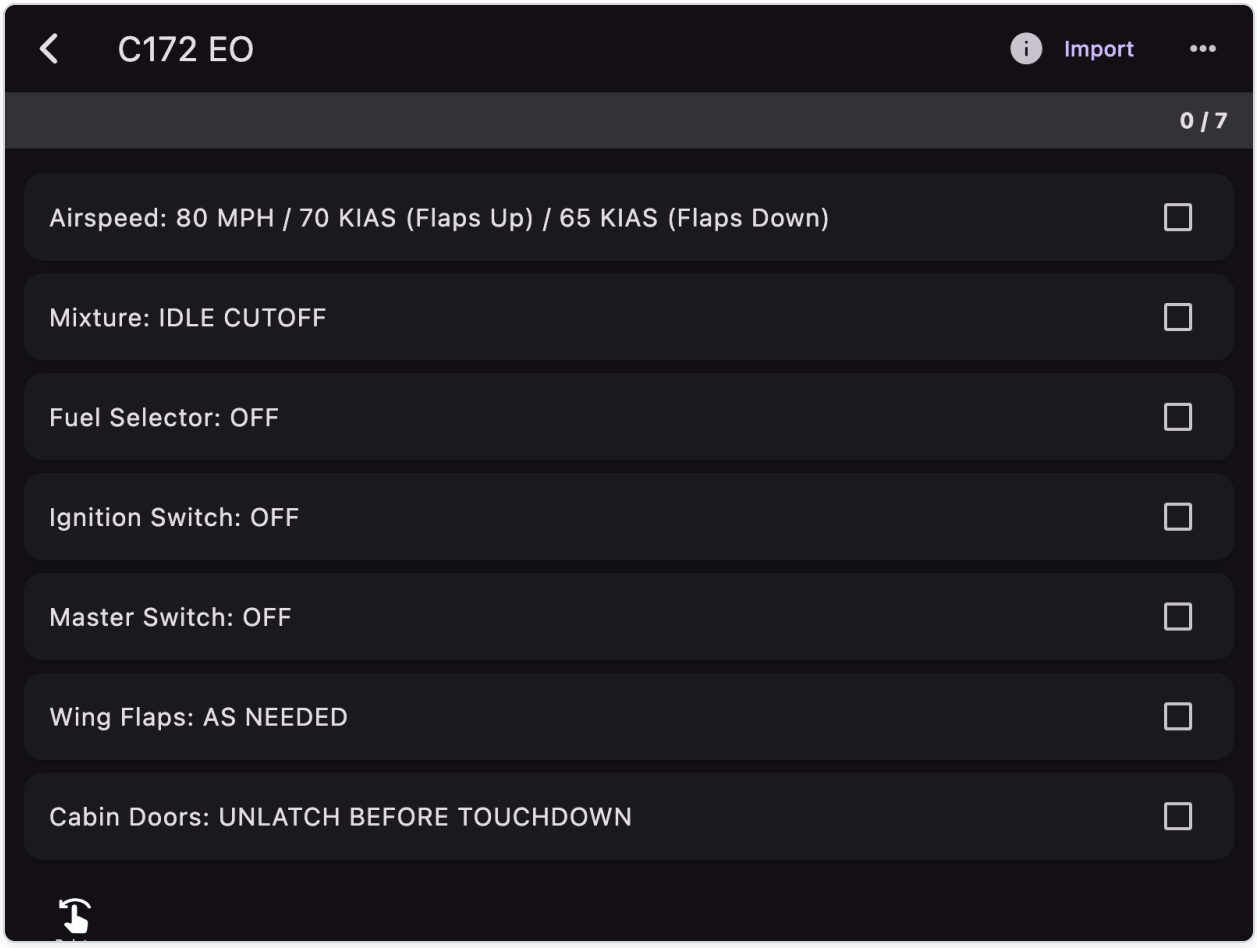
Performance Data:

- Max Gross Weight, Usable Fuel, Empty Weight
- Takeoff/Landing correction factors (headwind %, tailwind %, soft field %)
- Custom performance tables (altitude, temperature, weight entries)

Performance Tables: Add rows for takeoff, landing, and cruise performance at various conditions (omitted when **Aircraft Icon** is **Helicopter**). The app interpolates between entries for accurate calculations. Under each row, a read-only **density altitude** (ft) is shown from that row's altitude and temperature (takeoff/landing use °C OAT; cruise uses the same ISA temperature deviation convention as the Cruise tab).

9.4 Check Lists

Open: **Menu → Check Lists**



- **Progress bar**: shows completion percentage with completed/total count
- **Checkbox items**: tap to check off steps

- **Visual feedback**:
  - Checked items: green background, strikethrough text
  - All complete: green progress bar and background
- **Reset button**: reset all items to unchecked
- **Multiple checklists**: dropdown to switch between saved checklists
- Import checklist from `.txt` (first line is title, following lines are steps)
- Swipe delete to remove checklist
- Info tooltip explaining import format
- Persistent state during session

9.4.1 Profile

Open: **MAP → Map Settings → Profiles**

Each checkbox stores the current value with the profile. **Layers**, **Weather**, **Traffic volume**, and **Instruments** start checked. **Plan**, **Plate**, **Chart Type**, **Aircraft**, and **Checklist** start unchecked. Applying the profile restores the checked items. The value in use is shown under each checkbox.

- **Profile name**: text box for the profile name
- **Save**: stores the checked items under that name. Saving an existing name replaces it.
- **Profile list**: tap the three-dot menu on a profile and choose **Delete** to remove it. A profile shows the items it will restore under its name.
- Apply a saved profile from the same `Profiles` list in `MAP → Map Settings` by tapping its name. That list shows the profile name. The `Profiles` heading opens this screen.
- **Emergency** is a permanent profile at the top of the list. It cannot be saved over, renamed, or deleted. Its tile looks like the others, in red, and shows the checklist in use. Its three-dot menu lists the checklists imported under `MAP → Menu → Check Lists`. `Default` uses a built-in engine-out checklist. Choosing a checklist here changes only the Emergency profile, not the checklist chosen on the `Check Lists` screen.

9.4.2 Emergency Mode

Open: **MAP → Map Settings → Profiles → Emergency** (the red entry at the top of the list)

Emergency mode is for an engine failure, a fire, or running out of fuel. It opens a simple map and four cards, A, B, C, and D, that follow the usual emergency steps: **A**viate with best glide speed, pick the best place to land (**B**), run the **C**hecklist, and **D**eclare the emergency.

It is an aid. It does not replace flying the airplane: fly first, then navigate, then communicate.

The map

- The map shows the chart you were using, always north up, with ownship at the center and the glide ring around it.
- It opens at the same zoom as a long-press on `Center`.
- Pinch or scroll to zoom. The map cannot be panned or rotated, and ownship stays centered.
- Emergency mode changes none of your settings: layers, weather, chart type, zoom, plan, and your checklist choice all stay as they were. `EXIT`, or the Back button, asks for confirmation and returns to the map exactly as you left it.

The aircraft

The aircraft is the one chosen in the dropdown on `MAP → Menu → Aircraft & Performance`. The glide ring, the glide speed on A, and the landing length that B needs all use that aircraft. If nothing was ever chosen there, the Cessna 172SP is used. Check before a flight that the right aircraft is chosen and has a best glide speed.

A, top left: best glide speed

- Shows the best glide speed of the aircraft in your speed units, with your current indicated airspeed under it. Airspeed is estimated from GPS ground speed and the winds aloft.
- The card flashes red when the airspeed is more than 5 knots from best glide.

B, top right: best place to land

- **Which runways count**: any runway at an airport (public or private) whose landing end is inside the glide ring. The far end of the runway can be outside the ring. Helipads, heliports, and seaplane bases are never chosen.
- **How one is chosen**: among runways at least as long as the aircraft needs to land, B picks the one with the most headwind, and the nearest if tied. The length needed is the landing distance over a 50 ft obstacle at the field's elevation on a standard day, at maximum gross weight, from the aircraft's landing data. If the aircraft has no landing data, length is not checked.
- **On the map**: a red line runs from ownship to the runway, and a large red outlined arrow at the landing end points the landing direction. The outline lets you see the airport under it.
- **The card**: airport, runway, length (and the length needed), headwind or tailwind and crosswind, distance, bearing, and landing heading.
- **Short runway**: if no reachable runway is long enough, B shows the longest one and flashes with `SHORT RUNWAY`.
- **No runway data**: many fields, private ones especially, publish no usable runway. If no runway is in reach but such an airport is inside the glide ring, B flashes `NO RUNWAY DATA`

with that airport. The arrow points into the wind, or straight to the field if there is no wind data.
- **Nothing in range**: B flashes `No runway identified` and shows an arrow pointing directly into the wind. Without wind data the arrow shows the current track.
- **Lock**: tap `LOCK` on the B card to hold the current choice so it stops changing as you fly. The distance and bearing keep updating. Tap `LOCKED` to release it and choose again. If a locked runway is no longer inside the glide ring, B flashes and shows `OUTSIDE GLIDE RING`.

C, bottom right: checklist

- Shows the emergency checklist one step at a time, as `step/total`. Tap the card for the next step. After the last step it goes back to the first.
- `EXIT` is on this card. It asks for confirmation before leaving.
- The checklist is the one chosen for the Emergency profile (below), or a built-in engine-out checklist.

D, bottom left: declare

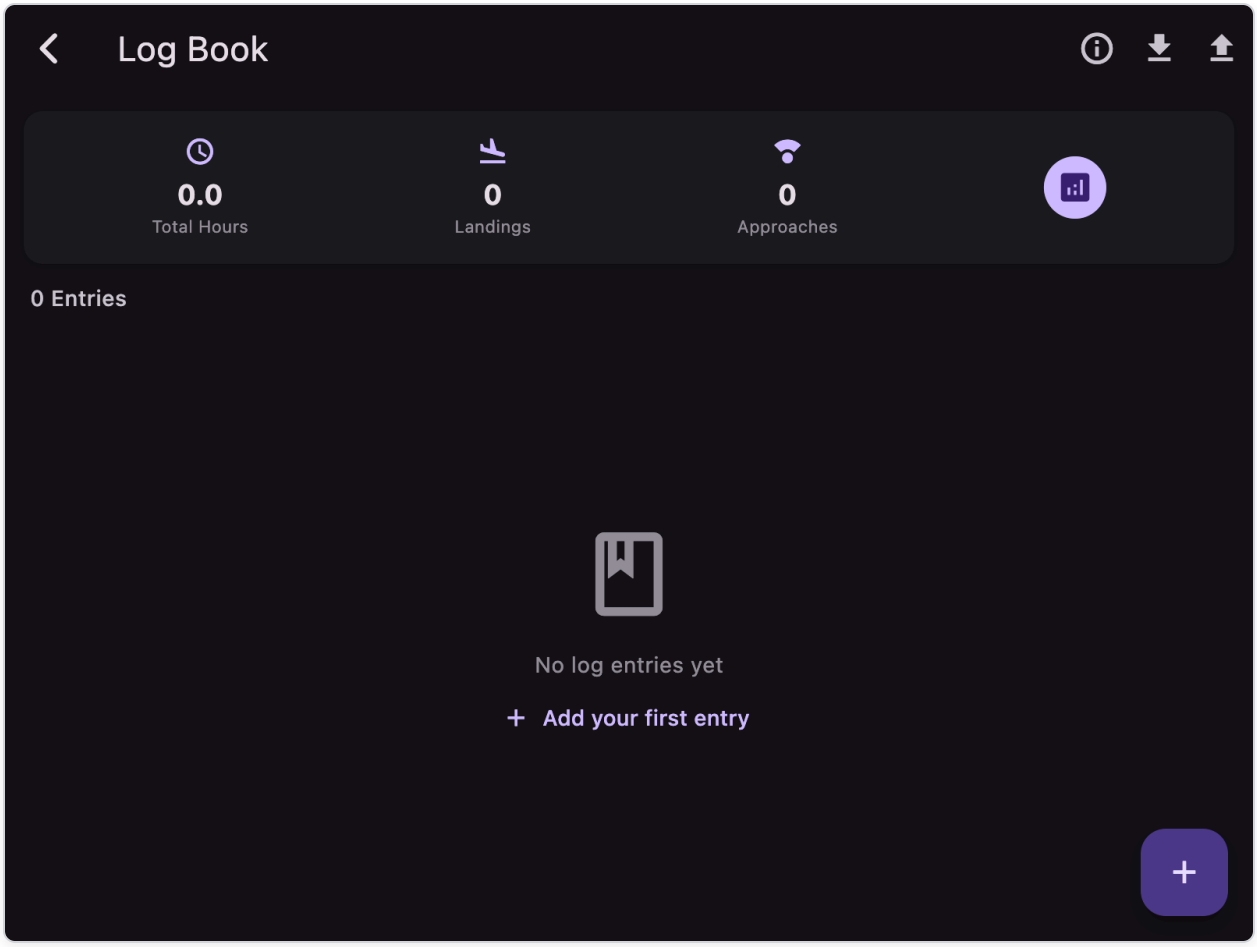
- The CTAF or UNICOM frequency of the airport of the runway shown on B, then `121.5` and squawk `7700`.

Before you need it

1. Choose your aircraft on `Aircraft & Performance`. Fill in its Best Glide Speed and Sink Rate under `My Aircraft`, and its landing table if you have one.
2. To use your own checklist, import it under `MAP → Menu → Check Lists`. Open the `Profiles` heading, tap the three dots on the `Emergency` tile, and pick it. `Default` uses the built-in checklist.
3. Open Emergency mode once on the ground to see the layout, then `EXIT`.

9.5 Log Book

Open: **Menu → Log Book**



Features:

- **Summary Card**:
  - Total Hours, Total Landings, Total Approaches
  - Analytics button to view dashboard
- **Flight List**:
  - Flight time (hours), aircraft make/model (tail)
  - Date, route
  - Day/Night landings, instrument approaches
  - Tap entry to edit, or use FAB to add new entry
- **Log Entry Form** (sections):
  - Flight Info: Date, Aircraft Tail, Type, Route
  - Flight Time: Total, Day, Night, Cross Country, Solo
  - Pilot Function: PIC, SIC, Dual Received, Instructor, Examiner
  - Instrument: Actual IMC, Simulated, Approaches, Holds
  - Landings: Day, Night
  - Training & Simulation: Ground Time, Simulator, Instructor Name/Certificate

    - Remarks: Free-form notes

- **CSV Import/Export**:

    - Import from CSV file (header row required)
    - Export to CSV (share via system share sheet, not Linux)
    - Info tooltip explaining format

Logbook Dashboard (Details)

Open by tapping **Details** button:

- **Currency Status Card**: Day currency (3 landings/90 days), Night currency (3 landings/90 days), IFR currency (6 approaches/6 months)
- **Summary Card**: Total hours, PIC, Night, XC, Landings, Approaches
- **Filters**: Year, Aircraft Type, Tail Number, Remarks keywords (Check Ride, IPC, Flight Review, Favorite)
- **Bar Charts**:
    - Flight Hours by Year
    - Hours by Aircraft Type
    - Hours by Tail Number
    - Hours by Category (Solo, Dual, PIC, SIC, Day, Night, IMC, XC, etc.)
    - Landings & Approaches
- **Filtered flight list**: scrollable list of filtered entries

9.6 IO (Bluetooth) — Android only

Open: **Menu → IO**

Functions:

- **Device Discovery**: auto-discovery starts when screen opens
- **Device List**: shows device name, address, signal strength (RSSI in dBm)
    - RSSI color-coded: green = strong, red = weak
    - Icons indicate connection status: connected (↔), paired/bonded (link icon)
- **Device Operations** (tap device for dialog):
    - **Pair**: bond with unpaired device
    - **Unpair**: remove device bond
    - **Connect**: establish Bluetooth SPP connection (for paired devices)
- **Connection Status Bar**: shows currently connected device with disconnect button

Connected stream feeds external data input into app parser for GPS, traffic, weather, AHRS. Also used for autopilot output of navigation data.

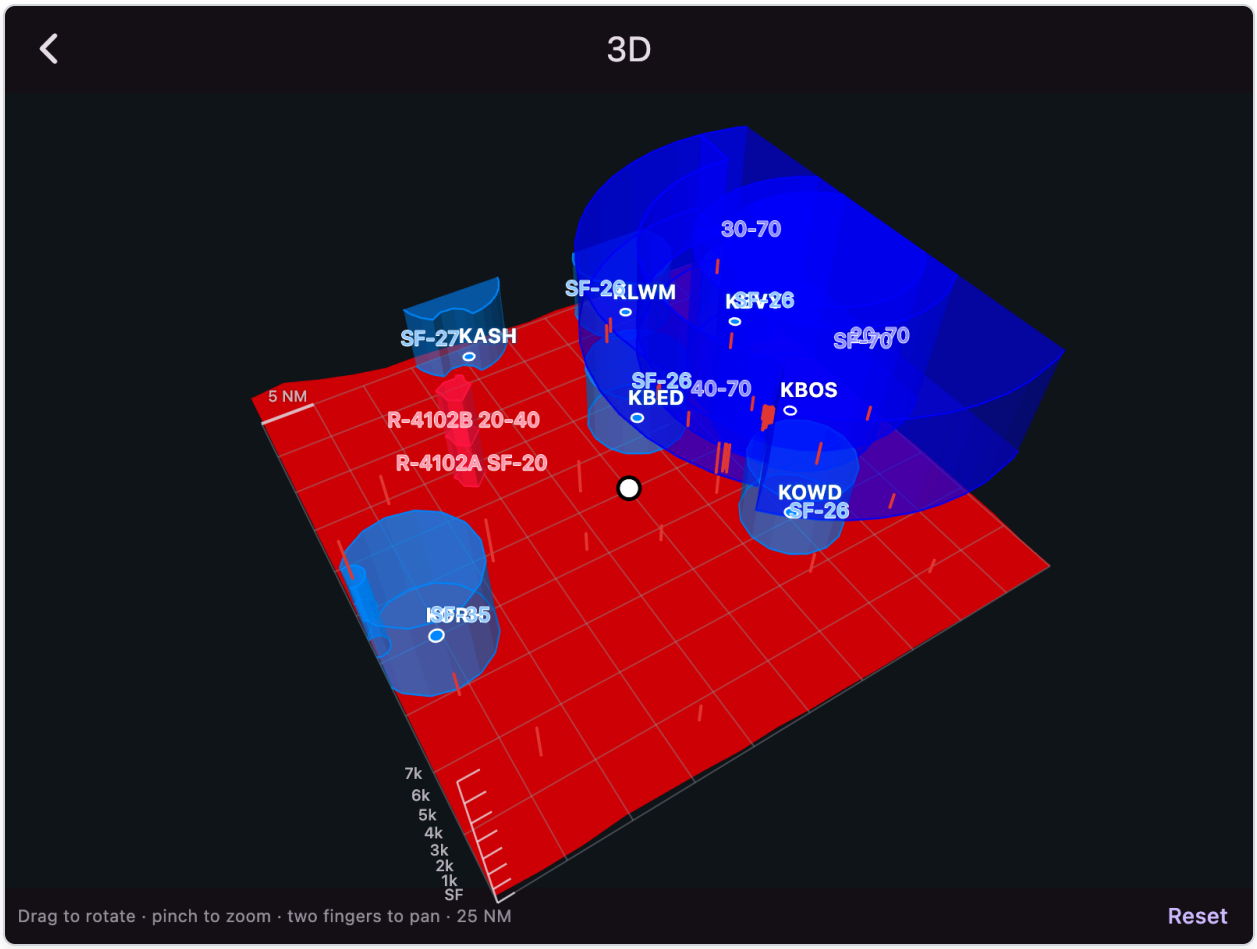
9.7 Help

Open: **Menu → Help**

- Copies the User Manual PDF that shipped with this version into `Documents`, then opens it. The app bar says `User Manual`. The file is named with this version, and a copy from an older install is removed at that time.
- Available on platforms that support PDF viewing (not Linux).

9.8 3D

Open: **Menu → 3D** (under `Write a Note`)



Same view as the destination popup `3D` tab, centered on the aircraft, without the yellow long-press mark.

- **Terrain**: downloaded Elevation tiles, brown, and red within 500 ft of altitude. Shelf floors sit on that surface instead of extending below it

- **Airports**: each airport inside 25 NM is a runway diagram on the ground at the field. Pavement matches the AD tab, with the runway numbers at the ends. A surface Class B, C, or D airport still has its name
- **Traffic**: ADS-B targets inside 25 NM, drawn as dots
- **Plan**: the active plan at each leg's altitude, cyan behind, purple on the current leg, and grey ahead. The line leaves a departure airport at field elevation and arrives at a destination airport at field elevation
- **Glide ring**: the map glide circle on the surface, and a cone from the aircraft down to it
- **TFR**: when the `TFR` layer is on, red if in effect and brown if later. `AGL` is height above the terrain, and the picture grows to include that ground. `MSL` is a fixed altitude, and the floor still sits on the ground when the terrain is higher. A floor above the picture, such as 15,000–17,999 ft MSL, is a dashed outline at the top of the scale with arrows pointing up. The map marker text uses the same `AGL` or `MSL` label
- **Game TFR**: when that layer is on, an orange 3 NM cylinder from the surface to 3000 ft above the ground
- **Filters**: `B`, `C`, `D`, `SUA`, `TFR`, `Game`, `Plan`, `Obstacles`, and `Terrain` show or hide those parts. `SUA` is restricted, warning, prohibited, and MOA. `TFR` and `Game` still follow the map layers
- **Drag** to rotate, **pinch** to zoom, **two fingers** to pan
- **Reset** restores the view
- Shelf colors, altitude labels, obstacles, and the ownship dot match the destination `3D` tab (Section 5.3)

9.8.1 3D Weather

Open: **Menu → 3D Weather** (under `3D`), or the destination popup `3DW` tab (after `3D`, before `ST`)

The same 3D box as `3D`, for weather instead of airspace. It shows the terrain, the runway diagrams, the aircraft, the winds aloft, and 3D NEXRAD radar. It has no airspace shelves, TFRs, plan, obstacles, glide ring, or traffic. The menu screen follows the aircraft. The popup tab is centered on that place, with the yellow long-press mark. The box is at least 12,000 ft tall, and grows to hold storm tops when radar is on.

- **Winds aloft**: on as soon as the screen opens. Winds inside 25 NM, drawn at their altitude and not under the terrain, from the 6-hour forecast. The arrow points downwind. The `Winds aloft` key at the bottom right shows the colors: blue is light wind, then cyan, green, yellow, and orange, and red is 60 kt or more
- **Build NEXRAD**: turns on 3D radar from the nearest WSR-88D radar. Nothing is downloaded until you press it. The button shows `Building 2/5` while it loads, and then

`Hide NEXRAD`. Press it again to hide the radar

- **10-minute life**: the radar keeps the time it was built. The NEXRAD key shows `Built 1542Z rebuild in 7 min`. Ten minutes after the build the radar hides, and the button changes to `Rebuild NEXRAD` with a red note, `NEXRAD is 10 min old. Rebuild for current radar.` Press it to download the newest scans and build again
- **dBZ**: sets the weakest echo shown, 10, 20, 30, 40, or 50 dBZ. The setting is kept between openings
- **NEXRAD key**: at the top left, with the radar, the scan time in UTC, and how many minutes it is behind the newest scan. The color bar runs from 10 to 70 dBZ, from light blue and green for rain to yellow, orange, and red for heavy rain, and pink for the strongest echoes
- **Loop**: the radar plays the last 30 minutes, one frame per radar scan (every 4 to 10 minutes), and rests on the newest frame before it starts over. The pause button stops it, and the slider steps to a frame
- **Radar data**: needs an internet connection. The first build downloads the last 30 minutes of scans from NOAA, usually 3 to 15 MB each, newest first, so use Wi-Fi when you can. The radar is a snapshot and does not update itself. A second radar fills in the gap above the antenna when you are close to the first. A radar sees up to about 230 NM away. A radar that is offline for 90 minutes is skipped, and a scan that stops at a low tilt can leave a gap overhead or at the top of a tall storm
- **Drag** to rotate, **pinch** to zoom, **two fingers** to pan
- **Reset** restores the view

9.9 PFD

Open: **Menu → PFD** (under `3D`)

The same `PFD` drawn by the map layer, sized to this screen. The map `PFD` layer is unchanged.

- Attitude, speed and altitude tapes, VSI, compass, CDI/VDI, AOA, and turn coordinator
- Hold the **OBS** knobs to set a course; double-tap either knob to return to the GPS course (Section 4)
- Elevation terrain and the lined-up runway are drawn the same way as on the map `PFD`

9.10 Link

Open: **Menu → Link** (under `Connectivity`). Not available on web.

Shares the active flight plan with other phones and tablets running AvareX on the same Wi-Fi. One device is the **Master**. The others are **Slave** and copy the master's plan, including which waypoint is active.

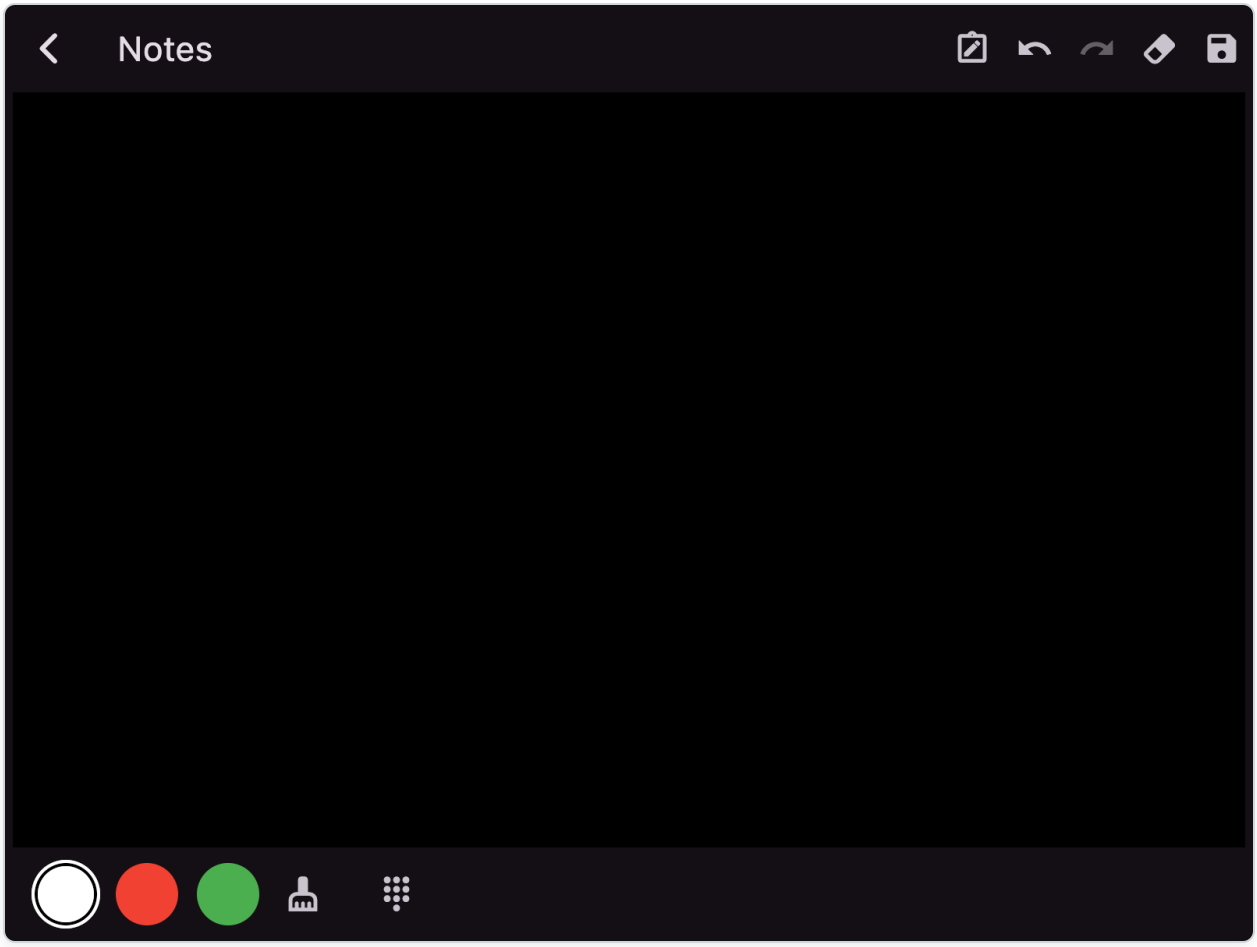
- **Off**: this device does not announce itself and does not follow a plan. Master, Slave, and Off are remembered after the app restarts.
- **Master**: sends this device's plan only when it changes. That includes a waypoint added, removed, or reordered, and a change of the active waypoint (including the fix inside an airway or procedure). Discovery still goes out about every 10 seconds.
- **Slave**: copies the master's plan and sets the same active waypoint. Edits made on a slave are replaced when the master sends the next update.
- **Name**: the name other devices show for this one.
- **Devices**: this device, and every other AvareX device heard in the last minute, with its role and address. Masters and slaves announce about every 10 seconds. Off does not. A device leaves the list after a minute with no message.
- **Messages**: debug log of the JSON going back and forth. **TX** is this device, **RX** is another. Tap a row to read the JSON. **Pause** freezes the log. **Clear** empties it. The antenna icon announces immediately.
- Every device must be on the same Wi-Fi. On iOS, allow **Local Network** for AvareX. On Windows, allow AvareX on private networks if the firewall asks.
- Use one master. If two devices are master, slaves follow whichever plan arrives last, and `Link` shows a warning.

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10) Notes Screen

10.1 How to access

On MAP tab, open **Menu** (bottom-left) and tap `Write a Note` under the `Flight` section.

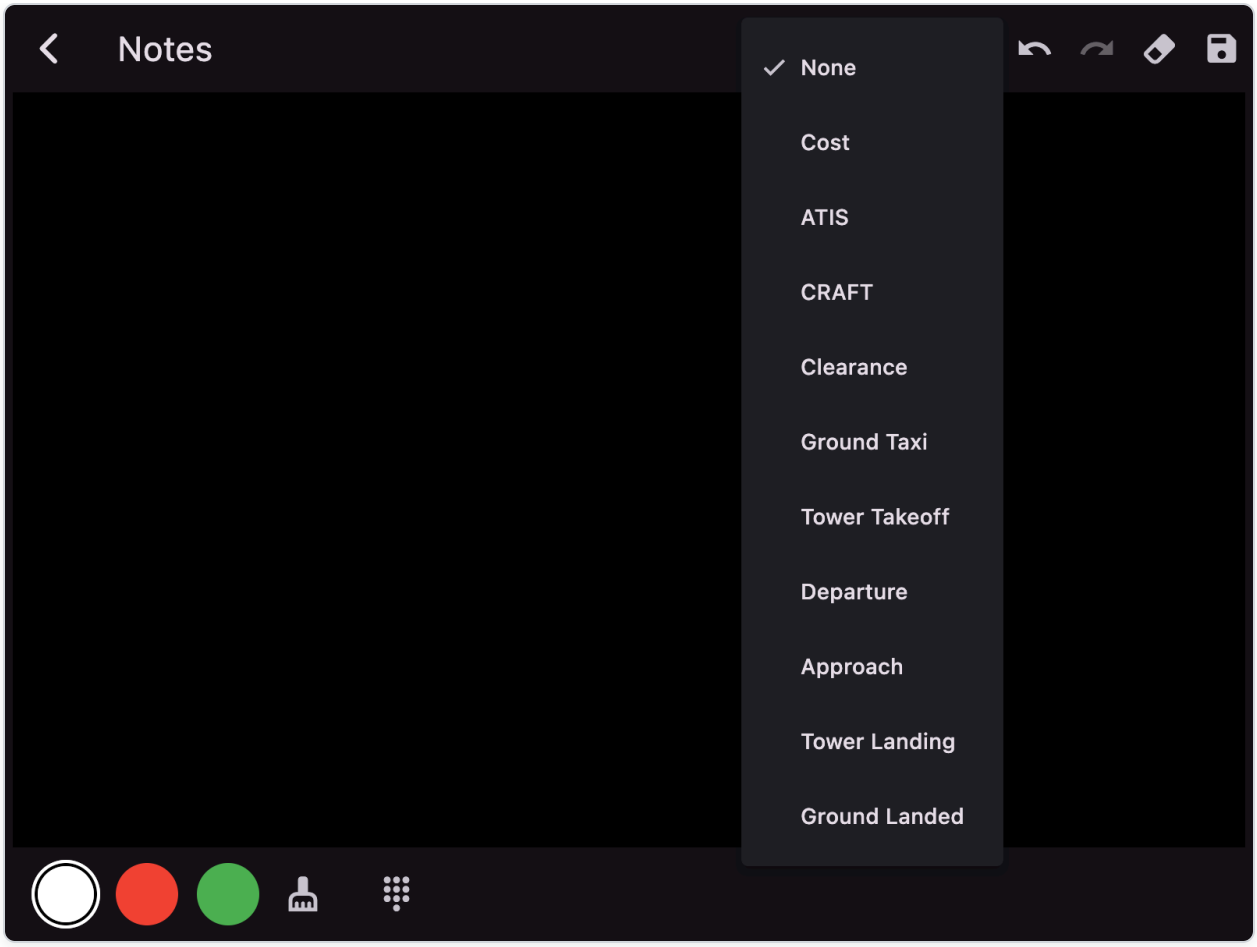


10.2 Features

- **Freehand drawing** with finger or stylus
- **Color choices**: Black/White (theme-adaptive), Red, Green
- **Eraser mode**: tap cleaning services icon in toolbar (wide stroke eraser)
- **Undo/Redo**: full stroke history
- **Clear canvas**: eraser icon in app bar
- **Save snapshot**: saves to Documents/notes as `notes_<timestamp>.png`
- **Auto-save**: current sketch autosaves on exit and reloads next time

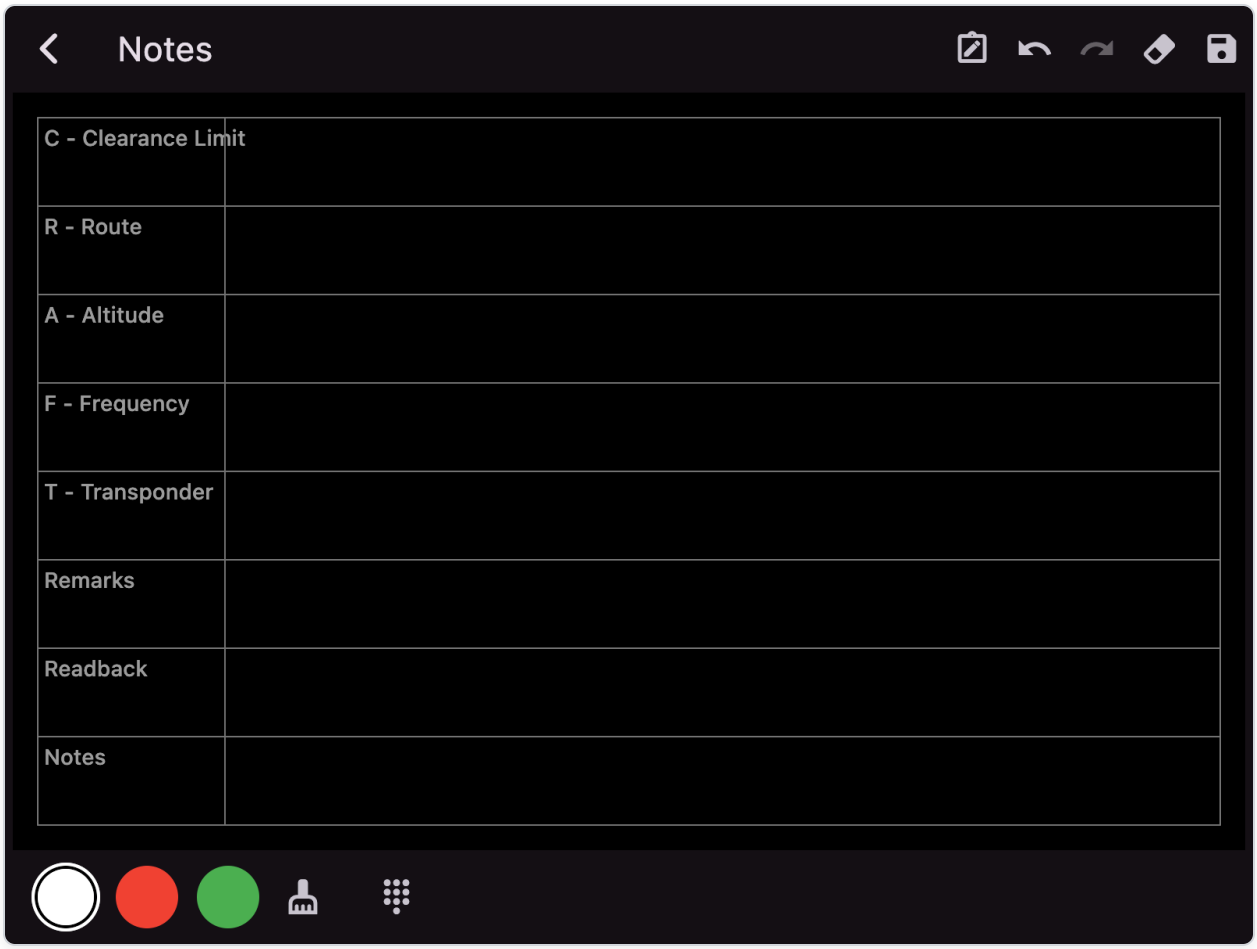
10.3 Aviation background sheets

Use the sheet icon in the app bar to select a background template for common aviation communications:



Sheet	Purpose
None	Blank canvas
Cost	Flight times (Hobbs, Tach, Fuel, Oil)
ATIS	VFR ATIS copydown fields (Airport, ATIS letter, weather, runways, NOTAMs)
CRAFT	IFR clearance (Clearance limit, Route, Altitude, Frequency, Transponder)
Clearance	VFR flight following clearance
Ground Taxi	Ground control taxi instructions
Tower Takeoff	Tower takeoff clearance
Departure	Departure control contact
Approach	Approach control contact
Tower Landing	Tower arrival/landing

Sheet	Purpose
Ground Landed	Ground control after landing



Each sheet maintains its own saved sketch state. Switching sheets saves the current sheet and loads the new one.

10.4 Number keypad

Tap the dialpad icon in the toolbar to display an on-screen keypad:

Number pad (default):

- Digits 0-9, decimal point
- Space, newline, backspace
- **C** (Clear): removes the last text entry; press repeatedly to remove more
- **ABC**: switches to QWERTY letter keyboard

QWERTY keyboard:

- Letters A-Z in standard QWERTY layout
- Space, backspace, newline
- **C** (Clear): removes the last text entry
- **123**: switches back to number pad

General usage:

- **Tap anywhere on canvas** while keypad is active to create a new text entry at that location
- Type to add text at the current location, then tap elsewhere to add text at another spot
- Supports multiple text entries at different positions on the same sheet
- All text entries are saved with the sheet and persist across sessions

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11) Account and Cloud Features

11.1 Access

Cloud features (Backup/Sync, Community, Scheduler, Airport Businesses) run on **iOS, Android, Windows, and macOS**. Flight Intelligence runs on **iOS and Android** only. Cloud features are not available on Linux or Snap.

They are reached from:

- **Account screen** — `MAP → Menu → Internet → Account`. Sign in, view profile, sign out, or delete the account. The screen title is `Account`. After you **Sign out** (or delete the account), the same `Account` screen switches to the sign-in form — you are not sent back to the map.
- **Pro icon** (iOS/Android) — `MAP top-right account icon`. Opens `Flight Intelligence` when signed in (paywall if needed), or the `Account` sign-in screen if not. Some features also route here on demand (e.g., translate actions).
- **`INTERNET` drawer section** — `MAP → Menu → Internet`. Holds `Account`, `Backup/Sync`, `Community`, `Scheduler`, and `Flight Intelligence`. `Account`, `Backup/Sync`, `Community`, and `Scheduler` are free (sign-in only). `Flight Intelligence` still uses the Pro paywall.

11.2 Login and subscription flow

- Sign in/register with email authentication (Firebase) on the `Account` screen
- While signed in, `Account` shows your profile (avatar, display name, sign-in methods), **Sign out**, and **Delete account**

- After sign-out, `Account` shows the sign-in form again (same route, title still `Account`)
- Cloud features (`Backup/Sync`, `Community`, `Scheduler`, Airport Businesses) need a signed-in account only — they are free on iOS, Android, Windows, and macOS
- `Flight Intelligence` uses the RevenueCat entitlement (`Pro`) paywall
- After login, open `Flight Intelligence` from the MAP top-right account icon or from `MAP → Menu → Internet → Flight Intelligence`

11.3 Flight Intelligence (AI)

Screen title: **Flight Intelligence**

Access: `MAP → Menu → Internet → Flight Intelligence`, or `MAP top-right account icon` (opens `Flight Intelligence` when signed in, or `Account` sign-in if not)

Powered by Gemini 3.8 Flash with Google Search integration.

Capabilities:

- Ask free-form aviation/trip questions (internet required)
- Question must be 256 characters or less
- Context limit: 10,000 tokens

Context toggles (tap to highlight, highlighted context sent with question):

- **Logbook icon**: include your 50 most recent logbook entries
- **Aircraft icon**: include tail number and type from first aircraft
- **Route icon**: include current flight plan
- **Cloud icon**: include winds aloft from departure and destination

Query/answer workflow:

- Type question in text box
- Tap **Ask** to submit
- Response appears in text box with disclaimer
- History drawer (clock icon): view previous questions/answers, choose question, show answer, delete history item
- **Clear (X)** button clears text box
- **NOTAM translate**: on the destination popup `NOTAM` tab, tap the **Translate** icon on a NOTAM to open this screen with a plain-English translation question for that NOTAM

Disclaimer: Responses are generated by an AI model and may not be accurate or reliable. Do not use when life, health, property are at stake.

11.4 Pilot Community

Screen title: **Pilot Community**

Access: `MAP → Menu → Internet → Community`

A pilot-only social network backed by Firebase. Pilots can build a public profile, create groups, and join existing ones. **Private** groups require owner approval; **public** groups are open to all signed-in users. This is a free Cloud feature (sign-in required, no Pro subscription).

Disclaimer (shown as a strip at the top of the screen and via the info icon in the app bar):

- Apps4Av is **not responsible for data loss** in the Pilot Community. Keep your own copies of anything you want to retain.
- **Do not share sensitive information** (passwords, government IDs, financial details, medical records, etc.). Anything you post may be visible to other pilots.
- Apps4Av **does not moderate** community activity. Group owners are responsible for their own groups.
- Apps4Av **will not share community data with third parties**. Data is stored only in the project's Firebase backend to operate this feature.

Tabs:

- **My Groups**: groups you own or have been approved to join.
- **Discover**: searchable list of public groups, sorted by member count.
- **Profile**: your shared pilot profile (display name, home airport, ratings, aircraft, bio).

Top action:

- **New Group** (floating action button): create a new group with name, description, optional home airport (ICAO), and visibility (Public or Private).

Group detail screen tabs:

- **Feed**: scrollable list of topics (top-level posts). Active members can **Post** (floating action button) up to 1000 characters and optionally attach:
    - **Photos** (up to 4 per post, JPEG/PNG, automatically downscaled to 1920px wide and re-encoded at quality 80, which also strips EXIF). Tap a photo to zoom.
    - **A flight plan** — tap **Attach plan** to share your current PLAN. Other members see the route inline with a **Load** button that imports the plan into their PLAN tab (after a confirmation dialog).
    - **An airport ICAO** quick-tag.

Threaded discussions: each topic keeps its own thread of replies. Tap **Reply** on a topic to add a reply (text and/or photos), or **tap the topic** to open its discussion thread — the topic at the top followed by all replies, oldest first, with a **Reply** button. The reply count is shown on each topic in the feed. Replies stay attached to their topic, so a conversation never gets mixed up with unrelated posts.

Unread topics: topics posted since you last opened a group's feed are shown in **bold**. Opening the feed marks it read, so they return to normal weight on your next visit (your own posts are never shown as unread).

Authors can delete their own posts; the group owner can delete any post. Deleting a topic also deletes all of its replies. Deleting a post removes its attached photos from cloud storage on a best-effort basis.

- **Members**: list of active members with their home airport. The owner sees a separate **Pending requests** section at the top with **Approve** / **Reject** controls and can **Remove** any non-owner member.

- **About**: owner, visibility, home airport, member/post counts, creation date.

Membership controls (shown as a banner inside each group):

- **Join** (public group): immediate active membership.
- **Request to Join** (private group): puts you in `pending`; owner must approve before you see the feed or can post.
- **Requested**: you have a pending request; tap to cancel.
- **Leave**: removes you from the group. Owners cannot leave — they must **Delete group** from the app bar (deletes all posts and memberships).

Visibility rules:

- Public groups are discoverable and any signed-in user can read the feed.
- Private groups are discoverable by name but the feed is hidden from non-members. Owners get a red badge over the **Members** tab when pending requests exist.

Notifications: in-app only (no push). A **bell icon** appears both in the `Pilot Community` app bar and on the `Community` row in the drawer's `INTERNET` section; it shows an unread count and opens the **Notifications** screen. You are notified when:

- someone **replies to a topic you started**, or
- someone **replies to any post in a group you own**.

On the Notifications screen, unread items are shown in **bold** with a dot. You can **tap** a notification to jump straight to that discussion thread (which marks it read), **swipe** a notification away to dismiss it, **mark all read** (`✓✓` icon), or open **notification settings** (gear icon). Settings let you turn reply notifications **off globally**, or **mute individual groups** — muted groups and a global "off" stop counting toward the unread badge and are hidden from the list. Pending join requests still appear as a red badge over the **Members** tab when you open a group you own.

11.5 Aircraft Scheduler

Screen title: **Aircraft Scheduler**

Access: `MAP → Menu → Internet → Scheduler`

This is a free Cloud feature (sign-in required, no Pro subscription).

A shared booking calendar and club dispatch board for flying clubs, partnerships, and flight schools, backed by Firebase. It works like the Pilot Community: an owner creates a scheduler, members request to join, and members reserve the resources the owner adds. **All schedulers are private** — they are discoverable by name, but the owner approves every member, and the schedule and roster are hidden from non-members.

Tabs:

- **My Schedulers**: schedulers you own or have been approved to join.
- **Discover**: type a name to find a scheduler to request to join. There is no public browse list because all schedulers are private.

Top action:

- **New Scheduler** (floating action button): create a scheduler with name, description, and optional home airport (ICAO). The creator becomes the owner. Every scheduler is private; set booking limits afterwards from the **Booking Rules** screen.

Scheduler detail screen tabs:

- **Schedule**: the booking timeline. Resources are stacked **vertically** (one row each) and time runs **horizontally** across the day. Colors indicate slot state:
  - **Green** = available (free) slot on an in-service resource.
  - **Blue** = booked. A solid block is the main reservation; a thin lighter-blue bar at the bottom of the row is a backup. Your own reservations have a white border.
  - **Red** = unavailable (the resource is marked out of service by the owner/dispatcher).

Use the date bar (**‹ / ›**, the date button to pick a day, and **Today**) to change days. Tap a **green** slot to open the **Book** dialog, which pre-fills the tapped day/time as the start; choose a **start date + time** and **end date + time**. Bookings may span **multiple days** (up to 14 days). A multi-day reservation appears as a booked (blue) block on every day it covers. If the resource is already booked for any of that time, you are added to the **backup queue** instead. When booking an aircraft you can optionally assign an **instructor** and/or link a **lesson pack**. Tap any reservation block to see details (including training assignment) and cancel it. Aircraft with an open **grounding squawk** or overdue **100-hour** hobbs cannot be booked.

- **Dispatch**: club ops board for the shared fleet (members only):
    - **Fleet**: status cards for each aircraft (READY / ATTENTION / MX DUE / GROUNDED / OUT OF SERVICE) with hobbs, tach, annual / 100-hour / transponder / ELT due, open squawk counts, and MX notes. Tap a card to view details. Owners and **dispatchers** can **Update meters / MX** and toggle availability; any member can **File squawk**.
    - **Squawks**: open and resolved discrepancy reports (Grounding / Caution / Info). An open **Grounding** squawk blocks booking and shows the aircraft as GROUNDED on the fleet board. The reporter, owner, or dispatcher can **Resolve** a squawk.
    - **Lessons**: prepaid **lesson packs** (hour blocks) assigned to students. Owners create packs and can cancel/delete them. The student, assigned instructor, owner, or dispatcher can **Log hours**; when used hours reach the total, the pack completes automatically.
- **Mine**: your own reservations in this scheduler, split into **Upcoming** and **Past**. Upcoming entries can be cancelled here directly. When the owner has set booking limits, chips at the top show your current usage (e.g. `2/3 active`, `1/1 weekend`). Weekend reservations and backups are tagged.
- **Members**: list of active members with **club roles** (Pilot / Student / Instructor / Dispatcher). The owner sees a **Pending requests** section with **Approve** / **Reject** controls, can **Remove** any non-owner member, and can set club role via the badge icon. Students can be linked to an **assigned instructor**; that instructor is pre-selected when the student books.
- **About**: owner, visibility, home airport, member/resource counts, booking limits, creation date, and a short explanation of how booking and dispatch work.

Owner app-bar actions:

- **Booking rules** (tune icon): open the **Booking Rules** screen to set limits.
- **Delete scheduler** (trash icon): remove the scheduler and all resources/reservations/memberships/squawks/lesson packs.

Resources (owner only): tap **Add resource** on the Schedule tab to add an **Aircraft** or **Instructor** (name plus optional tail number/identifier). Tap a resource's name in the left column to **toggle availability** (mark it out of service for maintenance — shown in red) or **Delete** it (which also removes its reservations). Hobbs/tach and MX due dates are edited from the **Dispatch → Fleet** board (owner or dispatcher).

Booking Rules screen (owner only): set limits that apply to members (the owner is exempt). A value of **0 means unlimited**.

- **Reservations per member**: the most current/upcoming reservations a member may hold at once.
- **Weekend reservations**: the most Saturday/Sunday reservations a member may hold at once.

Limits are enforced when a member books; exceeding a limit shows an explanatory message and the booking is blocked.

Booking and backups:

- Any active member can book an available resource that is not already taken, subject to the owner's booking limits and dispatch holds (OOS, grounding squawk, overdue 100-hour).
- If a resource is already booked for an overlapping time, the new request joins the **backup queue** (Backup #1, #2, ...).
- When a **main** reservation is cancelled, the next backup in line is automatically **promoted** to the main reservation.
- **Only the member who made a reservation or the scheduler owner can cancel it.** The owner can cancel any member's reservation.

Club roles:

Role	Typical use
Pilot	Default member; can book and file squawks
Student	Linked to an assigned instructor; bookings can carry that instructor and a lesson pack
Instructor	Appears in booking and lesson-pack instructor pickers
Dispatcher	Same fleet/MX/squawk resolve powers as the owner (without owning the scheduler)

11.6 Backup/Sync

Screen title: **Backup/Sync**

Access: `MAP → Menu → Internet → Backup/Sync`

This is a free Cloud feature (sign-in required, no Pro subscription).

Cloud operations for `user.db` using Firebase Storage:

- **Backup** (upload icon): upload local database to cloud storage
  - Warning: overwrites existing cloud backup
- **Restore** (download icon): download cloud backup over local database
  - Warning: overwrites existing local data
- Progress indicator shows transfer percentage
- Both actions show confirmation warnings before proceeding
- Data stored under user's Firebase UID

11.7 Airport Businesses & Reviews

A crowd-sourced directory of airport businesses/FBOs that pilots build together. Business names are pre-seeded from the app's built-in business database; everything else is contributed by signed-in pilots.

This is a free feature — it is **not** part of a Pro subscription. It only requires a signed-in account (the same Cloud `Account`) so that contributions are accountable rather than anonymous. It is available on iOS, Android, Windows, and macOS.

Access: destination popup → `Business` tab. When you are signed in, the airport's businesses are listed **directly inside the tab** — there is no separate screen to open. If you are not signed in, the tab shows a sign-in prompt; tapping it takes you to the sign-in screen (the only time this feature navigates away), and no subscription or paywall is involved. Once signed in the tab refreshes to show the businesses in place.

What you can do (all in place, without leaving the tab):

- **Add**: tap **Add** to create a new listing at the airport with its name, services, fuel types, operating hours, phone number, and radio frequency.
- **View details & reviews**: tap a listing to expand it inline and see its fuel, services, operating hours, phone number, radio frequency, and reviews.
- **Edit details**: on an expanded listing, tap **Edit details** to add/update its services, fuel available, operating hours, phone number, and radio frequency. The phone number and radio frequency fields use a numeric keypad. Details are community-owned, so any signed-in pilot can enrich a listing.

- **Set fuel prices**: on an expanded listing, tap **Set fuel prices** to enter a price per gallon for any fuel type. Each price is shown with the date it was last set; leave a field blank to clear that fuel's price.
- **Add Review**: on an expanded listing, tap **Add** next to *Reviews*, choose a **1–5 star** rating and optional text. You can post **one review per business** — reviews are permanent and can't be edited or deleted.

Ordering: listings that pilots have created, updated or reviewed are floated to the top of both the `Business` tab list and the `PLATE` business selector (most recently touched first), marked with a small verified-user icon. The remaining (seeded) listings follow **nearest-first** from the airport, so the most relevant businesses appear before distant ones. The "interacted" status is stored on the listing in the cloud, so that part of the ordering is the same for everyone.

Accountability: every listing and review is tagged with your profile display name — nothing is anonymous, and each display name is **unique** across pilots so it can't be used to impersonate someone else. To keep a durable, attributable record, **listings and reviews cannot be deleted or edited once posted** (only detail fields like services/fuel/hours can be updated), and each pilot may post **one review per business**. A listing's average rating and review count update automatically as reviews are added.

Offline: businesses are cached on your device, so airports you've already opened stay browsable without a connection. An **offline banner** appears when you're viewing cached data. While offline: airports you've never opened may show nothing until you reconnect, star ratings show **"Ratings offline"** (they're computed on the server), and anything you add or edit is **saved on your device and uploads automatically when you're back online** (you'll see a note instead of the usual confirmation).

On the airport diagram: listings that carry a location (e.g. those seeded from the built-in business database) can be pinned on the `PLATE` airport diagram using its business selector — see Section 6.6.

11.8 Messages from Apps4Av

On iOS, Android, Windows, and macOS, AvareX checks for a message from Apps4Av when you open the app and again when you bring it back to the foreground. If one is waiting, a popup shows its title and text. No account is required. These messages are separate from the Community reply notifications in Section 11.4.

Each current message is also kept on this device. Open `MAP → Menu → Messages`. The list stays until you delete a message. Swipe a row left to delete it on this device. Deleting does not change the message Apps4Av published.

- **OK**: closes the popup. The message stays in `Messages`.
- Select the title, the text, or the web address to copy. A web address, when the message includes one, appears under the text.
- An update from Apps4Av shows the popup again and replaces the saved copy.
- Nothing new appears in the popup when you have already tapped `OK`. A message that has expired is not shown in the popup. One already saved stays in `Messages` until you delete it.

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12) Warnings and Troubleshooting Drawer

When warning state is active, a red alert icon appears on MAP top-right. Tap it to open issues drawer.

Possible issue items:

- GPS permission denied (opens app settings)
- GPS disabled (opens location settings)
- No GPS lock
- Critical data/charts missing (shortcut to Download)
- Data expired (shortcut to Download)
- Runtime exception notices

iOS: traffic / ADS-B weather / external GPS not appearing

If AvareX shows ownship moving but no traffic, no FIS-B weather, and no external GPS lock from your receiver — even though other apps on the same phone work — iOS has blocked AvareX's local network access.

Fix: `Settings → Privacy & Security → Local Network → AvareX → ON`, then force-quit and relaunch AvareX. See FAQ-16 for full details.

This is the most common cause of "AvareX traffic doesn't work" on iPhone/iPad and only affects iOS, not Android. The same permission is required for `Link` (other AvareX devices on UDP `49090`).

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13) Data Lifecycle and Auto-Update Behavior

- On reaching the onboarding **Databases and Maps** page, AvareX automatically downloads the **DatabasesX** package plus the **Sectional**, **Plates**, **CSUP**, and **Elevation** for your current GPS region whenever any of them are missing/expired (skips items already current).
- Weather downloads refresh periodically (10-minute cycle in storage timer).
- GPS source modes (tap `SRC` tile to cycle):
  - **Auto**: Prefers external GPS when available, falls back to internal after 30s timeout. Shows `Internal-A` or `External-A`.
  - **Internal**: Uses only internal GPS, discards external GPS data. Shows `Internal` with green background.
  - **External**: Uses only external GPS, discards internal GPS data. Shows `External` with blue background.
- After the last GPS, ADS-B, or NMEA fix, ownship heading, speed, altitude, and position keep coasting until the **No GPS lock** warning (60 seconds with no fix from either source) so the aircraft symbol does not jump or spin at rest or during a brief signal loss. When that warning appears, coasting stops and ownship freezes.
- Flight status tracks taxi/airborne transitions and accumulates flight time.
- External autopilot/NMEA sentence output is generated continuously while app is running and IO connection exists.
- Track recording continues while Tracks layer is enabled; saves to KML when layer is turned off.

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14) Quick Feature Path Index

Feature	Path
Download charts/data	`MAP → Menu → Download`
Read weather docs / import files	`MAP → Menu → Documents`
View saved tracks (KML)	`MAP → Menu → Documents → tracks folder → tap KML file`
Create logbook from track	`Documents → tracks → tap KML → 2D Map → Log Flight`
Create folder for documents	`MAP → Menu → Documents → folder icon in app bar`
Build or modify plan	`PLAN tab`

Feature	Path
Delete entire plan	`PLAN tab` → trash icon (top-right of header)
File FAA plan	`PLAN → Brief & File`
Manage filed plans	`PLAN → Manage`
Load reversed plan	`PLAN → Load & Save → 3-dot menu → Load Reversed`
Create IFR preferred route	`PLAN → Create → Create IFR Route`
Show recent ATC routes	`PLAN → Create → Show IFR ATC Routes`
Send plan to / get plan from Avidyne IFD (Wi-Fi)	`PLAN → Transfer`
Destination details	Long-press on map or tap FIND result
3D Class B/C/D, restricted, warning, prohibited, and MOA airspace, plus obstacles and runway diagrams	Destination popup `3D` tab (after `Wind`, before `ST`)
3D winds aloft and NEXRAD radar, with terrain and runway diagrams	Destination popup `3DW` tab (after `3D`, before `ST`)
3D airspace around the aircraft, with terrain, runway diagrams, and ADS-B traffic	`MAP → Menu → 3D` (under `Write a Note`)
3D winds aloft and NEXRAD radar around the aircraft	`MAP → Menu → 3D Weather` (under `3D`)
PFD on its own screen	`MAP → Menu → PFD` (under `3D`)
Draw a holding pattern	Long-press a fix or navaid → `Main` → `Hold` → `Draw on Map`
Show plates for airport	Destination popup `Plates` or `PLATE tab`
Sync plates to the active airport	`PLATE` airport diagram → sync icon
Airport diagram after landing	`PLATE` airport diagram → landing icon
Get airport satellite view	`PLATE tab → Plate selector → APS-AERIAL VIEW (Get)`

Feature	Path
Delete airport satellite view	`PLATE tab → Plate selector → APS-AERIAL VIEW → trash icon`
Configure aircraft	`MAP → Menu → Aircraft & Performance → My Aircraft tab`
Set best glide / sink rate	`MAP → Menu → Aircraft & Performance → My Aircraft tab` (Best Glide Speed, Sink Rate)
Set best climb speed and rate	`MAP → Menu → Aircraft & Performance → My Aircraft tab` (Best Climb Speed, Best Climb Rate)
Best cruise altitude	`PLAN tab` → altitude icon (next to the forecast dropdown)
Change aircraft map icon	`MAP → Menu → Aircraft & Performance → My Aircraft tab → icon dropdown`
Checklist operations	`MAP → Menu → Check Lists`
Save map settings as a profile	`MAP → Map Settings → Profiles`
Apply a saved profile	`MAP → Map Settings → Profiles`
Emergency mode (glide speed, best runway, checklist, declare)	`MAP → Map Settings → Profiles → Emergency`
Choose the Emergency checklist	`MAP → Map Settings → Profiles` heading → `Emergency` tile three-dot menu
Weight and balance	`MAP → Menu → Aircraft & Performance → W&B tab`
Takeoff performance	`MAP → Menu → Aircraft & Performance → Takeoff tab` (fixed-wing icon only; hidden for helicopter)
Landing performance	`MAP → Menu → Aircraft & Performance → Landing tab` (fixed-wing icon only; hidden for helicopter)
Cruise performance	`MAP → Menu → Aircraft & Performance → Cruise tab` (fixed-wing icon only; hidden for helicopter)
Runway crossing alert	Automatic when `Audible Alerts` is on; uses closest-airport runways

Feature	Path
Fuel burn in plan	`PLAN tab` — set **GPH** manually or use the aircraft icon to load from performance data
Logbook + dashboard	`MAP → Menu → Log Book`
Logbook statistics	`MAP → Menu → Log Book → Details`
Currency status	`MAP → Menu → Log Book → Details` (currency card)
Bluetooth pairing/connection	`MAP → Menu → IO` (Android)
Share the plan with other AvareX devices	`MAP → Menu → Link`
Check ADS-B receiver status	`MAP → ADSB instrument tile` (movable; tap opens ADS-B Status screen)
Enable Stratus 3 Open ADS-B Mode	`MAP → ADSB instrument tile → antenna icon (top-right)`
Notes/drawing	`MAP → Menu → Write a Note`
Notes with aviation sheet	`MAP → Menu → Write a Note → sheet icon → select template`
Notes number keypad	`MAP → Menu → Write a Note → dialpad icon`
Pro AI	`MAP → Menu → Internet → Flight Intelligence` or `MAP top-right account icon` (iOS/Android)
Translate a NOTAM (Pro)	`Destination popup → NOTAM → Translate icon` on a NOTAM (iOS/Android)
Cloud account	`MAP → Menu → Internet → Account` (iOS/Android/Windows/macOS)
Sign out / delete account	`MAP → Menu → Internet → Account` → **Sign out** or **Delete account** (returns to the sign-in form on the same `Account` screen)
Community	`MAP → Menu → Internet → Community`
Create a pilot group	`MAP → Menu → Internet → Community → New Group`

Feature	Path
Join a public group	`MAP → Menu → Internet → Community → Discover → tap group → Join`
Approve pending members (owner)	`Community → group → Members tab → Approve / Reject`
Edit your pilot profile	`MAP → Menu → Internet → Community → Profile → edit icon`
Post a photo to a group	`Community → group → Post → Photo → choose source`
Share your current flight plan	`Community → group → Post → Attach plan`
Load a shared flight plan into PLAN	`Community → group → tap Load on a post with a route`
Reply to a topic	`Community → group → tap topic → Reply`
View a topic's discussion thread	`Community → group → tap a topic in the Feed`
Apps4Av message	Shown automatically when you open the app (iOS, Android, Windows, macOS). Tap `OK`. Read again at `MAP → Menu → Messages`
View notifications	`Community → bell icon` (also shown on the `Community` row in the drawer's `INTERNET` section)
Turn reply notifications off / mute a group	`Community → bell icon → gear icon`
Aircraft Scheduler	`MAP → Menu → Internet → Scheduler`
Create a scheduler (private)	`MAP → Menu → Internet → Scheduler → New Scheduler`
Add an aircraft/instructor (owner)	`Scheduler → scheduler → Schedule tab → Add resource`
Book a resource	`Scheduler → scheduler → Schedule tab → tap a green slot → Book`
View / cancel my reservations	`Scheduler → scheduler → Mine tab`
Set booking limits (owner)	`Scheduler → scheduler → tune icon → Booking Rules`

Feature	Path
Cancel a reservation	`Scheduler → scheduler → Schedule tab → tap reservation → Cancel`
Mark a resource out of service (owner)	`Scheduler → scheduler → Schedule tab → tap resource name → Available toggle`
Fleet / dispatch board	`Scheduler → scheduler → Dispatch tab`
Update hobbs/tach / MX due (owner/dispatcher)	`Scheduler → scheduler → Dispatch → Fleet → aircraft → Update meters / MX`
File a squawk	`Scheduler → scheduler → Dispatch → Squawks → File squawk`
Create a lesson pack (owner)	`Scheduler → scheduler → Dispatch → Lessons → New lesson pack`
Assign student → instructor (owner)	`Scheduler → scheduler → Members → badge icon`
Airport Businesses & Reviews	`Destination popup → Business tab` (inline; free; sign-in required)
Add an airport business	`Destination popup → Business tab → Add`
Add services/fuel/hours to a business	`Business tab → tap listing → Edit details`
Set fuel prices for a business	`Business tab → tap listing → Set fuel prices`
Review an airport business	`Business tab → tap listing → Reviews → Add`
Cloud backup/restore	`MAP → Menu → Internet → Backup/Sync`
User Manual (Help)	`MAP → Menu → Help`
PFD terrain and synthetic runway	`MAP → Layers → PFD > 0` (Elevation charts for terrain; runway when lined up within 10 NM)
Set an OBS course	`MAP → PFD → hold an OBS knob` (double-tap either knob to return to the GPS course)
Enable Game TFR overlay	`MAP → Layers → Game TFR slider > 0`

Feature	Path
CAP Grid overlay	`MAP → Layers → Nav > 0 → CAP Grid slider > 0` (zoom to level 9+)
Enable wind vectors	`MAP → Layers → Weather > 0 → Wind Vectors > 0` (use altitude slider)
Enable ceiling overlay	`MAP → Layers → Weather > 0 → Ceiling > 0` (use altitude slider)
Change traffic volume (S/M/L)	`MAP → Layers → Traffic > 0 → Volume`
Change GPS source mode	Tap `SRC` tile in instrument overlay
Measure distances	`MAP → Map Settings → Measure Distance Off → long-press map to add points`
Rubber band waypoints	`MAP → Map Settings → Rubber Banding Off → long-press and drag waypoints`
Mute audible alerts	`MAP → Map Settings → Audible Alerts On` (label flips to `Audible Alerts Off`)
North-up / track-up orientation	Tap the `MT` tile, or `MAP → Map Settings → North Up / Track Up` (olive tile = track up)
Insert waypoint at position	`PLAN → long-press waypoint row → FIND → select destination`
View winds & terrain analysis	`PLAN → analytics icon → tap diagram for details`

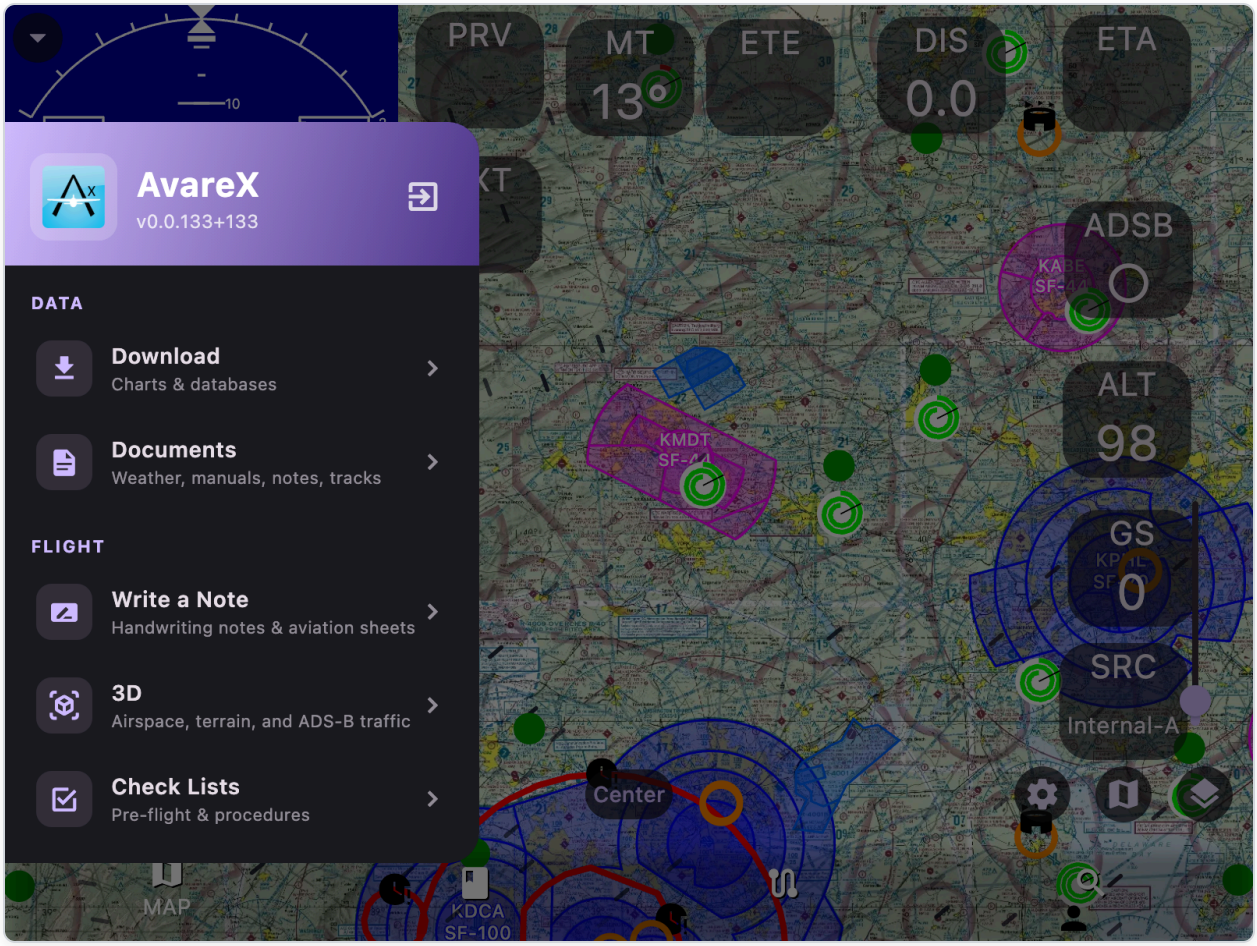
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15) Step-by-Step Use Cases (Common Scenarios)

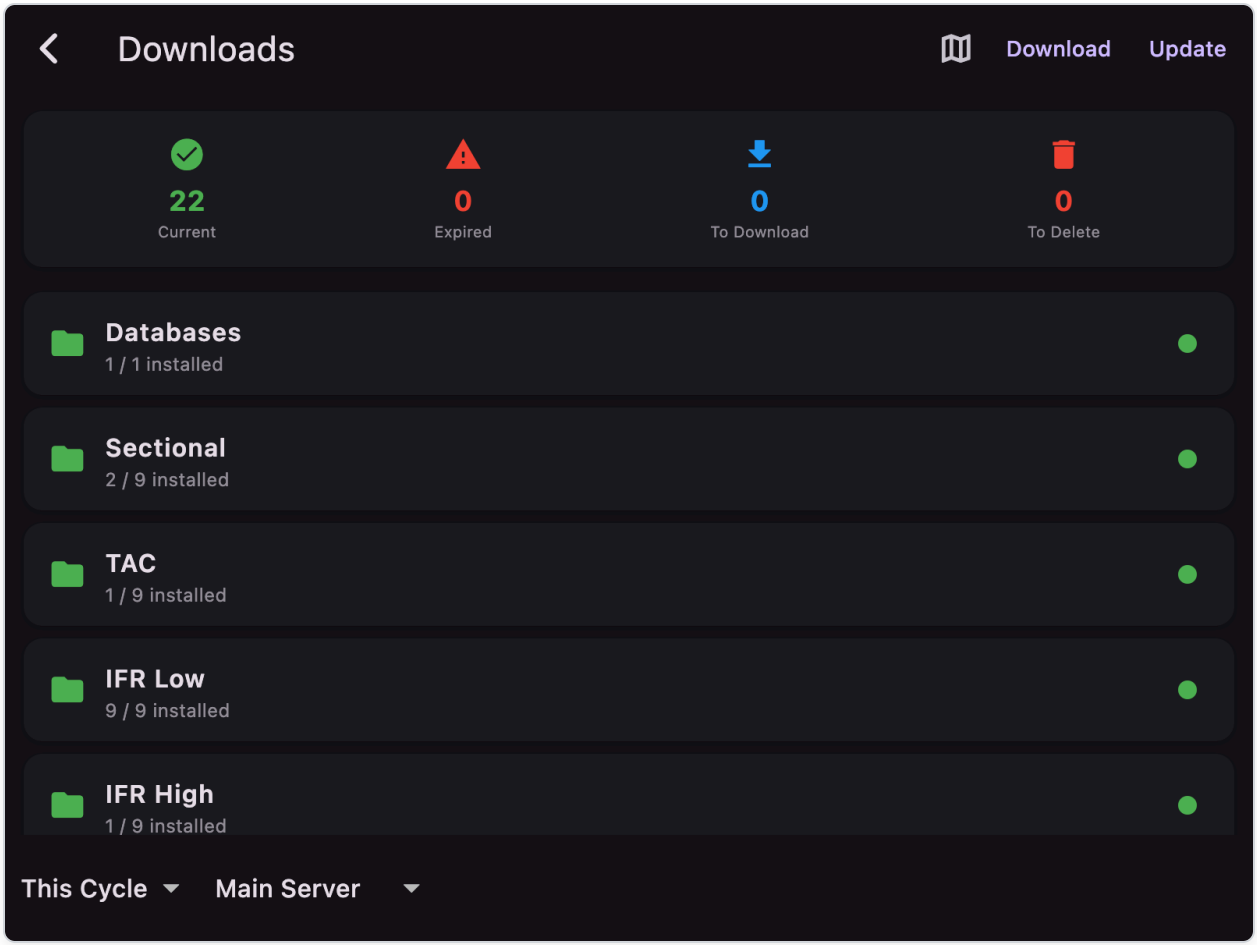
> Each use case is a numbered walkthrough you can follow end-to-end. Screen names match the labels in the app, and the screenshots above each step block show the exact place you should be.

UC-00: Download charts and databases

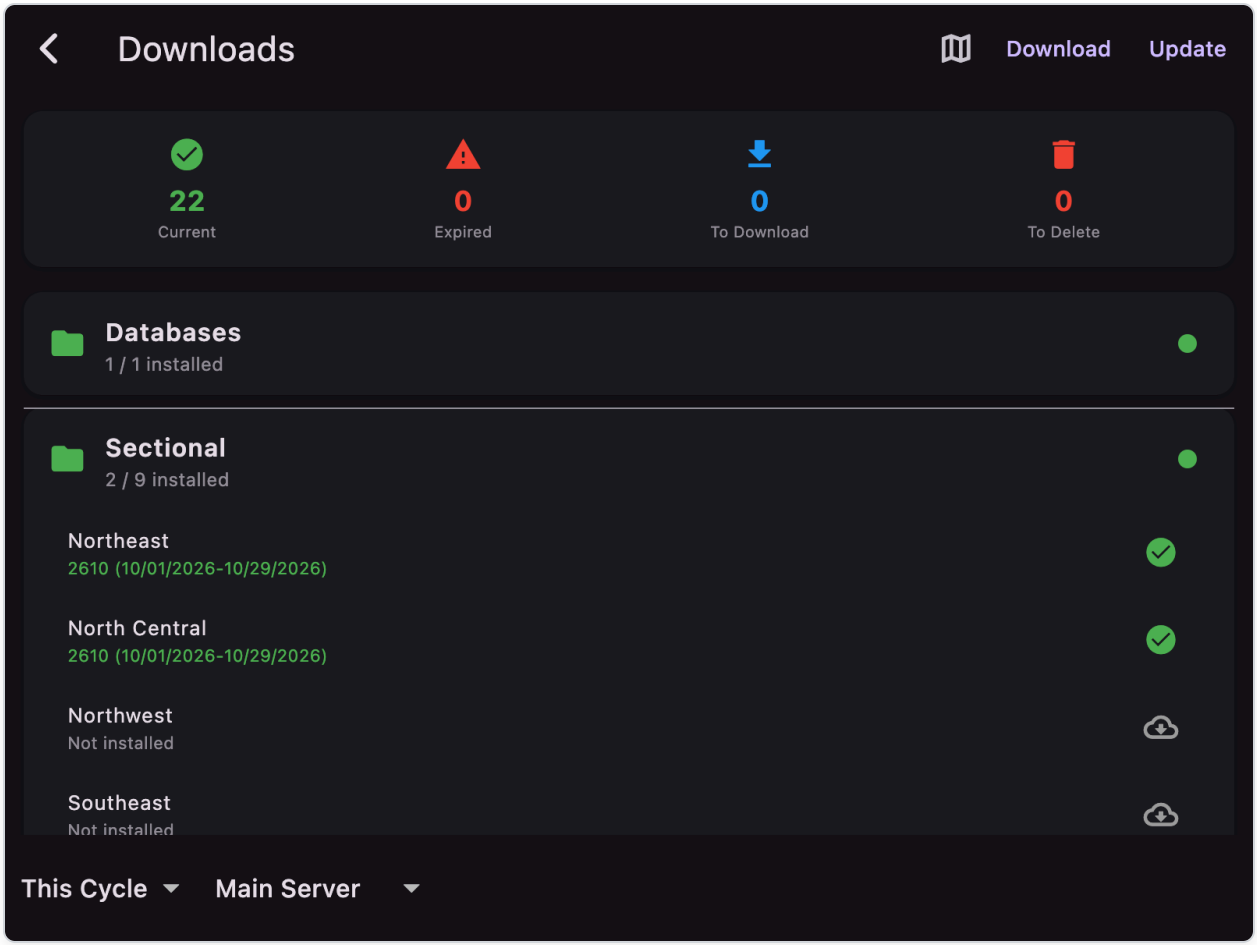
Without **DatabasesX** the airport database is empty and the FIND/PLAN tabs return no results. Onboarding already grabs the databases plus the Sectional, Plates, and CSUP for your GPS region **automatically** when they are missing. Use this walkthrough when you want to add or update **specific chart regions** manually.



1. On the **MAP** tab, tap **Menu** (bottom-left) to open the left drawer.
2. Tap **Download** under the **DATA** section.



3. Confirm the **cycle** dropdown is on **This Cycle** (use **Next Cycle** only if a new cycle is published and you want to pre-stage it). Use **Main Server** unless it is unreachable; then switch to **Backup Server**.
4. Tap **Databases** to expand it, then tap **DatabasesX** so it shows a download icon (queued).
5. (Optional) Tap **Sectional**, **TAC**, **IFR Low**, **IFR High**, **Plates**, **CSUP**, **Elevation**, etc. to queue the chart products you need.



6. Tap individual regions inside each category (e.g. *NE_SEC*, *SE_SEC*) — each tap toggles the queue state (download icon → trash icon → cleared).
7. Press **Download** in the app bar. A progress ring appears next to each item; the **Status Summary Card** updates the **Current / Expired / To Download / To Delete** counts.
8. Wait for downloads to complete before leaving the screen — exiting can abort an in-flight download.
9. To refresh expired data later, just press **Update** instead — it queues every expired item automatically.

> Tip: During **first-run onboarding**, the *Databases and Maps* page automatically downloads your GPS region's databases, Sectional, Plates, and CSUP. It also offers **Choose Manually** (opens this Download screen) if you prefer to pick data yourself. The manual flow is the same as above.

UC-01: Connect an external ADS-B/GPS receiver over Wi-Fi UDP

Best when your receiver broadcasts GDL90/NMEA over local network.

1. Power on your ADS-B/GPS receiver.
2. Connect your tablet/phone/computer to the receiver's Wi-Fi network (or same LAN).
3. Open AvareX and go to `MAP`.
4. **iOS only — first launch on a new device**: when iOS shows the *"AvareX would like to find and connect to devices on your local network"* prompt, tap **Allow**. Without this permission iOS will silently drop all ADS-B/GPS UDP packets and traffic/weather will never appear, even though ownship over internal GPS still works. If the prompt was previously dismissed or denied, enable it under `Settings → Privacy & Security → Local Network → AvareX`.
5. Wait for incoming data on supported UDP ports: `4000`, `43211`, or `49002` (automatic listener).
6. **Stratus 3 / 3i only**: if the receiver is not in Open ADS-B mode, open `MAP → ADSB` tile → **ADS-B Status**, then tap the antenna icon (top-right). Confirm the toast, then wait for the Receiver status to show Connected.
7. Turn on useful map layers:
   - `Traffic`
   - `Weather`
   - `Radar` (internet radar)
8. Verify data:
   - Ownship updates smoothly
   - Traffic symbols appear (if in range)
   - Weather products populate
9. If needed, open warning drawer from red icon and resolve GPS/data warnings.

UC-02: Connect an external ADS-B/GPS receiver over Bluetooth (Android)

1. Open `MAP → Menu → IO`.
2. Wait for device discovery to populate the list.
3. Select your receiver from list.
4. Pair (if needed), then tap **Connect**.
5. Confirm status line shows connected device.
6. Return to map and enable `Traffic` / `Weather` layers as needed.
7. To disconnect: `IO → Disconnect`.

UC-03: Rubber-band a route directly on the map

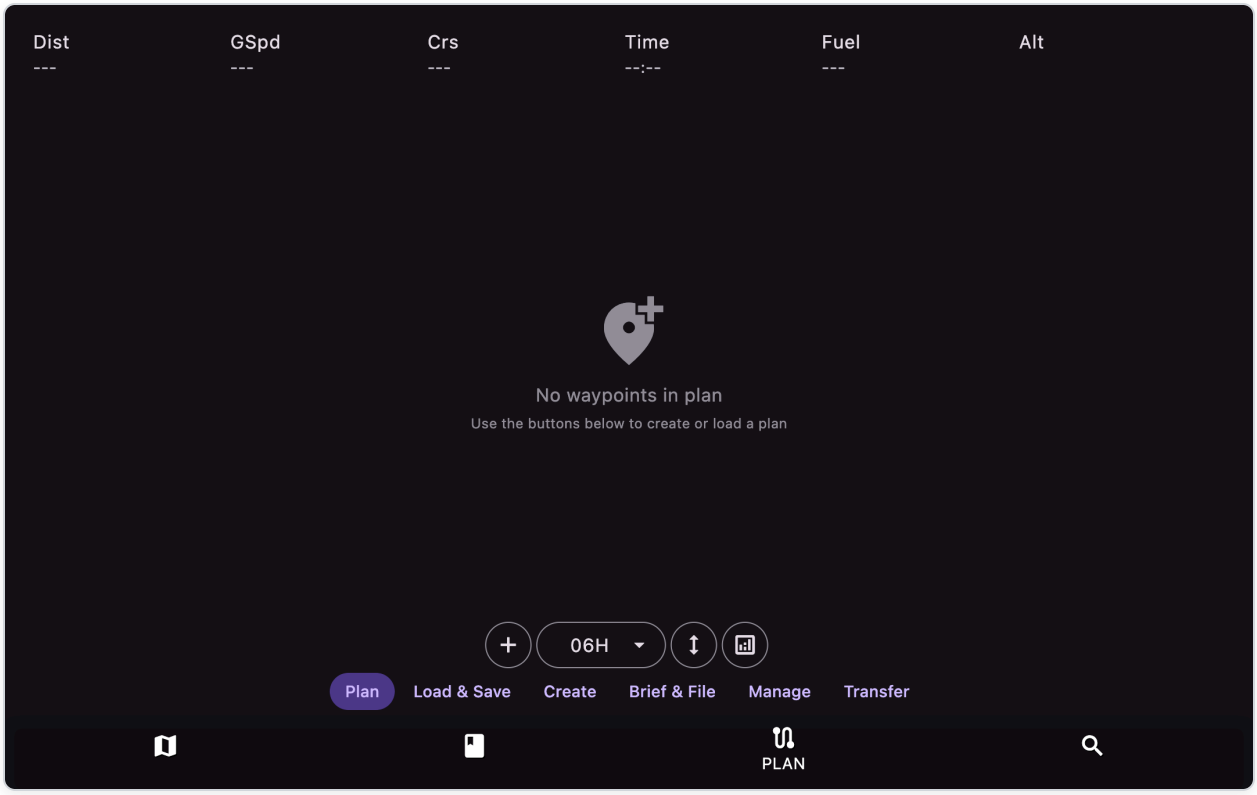
Use this to adjust waypoints graphically.

1. Ensure your route has waypoints (`PLAN` tab or destination popup `+Plan`).

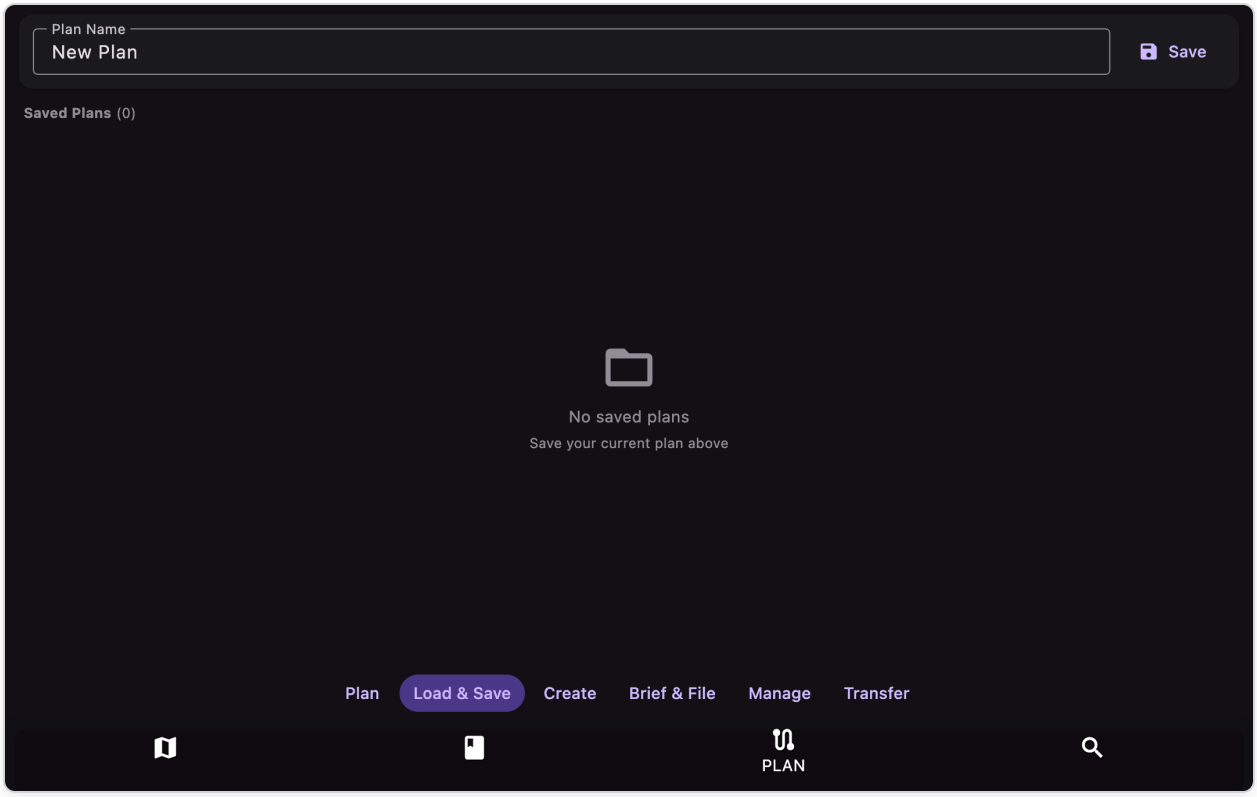
2. Go to `MAP`.
3. Tap the **gear** icon in the bottom-right controls, then tap `Rubber Banding Off` to enable it (the row changes to `Rubber Banding On` and is highlighted).
4. Long-press a waypoint marker or label and drag to a new position.
5. Release to snap/update from database lookup and rebuild route geometry.
6. Turn rubber banding off when done to avoid accidental edits.

UC-04: Build a flight plan from scratch (quick VFR workflow)

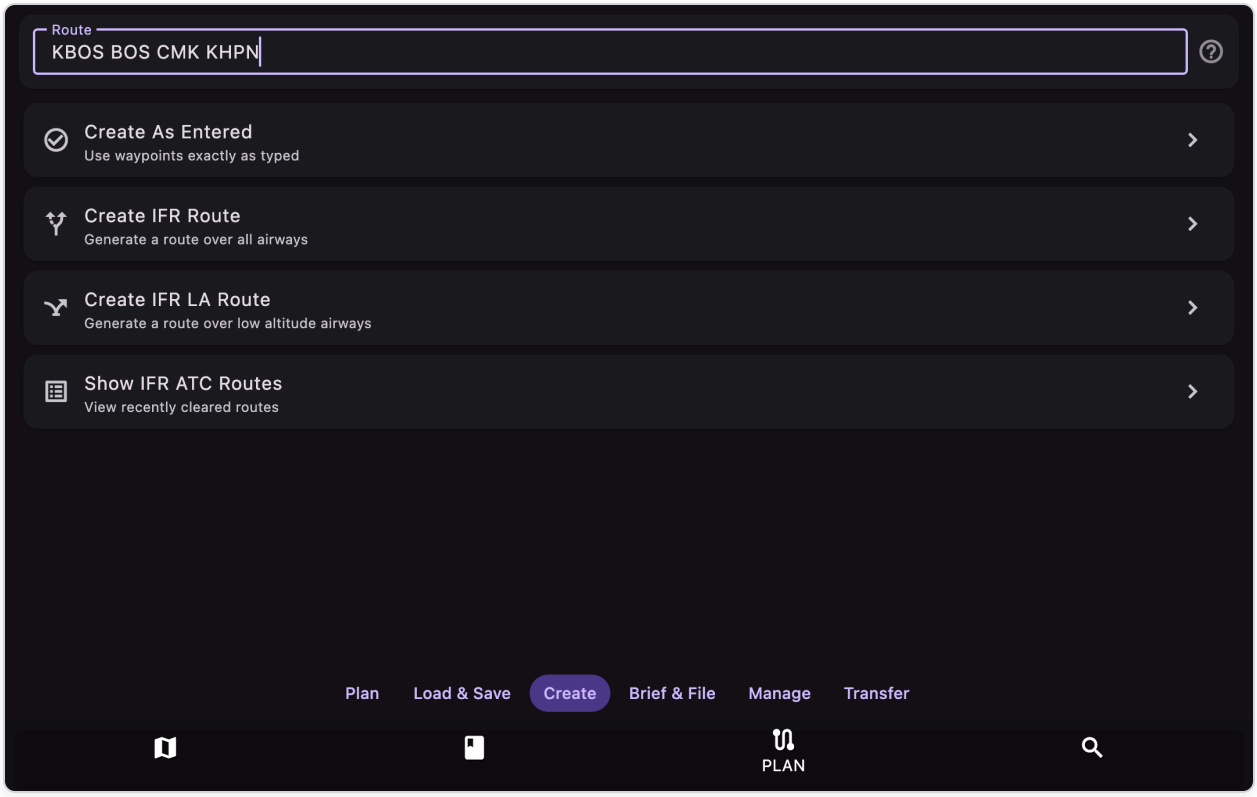
There are three ways to add a waypoint to the plan; the fastest for a brand-new VFR plan is **Plan → Create → Create As Entered**, shown below.



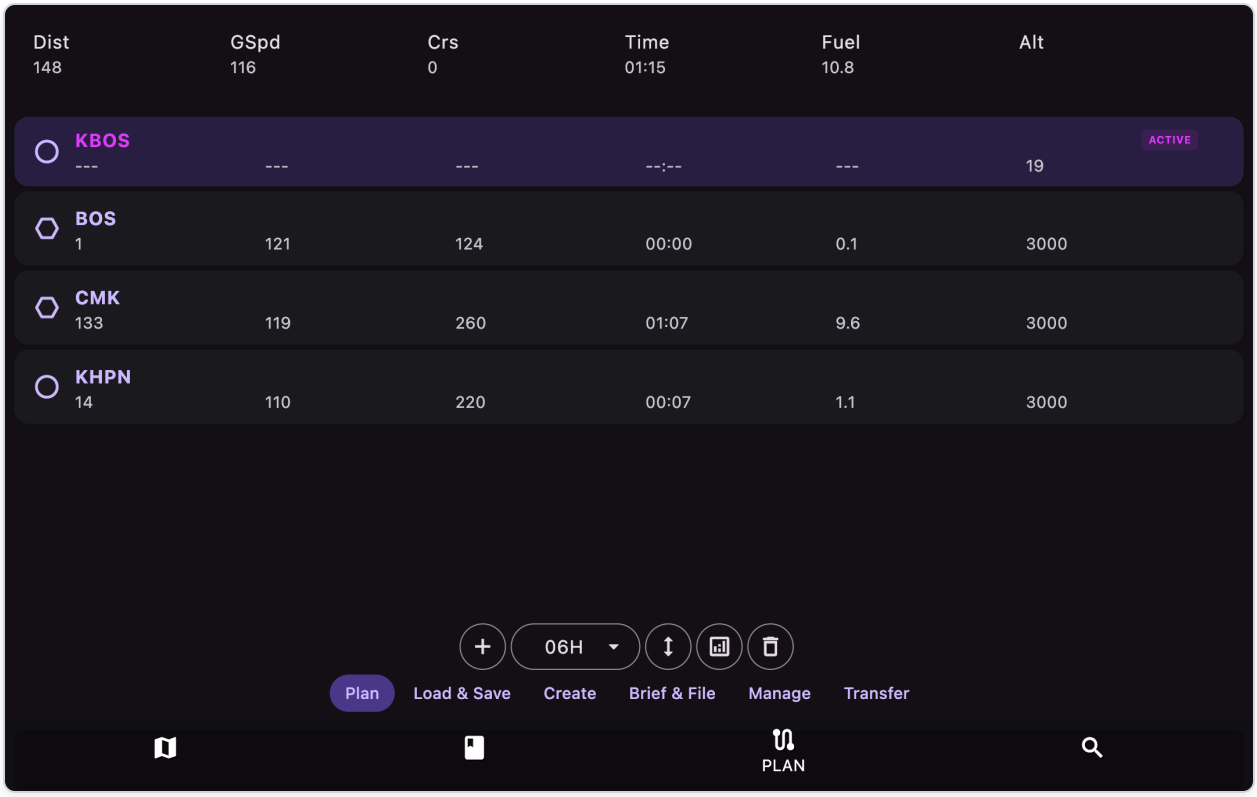
1. Tap the **PLAN** bottom tab. You'll see the empty plan view above (Distance / GSpd / Crs / Time / Fuel show `---`, and `No waypoints in plan` is displayed in the body).
2. Press **Create** at the bottom; the Create sub-page appears inline (the **Create** button is highlighted).



3. If a different sub-page is showing, tap **Create** in the bottom row to switch.



4. Tap into the **Route** text field and type your route as space-separated identifiers (airports, navaids, fixes, airways). Example: `KBOS BOS CMK KHPN`.
5. Tap **Create As Entered**. The app parses each identifier, looks it up in the database, and returns to the main PLAN view.



6. Verify the totals header (Dist 148, GSpd 91, Time 01:55, Fuel 18.6 in the example).
7. Adjust the bottom controls as needed:
    - Cruise speed and fuel flow come from the selected aircraft, read once when PLAN opens
    - Each waypoint **Alt** column is the altitude reached at that waypoint while the airplane is still climbing. After it levels off, Alt stays on that cruise altitude. A short leg keeps climbing on the next leg. Clear it to use the plan default (3000 ft, or the map altitude slider)
    - **Altitude icon** — next to the forecast dropdown. Picks the fastest altitude for the rest of the route, at least 1,000 ft above the terrain in the next 50 miles. A climb waits 50 miles, and a descent can begin 20 miles after the airplane is at that altitude. Climb and descent use best climb speed and rate, and time and fuel use climb speed while the altitude is changing. A short leg shows the altitude reached there. Odd and even thousands follow the course from the first waypoint to the last
    - **Forecast horizon** — `06H / 12H / 24H` (winds aloft window for nav-log calculations)
8. To **reorder** legs: long-press and drag rows. To **set a different leg active**, tap its row. To **delete** a leg, swipe right-to-left.
9. To **save** the plan: tap **Load & Save** at the bottom → press **Save** and give it a name.

Alternative add methods (used after the plan exists):

- From **MAP**: long-press near a destination → tap a row in the popup → `+Plan` (insert at current waypoint) or `↓Plan` (append to end).

- From **FIND**: search/select an item → on the destination popup tap `+Plan` or `↓Plan`.
- Long-press a waypoint row in **PLAN** to insert *after* that row (opens FIND; the next selection inserts there).

> See **UC-21** below for using the **navigation log** (analytics icon) to review leg calculations, winds, and terrain along the route.

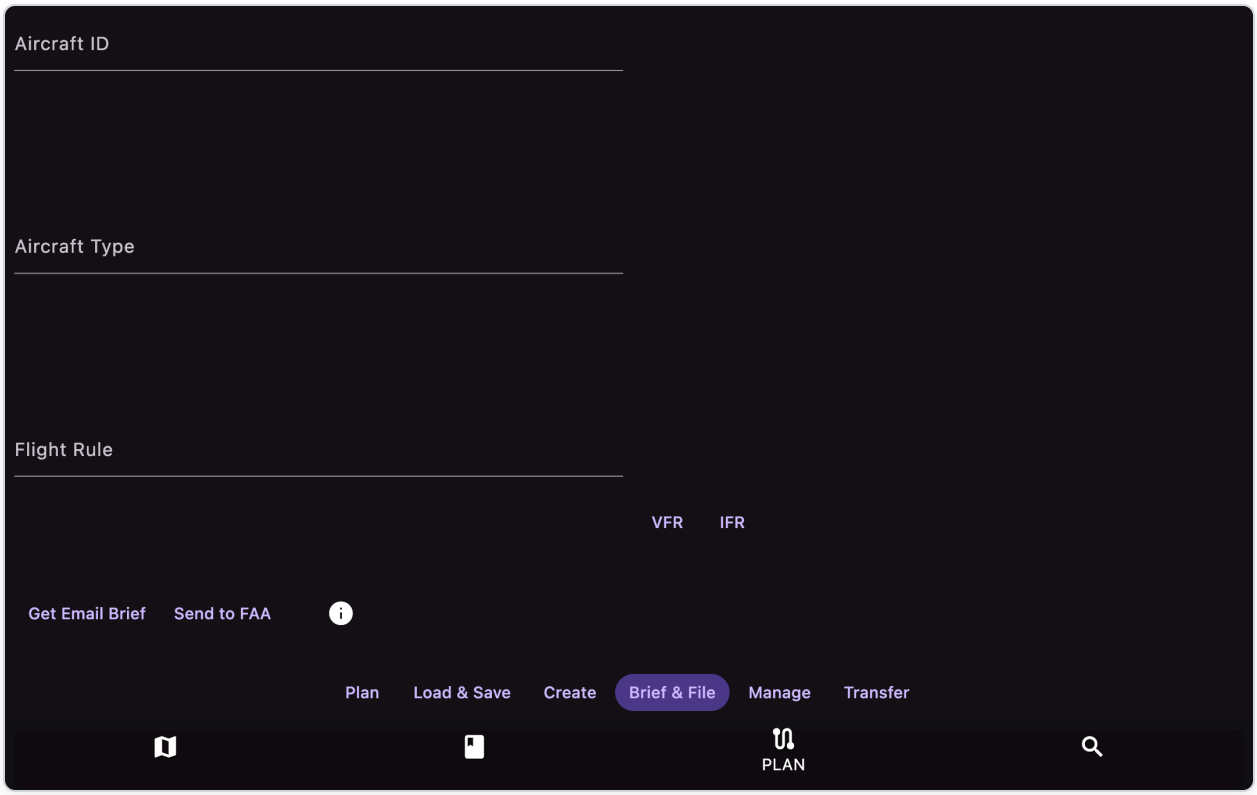
UC-05: Create an IFR route automatically

1. Open `PLAN → Create`.
2. In Route field:
   - Enter route text and use **Create As Entered**, or
   - Enter `DEPART DEST` and use **Create IFR Route**, or
   - Use **Show IFR ATC Routes** to view recent ATC route options.
3. When route is loaded, return to `PLAN` to review/reorder waypoints.

UC-06: File a flight plan with the FAA

Prereq: set your 1800wxbrief-compatible email in onboarding.

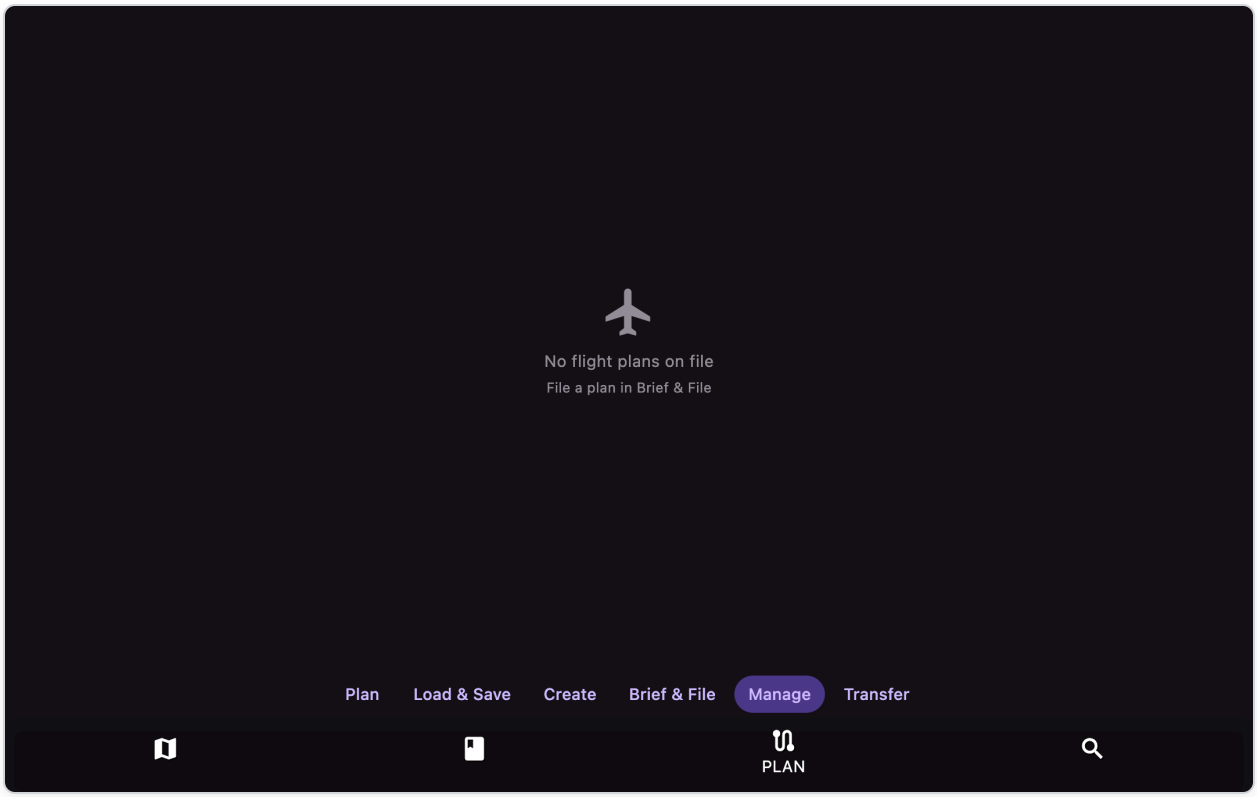
1. Build/verify your route in `PLAN` (see UC-04).
2. Open `PLAN → Brief & File`.



3. Fill required fields (aircraft ID/type, rule, departure, destination, route, times, fuel, POB, etc.).
4. Use quick-fill buttons:
    - "Planned" buttons for departure/destination/route from current plan
    - Aircraft buttons (e.g. `N80251`) to fill from stored aircraft profiles
5. Tap:
    - **Get Email Brief** for briefing email, and/or
    - **Send to FAA** to file.
6. Check status indicator (checkmark=success, info icon=error with tooltip).

UC-07: Activate, close, or cancel an FAA flight plan

1. Open `PLAN → Manage`.



2. Find your plan in the list (color-coded by status).
3. Use action by state:
    - `PROPOSED` (blue): **Depart** (choose time with sunrise/sunset reference) to activate
    - `ACTIVE` (green): **Close** after landing
    - Non-active: **Cancel** if no longer needed

UC-08: Quickly divert to a nearby airport

1. On `MAP`, long-press near destination area.
2. In popup, review nearby list and tap a candidate airport.

3. Tap `→D` to set Direct-To and center map.
4. Optional:
    - Tap `Plates` for immediate airport diagrams/procedures
    - Add to full route with `+Plan`

UC-09: Use plates with procedure profile and add procedure to plan

1. Open `PLATE` tab.
2. Select airport (bottom-right selector).
3. Select desired plate (bottom-left selector, color-coded by type).
4. Use procedure menu (plus/road icon):
    - Choose a procedure to show its profile card, draw the **CIFP route** on georeferenced plates (green→yellow→brown→blue→red→orange from final back, matched on the vertical profile), and append it to the plan. Each fix that has an altitude on that profile gets that altitude in the plan **Alt** column.
5. If other IAFs exist for the same chart, tap a **black IAF chip** to switch transition (route and profile update; plan is not duplicated).
6. Fly the approach using the colored route overlay and vertical profile.
7. Keep instruments visible on plate using top-right show/hide toggle.
8. Close profile card with X button when done (clears the route overlay).

UC-10: Save your flown track and retrieve it from Documents

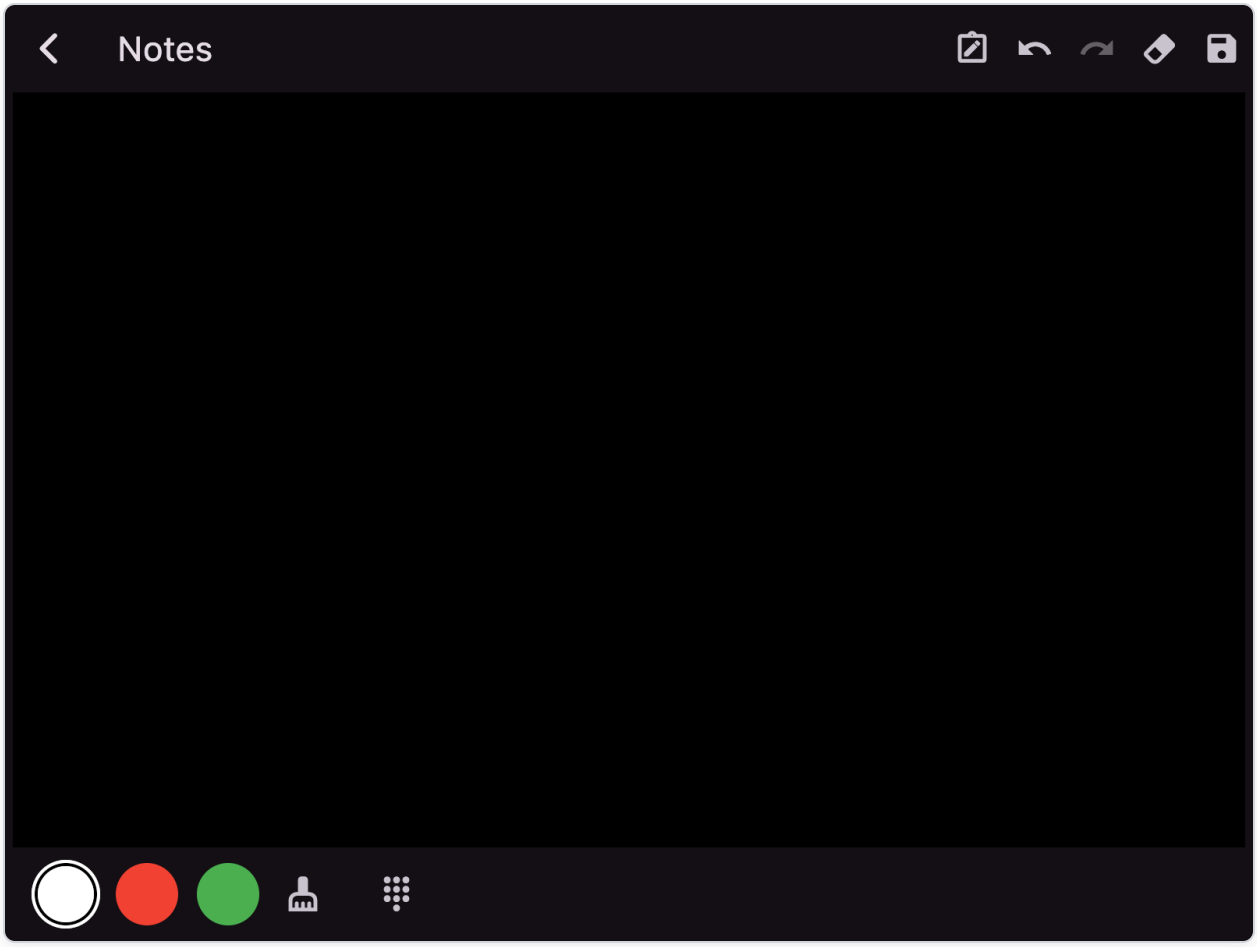
1. In `MAP`, keep `Tracks` layer on during flight (opacity > 0).
2. When done, set `Tracks` layer opacity to `0` in Layers menu.
3. App saves track automatically as KML in Documents/tracks folder.
4. Open `MAP → Menu → Documents → tracks` to access/share file.

UC-10a: View a saved track and create a logbook entry

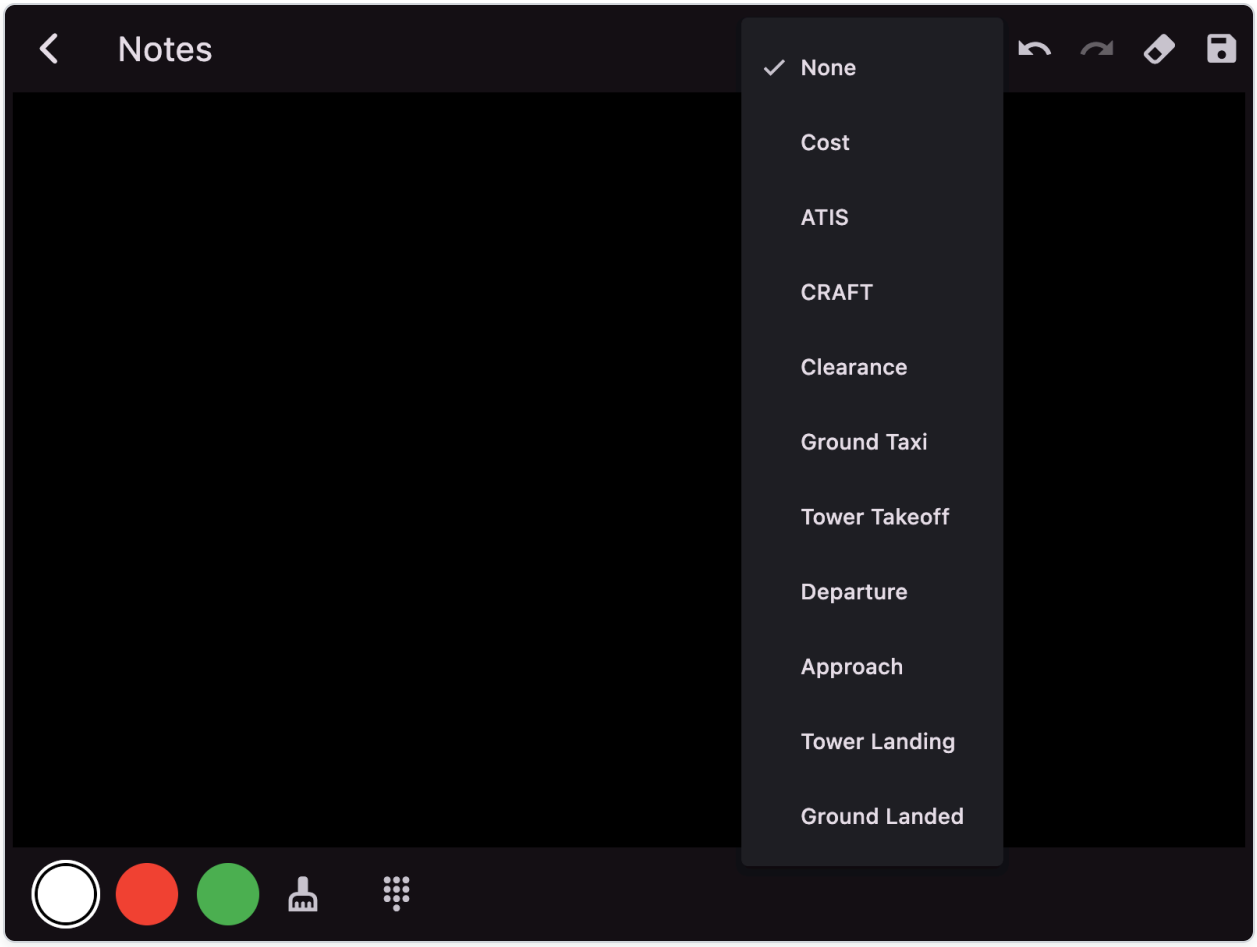
1. Open `MAP → Menu → Documents → tracks`.
2. Tap a KML file to open the Track Viewer.
3. Use view icons to switch between 2D Map, Altitude Profile, and 3D View.
4. In 2D Map view, tap **Log Flight** to create a logbook entry from the track.
5. The entry auto-fills route (nearest airports), time (from timestamps or distance/TAS), and aircraft info from selected profile.

UC-10b: Use aviation sheets for IFR clearance copydown

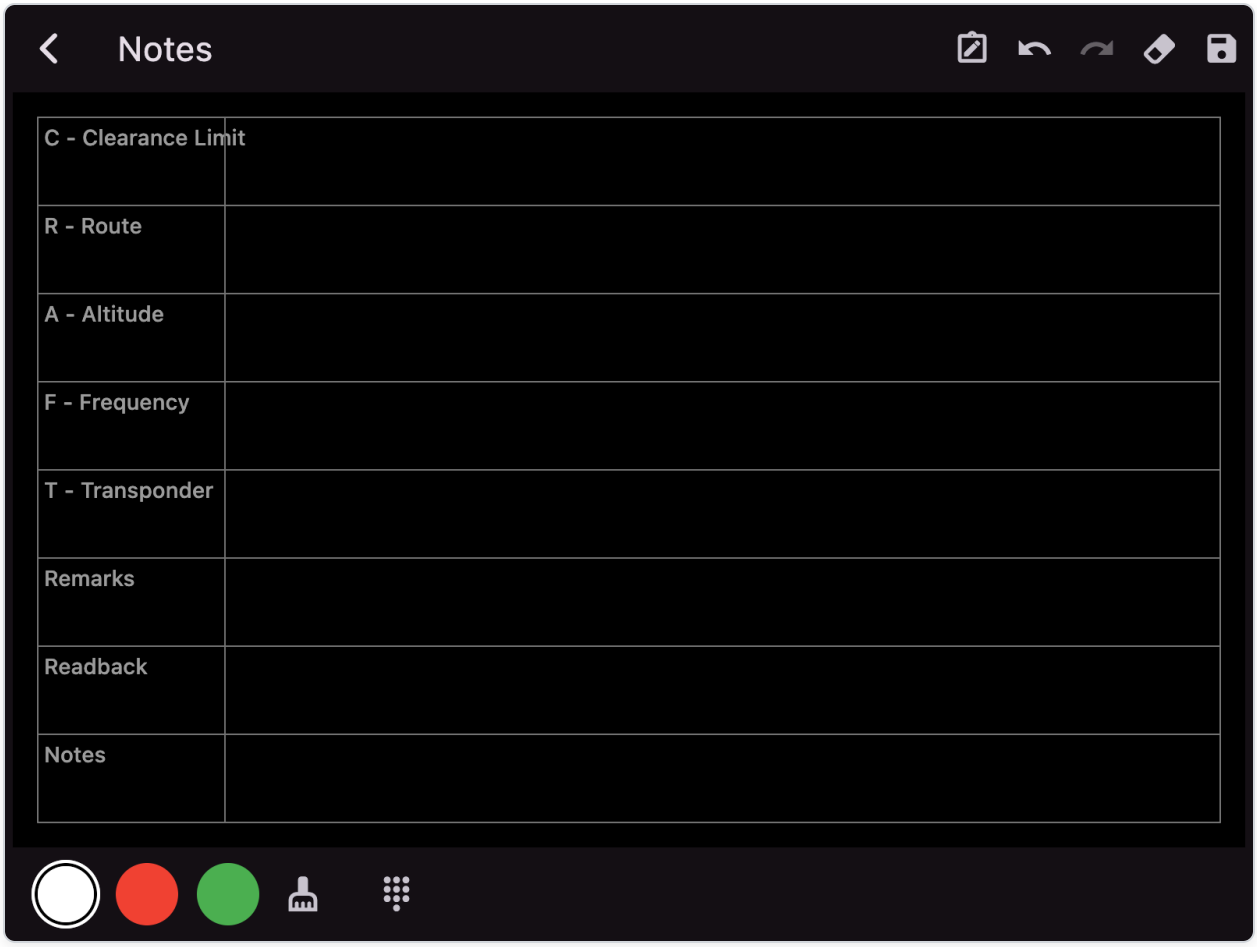
1. On `MAP`, open **Menu** and tap `Write a Note` (under `Flight`).



2. Tap the sheet icon (note_alt) in the app bar.



3. Select **CRAFT** for IFR clearance fields.



4. Use the number keypad (dialpad icon) to type frequencies, altitudes, or squawk codes.
5. **Tap on the canvas** to position the typed text exactly where you want it (e.g., next to the frequency field).
6. Use freehand drawing to fill in route or other notes.
7. The sheet auto-saves when you exit or switch sheets.
8. Return anytime to continue where you left off; each sheet maintains its own state.

UC-11: Import a GeoJSON overlay and display it on map

1. Open `MAP → Menu → Documents`.
2. Tap **Import** and select a `.geojson` file.
3. Open `MAP` and set `GeoJSON` layer opacity > 0.
4. Imported polygons/markers now draw on the map.

UC-12: Send your plan to, or get a plan from, an Avidyne IFD over Wi-Fi

Exchange flight plans with a panel-mounted Avidyne IFD440/540/550 in either direction.

1. Connect this device to the **same Wi-Fi network** as the IFD (configure the network on the IFD's Maintenance-mode page if needed). **iOS**: grant the **Local Network** permission.

2. Open `PLAN → Transfer`.
3. Under **Avidyne IFD (Wi-Fi)**, wait for the IFD to appear in the list.
4. To **send**: build your route with at least 2 waypoints, then tap **Send** next to the IFD. Wait for the success toast, then **review and activate** the uploaded route on the IFD.
5. To **get**: tap **Get** next to the IFD. AvareX downloads the IFD's flight plan and loads it into the active plan (replacing the current one). Matched waypoints keep their database type; unmatched ones become GPS coordinate points.

UC-13: Back up and restore app data

1. Open `MAP → Menu → Internet → Backup/Sync`.
2. Sign in if prompted; the `Backup/Sync` screen opens right after.
3. Use:
   - **Backup** (upload icon) to upload local `user.db`
   - **Restore** (download icon) to overwrite local data from cloud copy
4. Confirm prompts carefully (both operations overwrite existing data).
5. Progress percentage shows during transfer.

UC-14: Get flight help from community resources

1. In onboarding or browser, open Apps4Av forum:
   - `https://groups.google.com/g/apps4av-forum`
2. Search existing threads for similar workflows/devices.
3. Post issue details:
   - Platform/device
   - Receiver type
   - What screen/action failed
   - Any warning drawer messages

UC-15: Use the CAP Grid overlay for search and rescue

1. Go to `MAP → Layers`.
2. Make sure **Nav** opacity > 0, then set **CAP Grid** opacity > 0 in the sub-list under **Nav**.
3. Zoom to level 9 or higher (grid only renders at sufficient zoom).
4. Grid squares appear with identifiers (e.g., BOS42, SEA123).
5. Use grid coordinates for communication with ground teams or other aircraft.

UC-16: Choose a different aircraft icon on the map

1. Open `MAP → Menu → Aircraft & Performance`.

2. Go to the **Custom** tab.
3. Select or create an aircraft profile.
4. Use the icon dropdown to select from: plane, helicopter, or canard.
5. Save the aircraft profile.
6. The map will display your selected icon for ownship.

UC-17: Load a plan in reverse order for a return flight

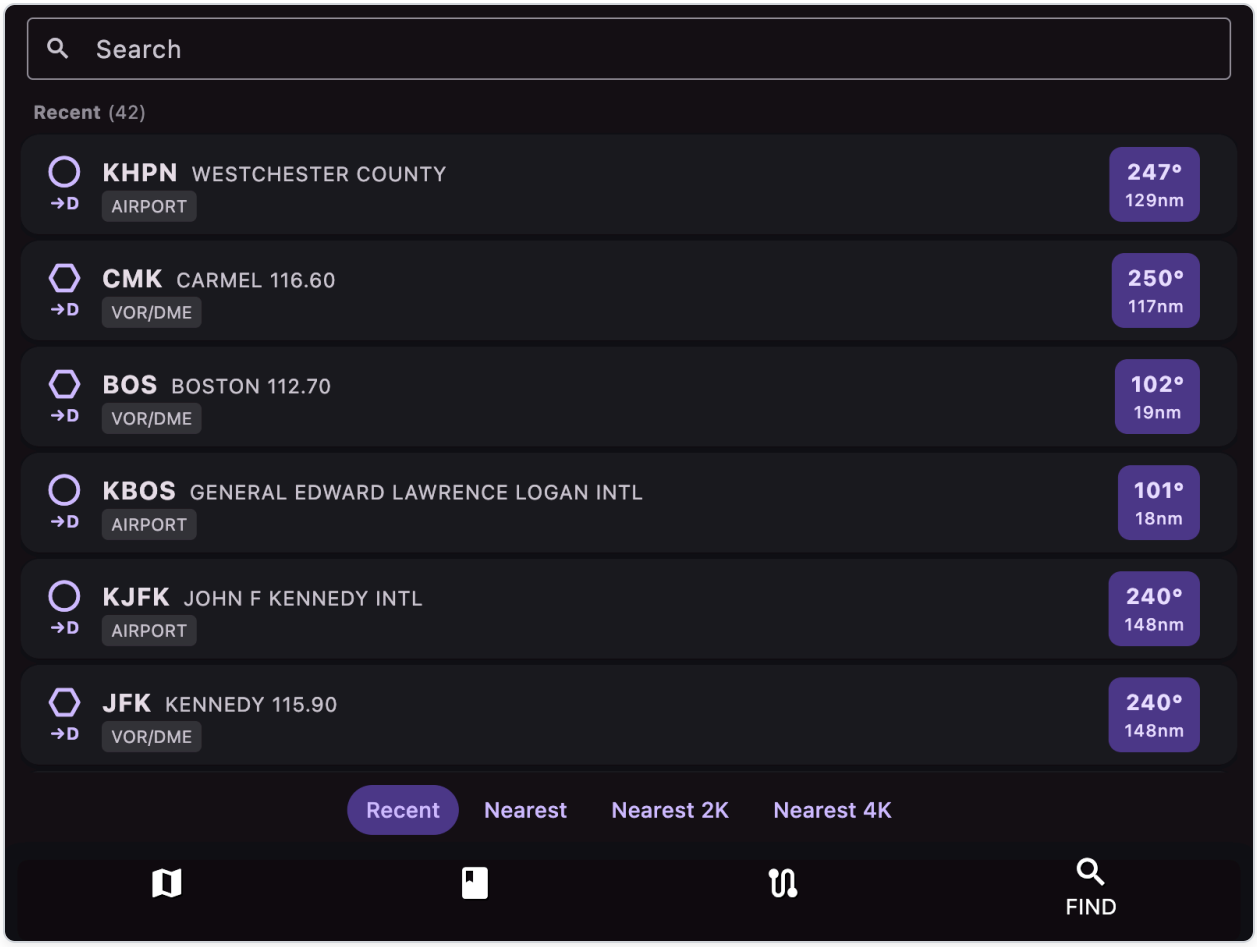
1. Open `PLAN → Load & Save`.
2. Find your saved outbound plan.
3. Tap the 3-dot menu on the plan.
4. Select **Load Reversed**.
5. The plan loads with waypoints in reverse order, ready for your return trip.

UC-18: Check your pilot currency status

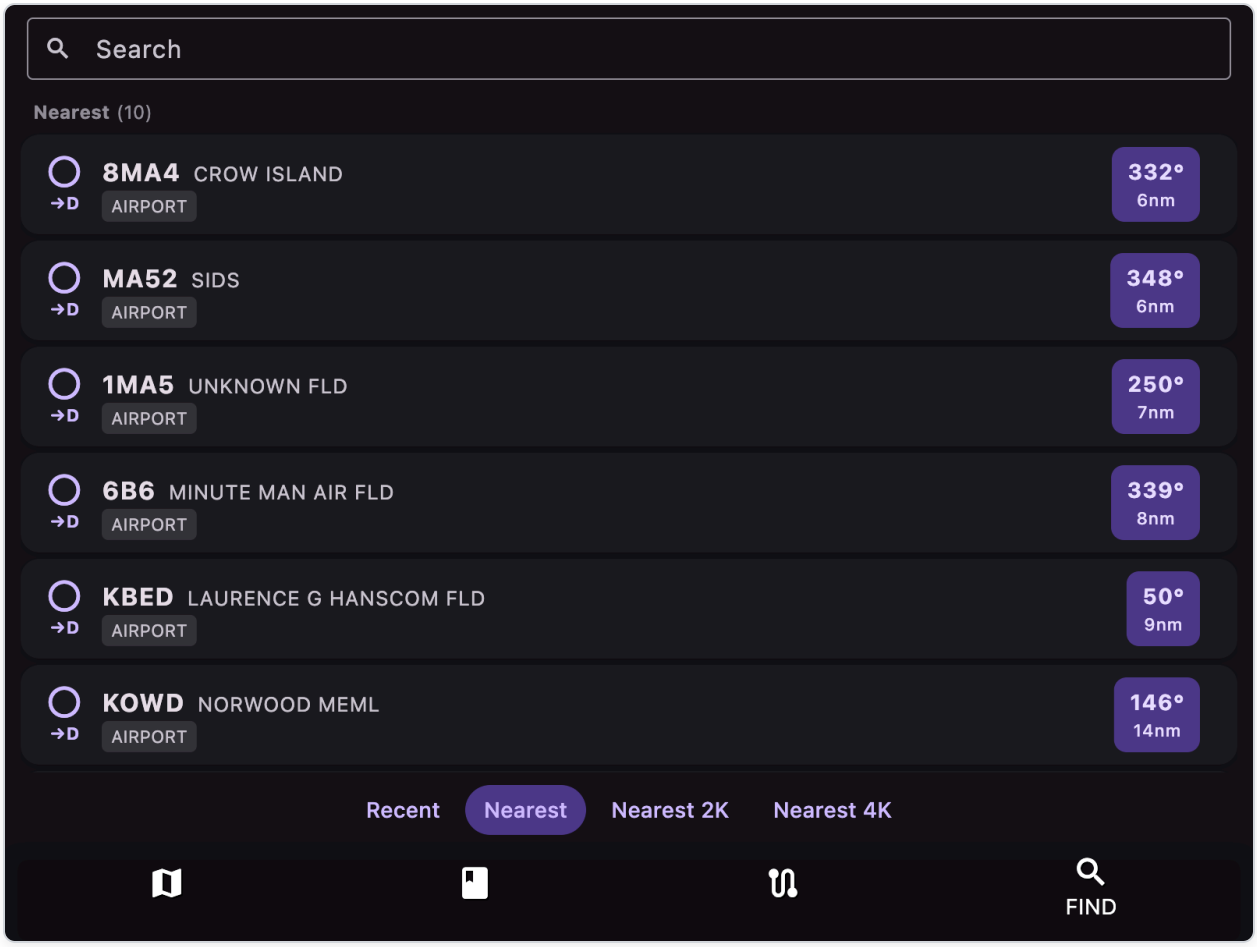
1. Open `MAP → Menu → Log Book`.
2. Tap the **Details** button.
3. View the Currency Status Card at the top showing:
    - Day currency: 3 landings in last 90 days
    - Night currency: 3 night landings in last 90 days
    - IFR currency: 6 approaches in last 6 months
4. Green = current, Red = not current

UC-18a: Search and find a destination

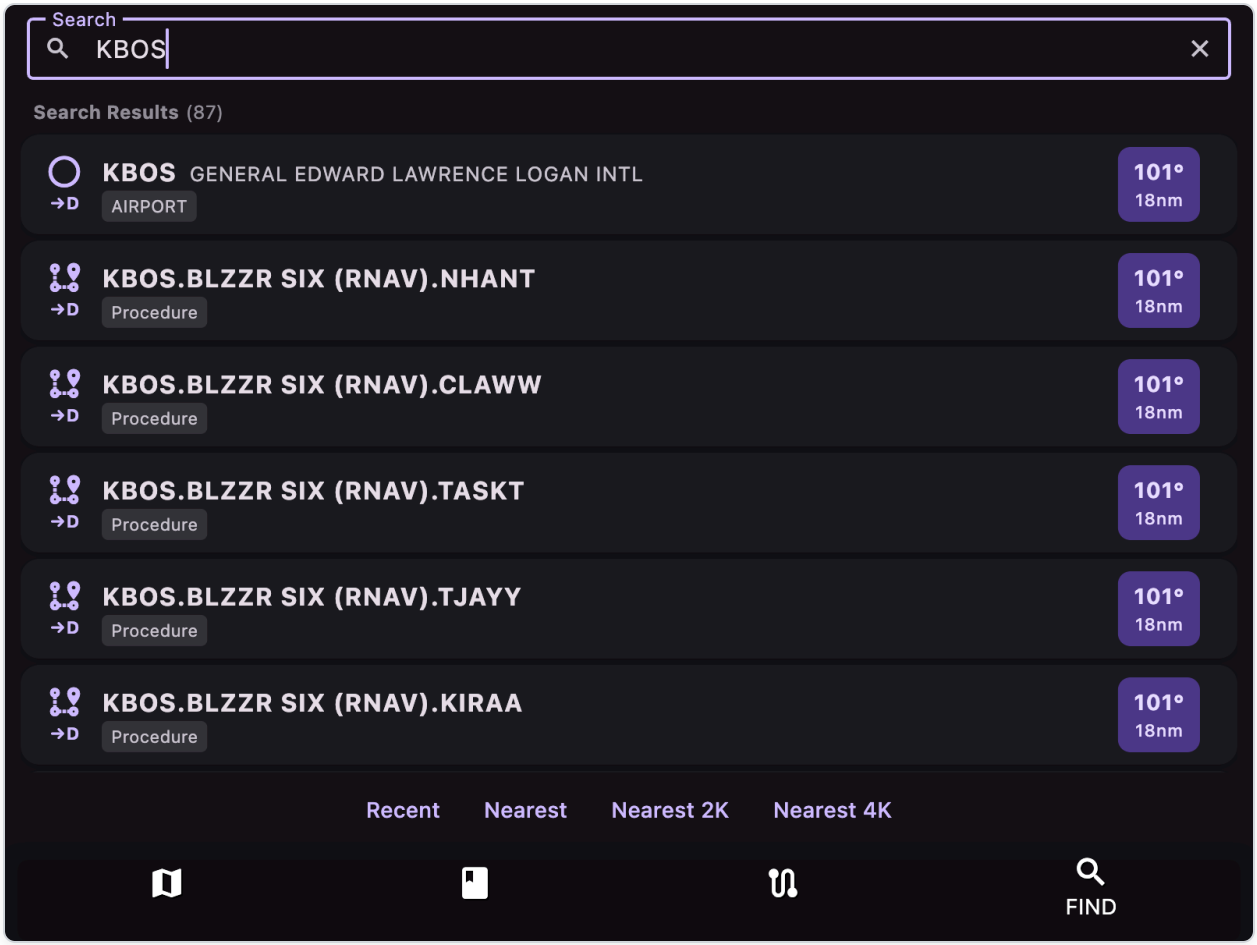
The **FIND** tab is the fastest way to locate any airport, navaid, fix, or procedure and act on it.



1. Tap the **FIND** bottom tab.
2. Use one of the four bottom filter buttons:
    - **Recent** — destinations you have viewed before
    - **Nearest** — every airport within range, sorted by distance from current ownship position
    - **Nearest 2K** — same, but only airports with ≥ 2000 ft of usable runway
    - **Nearest 4K** — same, but ≥ 4000 ft of runway



3. To search by identifier or text, tap into the **Search** field and start typing. Results appear immediately.



4. Each row supports several actions:
    - **Tap row** — open the destination popup with airport info, plates, METAR/TAF, NOTAMs, etc.
    - **Tap the bearing/distance pill on the right** — center the map on that destination and switch to MAP.
    - **Swipe right-to-left** — delete an entry from the Recent list.
    - For GPS-coordinate recent entries, an **edit (pencil) icon** lets you set a custom facility name.
5. From the destination popup you can:
    - **→D** Direct-To this destination
    - **+Plan** insert into the current plan after the active waypoint
    - **Hold** beside the name and type on the Main tab of a fix or navaid opens the holding-pattern dialog
    - **↓Plan** append to the end of the plan
    - **Plates** (airports) jump to the PLATE tab for that airport, even when it is not the active waypoint

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UC-19: Measure distance and bearing on the map

1. On `MAP`, tap the gear icon, then tap `Measure Distance Off` to enable measure mode (the row changes to `Measure Distance On` and is highlighted).
2. Long-press on the map to place your first point.
3. Long-press again to add additional points.
4. Distance (NM) and bearing (degrees) are shown between points.
5. Tap `Measure Distance On` to disable and clear measurements.

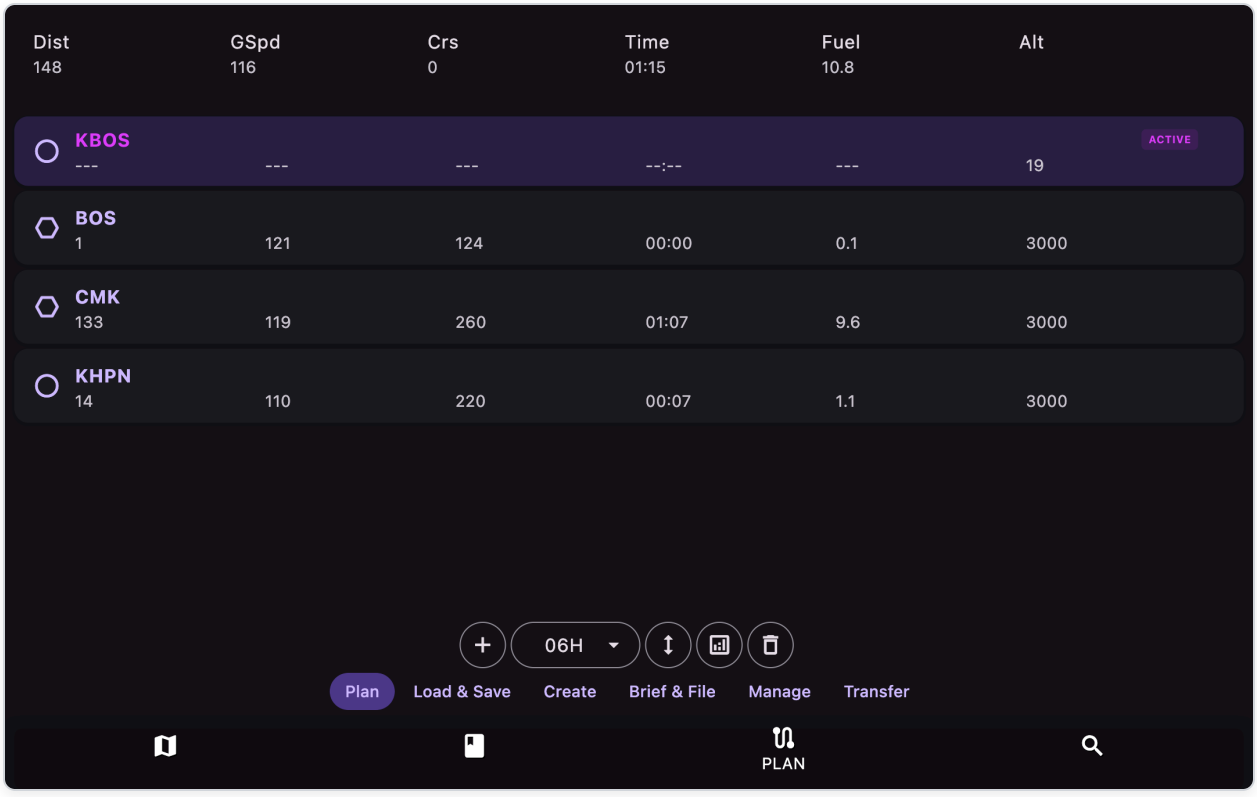
UC-20: Insert a waypoint at a specific position in your plan

1. Open `PLAN` tab with an existing route.
2. Find the waypoint you want to insert after.
3. **Long-press** that waypoint row.
4. The app switches to `FIND` tab.
5. Search for or select the waypoint you want to insert.
6. Tap `+Plan` or `↓Plan` — the new waypoint inserts after the long-pressed position.

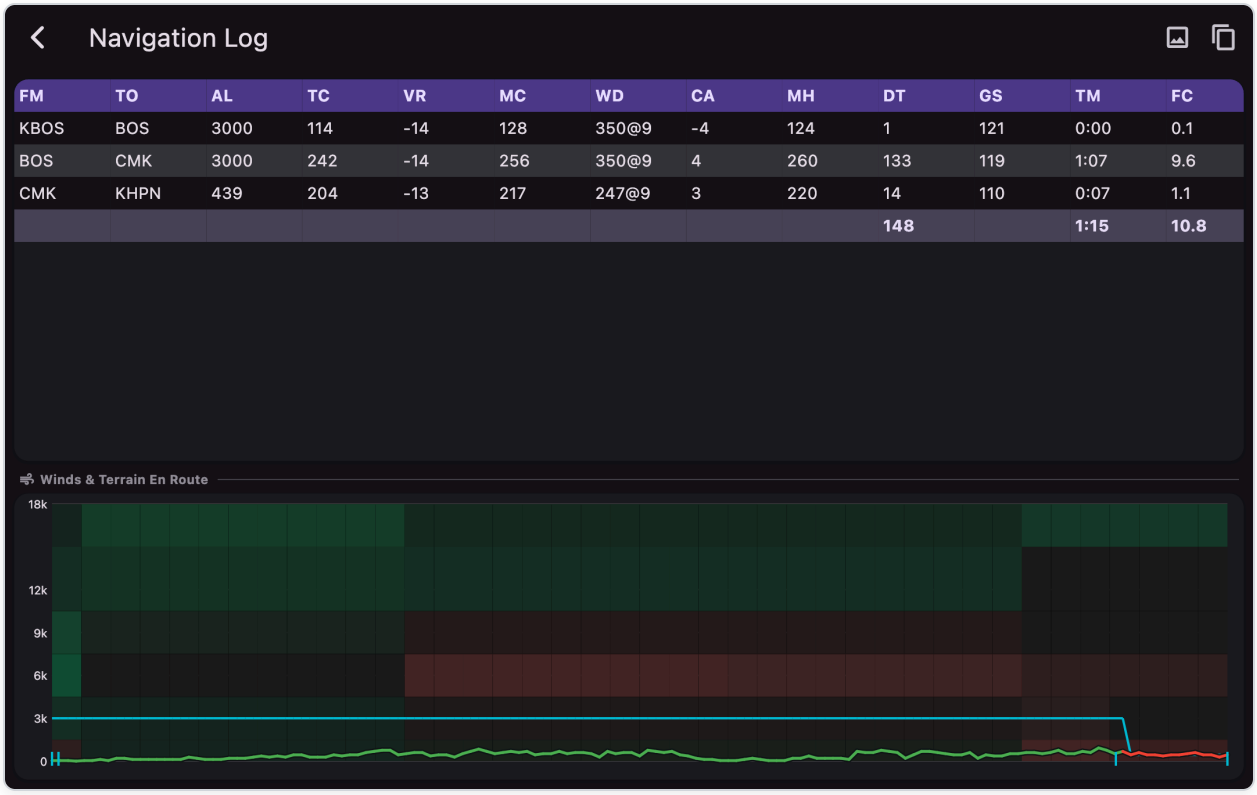
UC-21: Get the navigation log (winds & terrain analysis)

The navigation log gives you per-leg headings, magnetic variation, wind correction, ground speed, time, and fuel; plus a wind-and-terrain heat map at planning altitudes.

1. Open the **PLAN** tab.
2. Ensure your route has at least 2 waypoints (build one with **UC-04** if needed).



3. Set each leg's altitude in the **Alt** column. An empty cell uses the plan default (3000 ft, or the map altitude slider).
4. Set the **Forecast horizon** dropdown (`06H / 12H / 24H`) so wind data matches your departure window.
5. Cruise speed and fuel flow are read once from the selected aircraft when PLAN opens.
6. Tap the **analytics icon** (graph, to the right of the altitude icon, tooltip *"Navigation log and terrain"*). The **Navigation Log** opens as a full-screen dialog.



7. The **per-leg grid** at the top has 13 columns:

Col	Meaning
FM	From waypoint
TO	To waypoint
AL	Cruise altitude for that leg (ft). A waypoint **Alt** when set, otherwise the plan **Alt**
TC	True course (°)
VR	Magnetic variation (°)
MC	Magnetic course (°)
WD	Wind direction@speed (e.g. `210@53`)

Col	Meaning
CA	Wind correction angle (°)
MH	Magnetic heading to fly (°)
DT	Distance for this leg (NM/SM)
GS	Ground speed for this leg (kt/mph)
TM	Leg time (HH:MM)
FC	Fuel consumed on this leg (gal)

The bottom row of the grid is the **plan total** (sum of DT, TM, FC across all legs).

Double-tap the grid to reset zoom; pinch/scroll to focus on a specific leg.

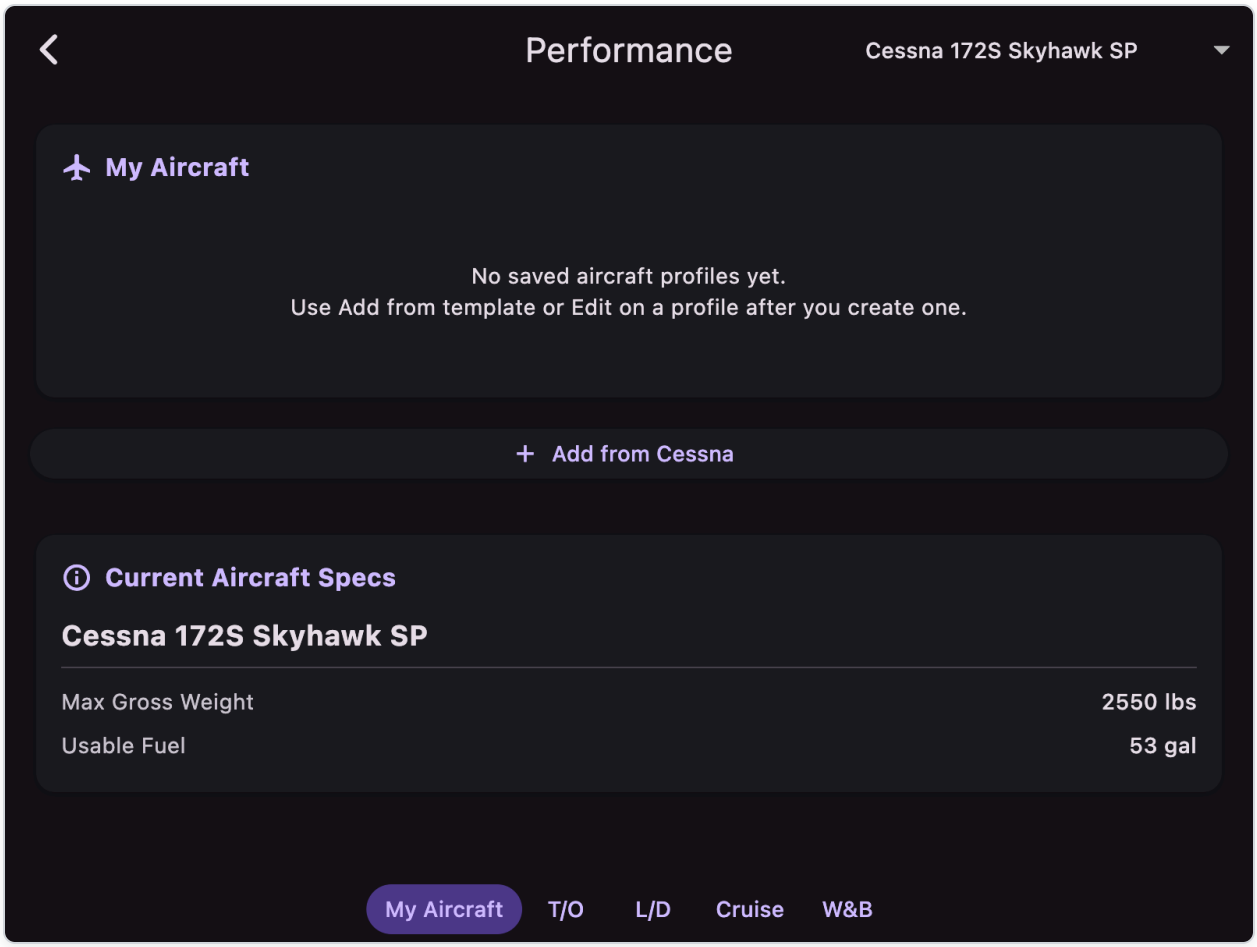
8. Below the grid, the **Winds & Terrain En Route** diagram is a single combined visualization for the whole route:
    - Y-axis: altitude bands (0, 3000, 6000, 9000, 12000, 18000 ft)
    - X-axis: position along the route from departure (left) to destination (right)
    - **Wind heat map**: each cell colored by tailwind/headwind component — **green = tailwind**, **red = headwind** (darker = lighter winds, brighter = stronger winds, scaled up to 50 kt)
    - **Cyan altitude line**: the altitude flown. After the altitude icon runs, it climbs and descends at best climb speed and rate and ends at the arrival field elevation
    - **Green/red terrain curve**: terrain elevation; green where it is below that leg's altitude, red where it is at or above it
    - **Cyan tick marks** along the bottom edge: waypoint positions
9. **Tap anywhere on the diagram** to display a tooltip showing:
    - Nearest waypoint (only if your tap is within ~5% of that waypoint's position)
    - Altitude at the tap (snapped to one of the altitude bands)
    - Terrain elevation at that point along the route
    - Course direction
    - Wind direction and speed at that altitude
    - Tailwind/headwind component (kt)
10. Use the **Copy plan to clipboard** icon in the app bar (top-right) to share the per-leg data outside the app.

11. Press **Back** (top-left chevron) to dismiss the dialog and return to the PLAN editor.

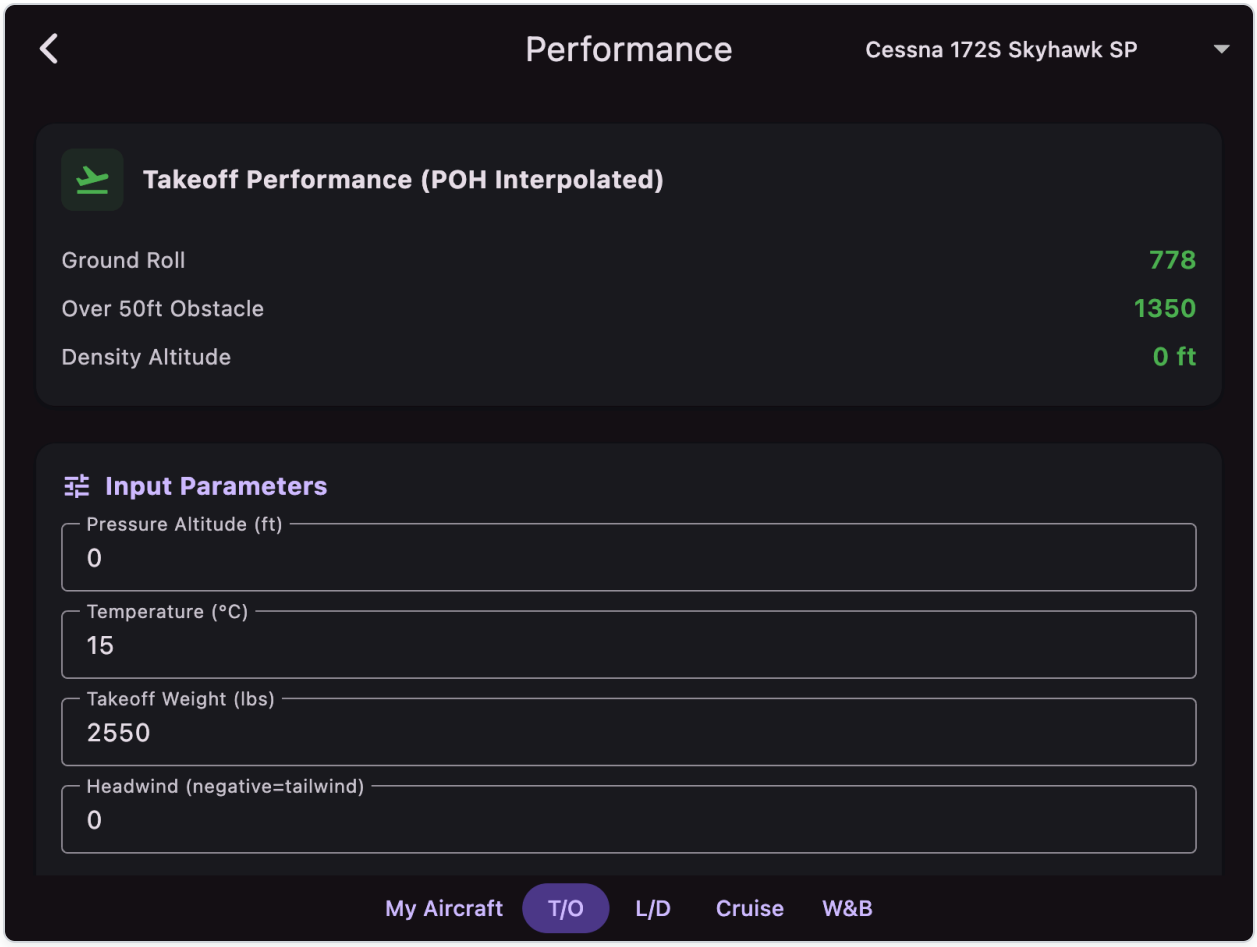
> Tip: The route in the screenshot above (`KBOS → BOS → CMK → KHPN` , plan altitude 3000 ft) shows mostly **red** at the cruise band — this means a headwind across the whole route at the chosen altitude. Use the diagram to see if a higher altitude band gets you a tailwind, or change the **Forecast horizon** dropdown to align with your actual departure window.

UC-22: Calculate takeoff and landing performance

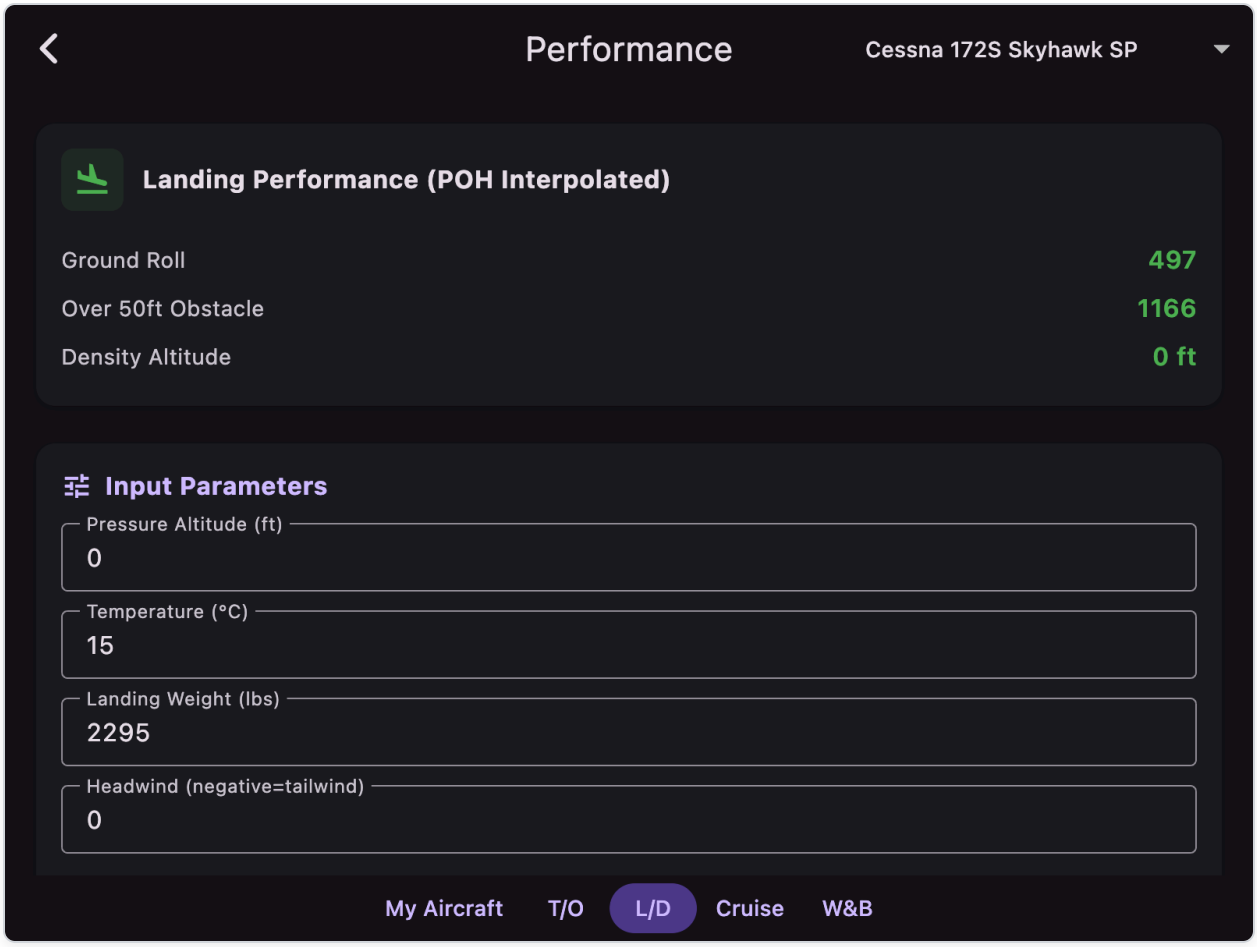
1. Open `MAP → Menu → Aircraft & Performance` .



2. Select your aircraft from the dropdown (built-in or custom). The dropdown is at the top-right of the screen.
3. Go to the **T/O** tab.



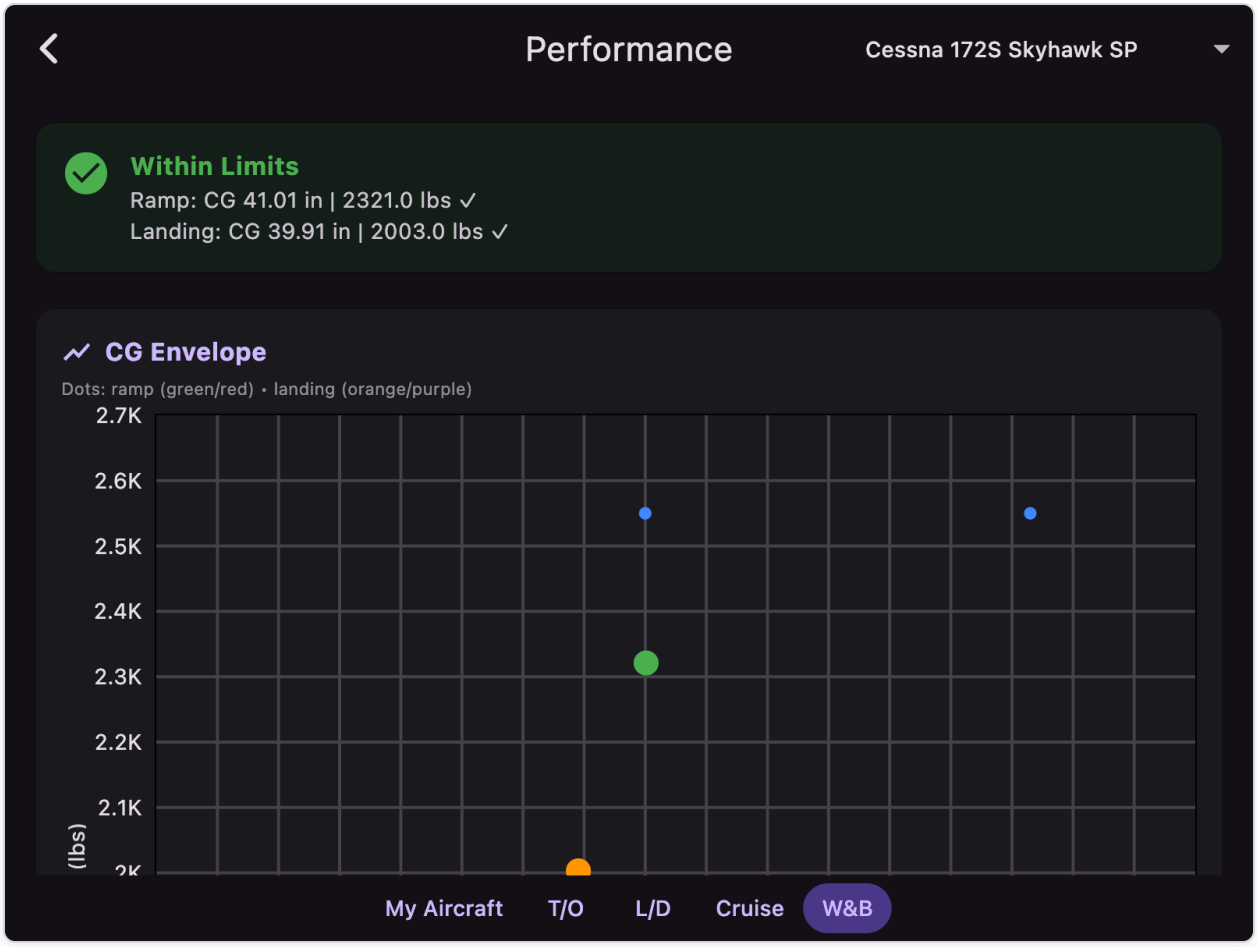
4. Enter current conditions:
    - **Pressure Altitude** (ft)
    - **Temperature** (°C)
    - **Takeoff Weight** (lbs)
    - **Headwind** (use a negative value for a tailwind)
5. View calculated **Ground Roll**, **Over 50ft Obstacle**, and **Density Altitude** distances at the top of the screen.
6. Switch to the **L/D** tab and repeat for landing performance.



7. Results account for aircraft POH data and any configured correction factors (headwind %, tailwind %, soft-field %) on the **My Aircraft** profile.

UC-23: Check weight and balance before flight

1. Open `MAP → Menu → Aircraft & Performance`.
2. Select your aircraft from the dropdown.
3. Go to the **W&B** tab.



4. Enter weights for each station in the table below the chart (passengers, baggage, fuel, plus Landing fuel).
5. The status card shows **Within Limits** (green) or **Outside Limits** (red), with a ✓ or ✕ for ramp and landing rows.
6. Read the CG Envelope chart:
    - Blue dots = envelope boundary
    - Large green/red dot = ramp/takeoff CG
    - Large orange/purple dot = landing CG
7. Adjust loading until both dots are inside the envelope.
8. For custom aircraft, use **Edit** to modify envelope points and station definitions.
9. If your aircraft uses the **helicopter** map icon, check the lateral CG chart and lateral columns as well; all four CG checks must pass for a green status.

UC-24: Create a pilot community group

1. Open `MAP → Menu → Internet → Community`.
2. Sign in if prompted.
3. Tap the **New Group** floating action button.
4. Fill in:
    - **Group name** (3–60 characters)

    - **Description** (optional, up to 280 characters)
    - **Home airport** (optional ICAO, e.g. `KBED`)
5. Choose **Visibility**:
    - **Public** — anyone can join immediately.
    - **Private** — discoverable by name, but you approve every new member.
6. Tap **Create Group**. You land on the new group's **Feed** tab as the owner.
7. Tap **Post** to share your first message; optionally attach an airport ICAO to the post.

UC-25: Join a private pilot group

1. Open `Community → Discover` and search for the group by name.
2. Tap the group to open it. Private groups show a lock icon and the feed is hidden.
3. Tap **Request to Join**. The banner switches to **Requested** and your status is `pending`.
4. The group owner will see your request in the **Members** tab under **Pending requests**.
5. When approved, the **Feed** unlocks and you can post immediately. You can leave at any time with the **Leave** button.
6. To cancel a pending request, tap **Requested** to withdraw it.

UC-26: Set up an aircraft scheduler and book a flight

1. Open `MAP → Menu → Internet → Scheduler`.
2. Sign in if prompted.
3. Tap the **New Scheduler** floating action button, fill in the name (and optional description/home airport), then tap **Create Scheduler**. Every scheduler is private; you land on the new scheduler as the owner.
4. On the **Schedule** tab, tap **Add resource** to add your aircraft (e.g. `Cessna 172`, tail `N12345`) and any instructors. They appear as rows, with time across the top.
5. (Optional) Tap the **tune** icon in the app bar to open **Booking Rules** and set how many reservations — and how many weekend reservations — each member may hold at once (0 = unlimited). The owner is exempt from these limits.
6. Share the scheduler name with your members; they open `Scheduler → Discover`, search the name, and **Request to Join**. Approve them on the **Members** tab. Use the badge icon to set **Student** / **Instructor** / **Dispatcher** roles and assign students to instructors.
7. On **Dispatch → Fleet**, tap an aircraft and **Update meters / MX** to set hobbs, tach, and inspection due dates for the shared fleet.
8. (Optional) On **Dispatch → Lessons**, create a **lesson pack** of prepaid hours for a student.
9. To book, a member taps a **green** slot on a resource row, picks a **start date/time** and **end date/time** (bookings can span multiple days, up to 14), optionally picks an instructor / lesson pack, then taps **Book**. The slot turns **blue**. Members can review their bookings on the **Mine** tab.

10. If that time is already taken, the member is added to the **backup** queue. If the main reservation is later cancelled, the first backup is promoted automatically.
11. To cancel, tap the reservation block (or use the **Mine** tab) and choose **Cancel**. Only the member who booked it or the owner can cancel a reservation.
12. For discrepancies, use **Dispatch → Squawks → File squawk**. A **Grounding** squawk blocks booking and shows as GROUNDED on the fleet board. For a hard maintenance hold, turn off **Available for booking** from Schedule or Dispatch (owner/dispatcher).

UC-27: Keep the same plan on a second phone or tablet

1. Join every device to the same Wi-Fi network. On iOS, allow **Local Network** for AvareX.
2. On the device that will be edited, open `MAP → Menu → Link` and tap **Master**.
3. On each other device, open `Link` and tap **Slave**. They appear under **Devices**.
4. Build or change the plan on the master, and tap the waypoint to fly. Each slave loads that plan and sets the same active waypoint, including the current fix on an airway or procedure.
5. Open **Messages** on either device to see the JSON sent (**TX**) and received (**RX**). Tap a row to read it.
6. Tap **Off** on a device to stop following, or to stop sending.

UC-28: Use Emergency mode

1. Before the flight, choose your aircraft on `Aircraft & Performance` and, if you want your own checklist, pick it on the `Emergency` tile (`Profiles` heading, three-dot menu).
2. If the engine quits, catches fire, or runs out of fuel, fly the airplane first. Then open `MAP → Map Settings → Profiles → Emergency`.
3. Read **A** and fly the best glide speed. The card flashes if you are more than 5 knots away.
4. Read **B**: it shows the runway to glide to, with the red arrow on the map for the landing direction. Tap `LOCK` once you have decided, so it stops changing.
5. Work through **C**, tapping the card for each next step.
6. Use **D** to call on the CTAF or UNICOM shown, or on `121.5`, and set the transponder to `7700`.
7. Tap `EXIT` and confirm to return to the normal map.

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16) Resources and Support

- Apps4Av Forum (Google Groups):
  `https://groups.google.com/g/apps4av-forum`
- Store links and platform download details: see `README.md`.
- For in-app issues, always check the MAP warning drawer first (red icon).

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17) Forum-Derived FAQ and Scenarios (Google Groups)

The following FAQs are derived from recent threads in the Apps4Av forum and mapped to the current AvareX workflows.

FAQ-01: How do I connect an external ADS-B / GPS receiver?

- For Wi-Fi receivers: connect the device to receiver network and AvareX auto-listens on UDP `4000`, `43211`, `49002`.
- **Stratus 3 / 3i**: if traffic/weather never appear, put the unit in Open ADS-B mode from `MAP → ADSB` tile → antenna icon (top-right) (or via Appareo Stratus Horizon Pro).
- For Bluetooth receivers (Android): `Menu → IO`, pair/connect, then return to map.
- **iOS only**: AvareX must have **Local Network** permission enabled (`Settings → Privacy & Security → Local Network → AvareX`). Without it, iOS silently blocks all ADS-B/GDL90 broadcasts and traffic/weather will never show, even though ownship from the internal phone GPS still works. The first time you launch AvareX after install you should see a prompt — tap **Allow**. See FAQ-16.
- Source threads:
  - GPS source discussion:
    `https://groups.google.com/g/apps4av-forum/c/e0ujCJQX1s8`
  - In-flight ADS-B issues:
    `https://groups.google.com/g/apps4av-forum/c/ZP2E9l35Dk8`
  - Android IO functionality:
    `https://groups.google.com/g/apps4av-forum/c/WI-BcDK7rT0`

FAQ-02: How do I transfer a plan to or from an external device?

- Connect this device to the **same Wi-Fi network** as your Avidyne IFD.
- Open `PLAN → Transfer` and wait for the IFD to appear.
- To upload: build a route with at least 2 waypoints, then tap **Send**.
- To download the IFD's plan into AvareX: tap **Get** (this replaces the active plan).

Supported avionics: Panel-mounted Avidyne IFD440/540/550 units with flight-plan input enabled. On iOS, grant the **Local Network** permission for discovery and upload to work.

If the IFD has an ADS-B receiver, AvareX also picks up its **Capstone** ADS-B traffic and weather automatically over the same Wi-Fi network (Capstone is GDL90 data received on UDP `4000`). See FAQ-16 for the iOS Local Network requirement.

FAQ-03: Where do I enable new wind/ceiling map features?

- `MAP → Layers → Weather` (opacity > 0)
- In the sub-list that appears under **Weather**, raise **Wind Vectors** or **Ceiling** above 0%
- Use the altitude slider to change displayed altitude context
- Source threads:
  - New features v83/v84:
    `https://groups.google.com/g/apps4av-forum/c/qac4-Bb-gVE`
    `https://groups.google.com/g/apps4av-forum/c/Xvhc9yeE42A`

FAQ-04: My track logs are empty. How do I save valid KML logs?

- Keep `Tracks` layer ON (opacity > 0) during flight.
- After flight, turn `Tracks` layer OFF (opacity = 0) — this saves KML to Documents/tracks folder and clears active track buffer.
- Open `Documents → tracks` and verify non-empty KML before sharing.
- Tap the KML file to view it in Track Viewer with 2D map, altitude profile, and 3D terrain views.
- Use **Log Flight** button to create a logbook entry directly from the track.
- Source thread:
  `https://groups.google.com/g/apps4av-forum/c/goPy8KQpfik`

FAQ-05: Where are FBO/business details?

- In the destination popup, open the **Business** tab (free; sign-in required). The airport's businesses are listed inline; tap one to view and contribute services, fuel, hours, and star reviews per airport (see Section 11.7).
- On the `PLATE` screen, when an airport diagram is active, use the right-side business selector (business icon) to mark a business's location on the diagram. It draws from the same crowd-sourced directory (iOS/Android/Windows/macOS, sign-in required).
- Source thread:
  `https://groups.google.com/g/apps4av-forum/c/oMOR-gaIqis`

FAQ-06: What are the colored rings on map?

- In the `Circles` layer:
  - **Black rings**: fixed 2/5/10 NM reference rings; the 10 NM ring is marked with a magnetic compass rose (ticks and labels at the eight major points: `N` / `NE` / `E` / `SE` / `S` / `SW` / `W` / `NW`)
  - **Blue ring with a yellow halo**: speed-based 1-minute travel distance
  - **White ring with an orange halo**: glide circle based on the selected aircraft's best glide speed and sink rate, plus winds aloft, terrain, and altitude. The same ring is drawn on the surface in the 3D view, with a cone from the aircraft down to it
- Source thread:
  `https://groups.google.com/g/apps4av-forum/c/VJ0S3ejWPC8`

FAQ-07: My waypoints appear in wrong order. Can I reorder or insert quickly?

- **Reorder**: In `PLAN`, long-press and drag rows to reorder legs.
- **Insert at specific position**: Long-press a waypoint row to insert after it (opens FIND; next selection inserts after that waypoint).
- **Set current leg**: Tap a waypoint row.
- Source thread:
  `https://groups.google.com/g/apps4av-forum/c/T-m4BWZynMg`

FAQ-08: How do I enter an ATC reroute quickly?

- Open `PLAN → Create`.
- Use **Create As Entered** and paste/type reroute string (space-separated waypoints/airways).
- Then review/reorder in PLAN list if needed.
- Source thread:
  `https://groups.google.com/g/apps4av-forum/c/ukaXEEhpvS0`

FAQ-09: Download/cycle issues - what to try first?

- In `Download` screen:
  - Try **This Cycle** vs **Next Cycle**
  - Try **Main Server** vs **Backup Server**
  - Keep download screen open until completion (exiting can abort incomplete downloads)
- Source threads:

    - Cycle download issue:
      `https://groups.google.com/g/apps4av-forum/c/LS_VaqKEJxw`
    - Background download request:
      `https://groups.google.com/g/apps4av-forum/c/XcoQzwpCxlw`

FAQ-10: FAA IFR preferred route tools not working - what now?

- Ensure app is updated to latest release and internet is available.
- Confirm your 1800wxbrief-compatible email is configured.
- Retry via `PLAN → Create` and `Brief & File`.
- Source thread:
  `https://groups.google.com/g/apps4av-forum/c/0wHqcJT-WiY`

FAQ-11: Is Flight Intelligence (AI) available on desktop?

- Current Pro AI workflows are targeted for iOS/Android only.
- Access from `MAP → Menu → Internet → Flight Intelligence` or the map account icon (opens Flight Intelligence when signed in, or Account sign-in if not).
- Uses Gemini 3.8 Flash with Google Search integration.
- Source threads:
    - AI feature thread:
      `https://groups.google.com/g/apps4av-forum/c/wWZUn6TNG1w`
    - Desktop/pro services request:
      `https://groups.google.com/g/apps4av-forum/c/XcoQzwpCxlw`

FAQ-12: Can I use AvareX with X-Plane/simulator feeds?

- AvareX accepts external NMEA/GDL90-like streams via UDP listener ports (4000, 43211, 49002).
- For simulator setups, configure simulator/network output accordingly.
- Source thread:
  `https://groups.google.com/g/apps4av-forum/c/QOfnQ-pkT0w`

FAQ-13: What is the CAP Grid layer?

- Civil Air Patrol grid overlay for search and rescue operations.
- Shows grid identifiers (e.g., BOS42, SEA123) at zoom level 9+.
- Enable via `MAP → Layers → Nav` opacity > 0, then **CAP Grid** opacity > 0 in the sub-list under **Nav**.
- Covers all US sectional chart areas.

FAQ-14: How do I view my flight in 3D?

- Record your flight with `Tracks` layer enabled.
- Turn `Tracks` off to save the KML file.
- Open `Documents → tracks` and tap the KML file.
- Use the 3D View icon (AR) to see interactive 3D visualization with terrain and topo map texture.

FAQ-15: How do I check if I'm current to fly?

- Open `MAP → Menu → Log Book → Details`.
- The Currency Status Card shows your currency for:
  - Day VFR (3 landings in 90 days)
  - Night (3 night landings in 90 days)
  - IFR (6 approaches in 6 months)
- Status is automatically calculated from your logbook entries.

FAQ-16: On iOS, my ownship moves but I never see traffic from Stratux/Stratus/Echo, even though FltPlanGo and ForeFlight see it on the same phone

This is a Local Network permission issue, not a bug in your receiver or in AvareX's traffic engine.

Since iOS 14, Apple silently blocks all incoming UDP broadcasts (including GDL90 traffic on UDP `4000`, ForeFlight on `43211`, X-Plane on `49002`, and AvareX Link on `49090`) until the user grants the **Local Network** permission. The internal phone GPS keeps working, so ownship still moves and the map looks correct — that's the trap. ADS-B traffic, ADS-B weather (NEXRAD/METAR/TAF/AIRMET/SIGMET via FIS-B), and external GPS lock from your receiver will all be missing.

Fix:

1. On the iPhone/iPad, open `Settings → Privacy & Security → Local Network`.
2. Find **AvareX** in the list and turn the toggle **ON**.
3. Force-quit AvareX (swipe up from app switcher) and relaunch it.
4. Make sure the device's Wi-Fi is joined to the receiver's network (e.g. `stratux`, `Stratus-XXXX`, `EchoUAT-XXXX`).
5. Open `MAP` and turn the `Traffic` layer on. Traffic should appear within a few seconds.

If **AvareX** is not listed at all under `Local Network`, you are running an older build that predates the Local Network entitlement — update to the latest AvareX from the App Store.

Notes:

- This setting is per-app and per-device; each iPhone and iPad must be configured separately.
- A factory reset of network settings or "Reset Location & Privacy" will clear this — you'll need to re-enable it.
- This does not affect Android. Android handles Wi-Fi UDP without an extra runtime permission, which is why the same Stratux works fine with AVARE on Android.
- AvareX always listens on UDP `4000` (Stratux/Stratus/Echo GDL90, and Avidyne IFD Capstone), `43211` (ForeFlight), and `49002` (X-Plane). There is no port to configure inside the app — if data is reaching the phone over Wi-Fi and Local Network is allowed, AvareX will pick it up.
